RQF LEVEL 3



NITCI301

NETWORKING AND INTERNET TECHNOLOGIES

CCTV System Installation

TRAINEE'S MANUAL

October, 2024

CCTV SYSTEM INSTALLATION

KOICA Korea International Cooperation Agency

TQUM TVET Quality Management Project

2024

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ACRONYMS

AHD: Analogue High Definition

BNC: Bayonet Neill–Concelman

Cat3,4,5,6,7 Category3,4,5,6,7

CCD: Closed Coupled Device

CCTV: Closed Circuit Television

CMOS: Complimentary Metal-Oxide Semi-Conductor

CoC: Control over Coax

CVBS: Composite Video Baseband Signal

CVI: Composite Video Interface

DSP: Digital Signal Processor

DVR: Digital Video Recorder

FPS: Frames Per Second

GOP: Group of Pixels

GUI: Graphic User Interface

HVEC: High Efficiency Video Coding

HD: High Definition

HDD: Hard Drive

IB: Infrablack

IP: Internet Protocol

IR: Infrared

LED: Light Emitting Diode

MP: Mega Pixel

NTSC: National Television System Committee

NVR: Network Video Recorder

PAL: Phase Alternating Line

PoC: Power over Coax

PoE: Power Over Ethernet

RCA: Radio Corporation of America

RF : Radio Frequency

RG59: Radio Guide 59

RJ: Registered Jack

RTB : Rwanda TVET Board

SDI: Serial Digital Interface

SSD: Solid State Drive

TVET: Technical and Vocational Education and Training

TVI: Transport Video Interface

TVL: Television Lines

INTRODUCTION

This trainee's manual includes all the knowledge and skills required in Networking and internet technology specifically for the module of **"CCTV System Installation".** Trainees enrolled in this module will engage in practical activities designed to develop and enhance their competencies. The development of this training manual followed the Competency-Based Training and Assessment (CBT/A) approach, offering ample practical opportunities that mirror real-life situations.

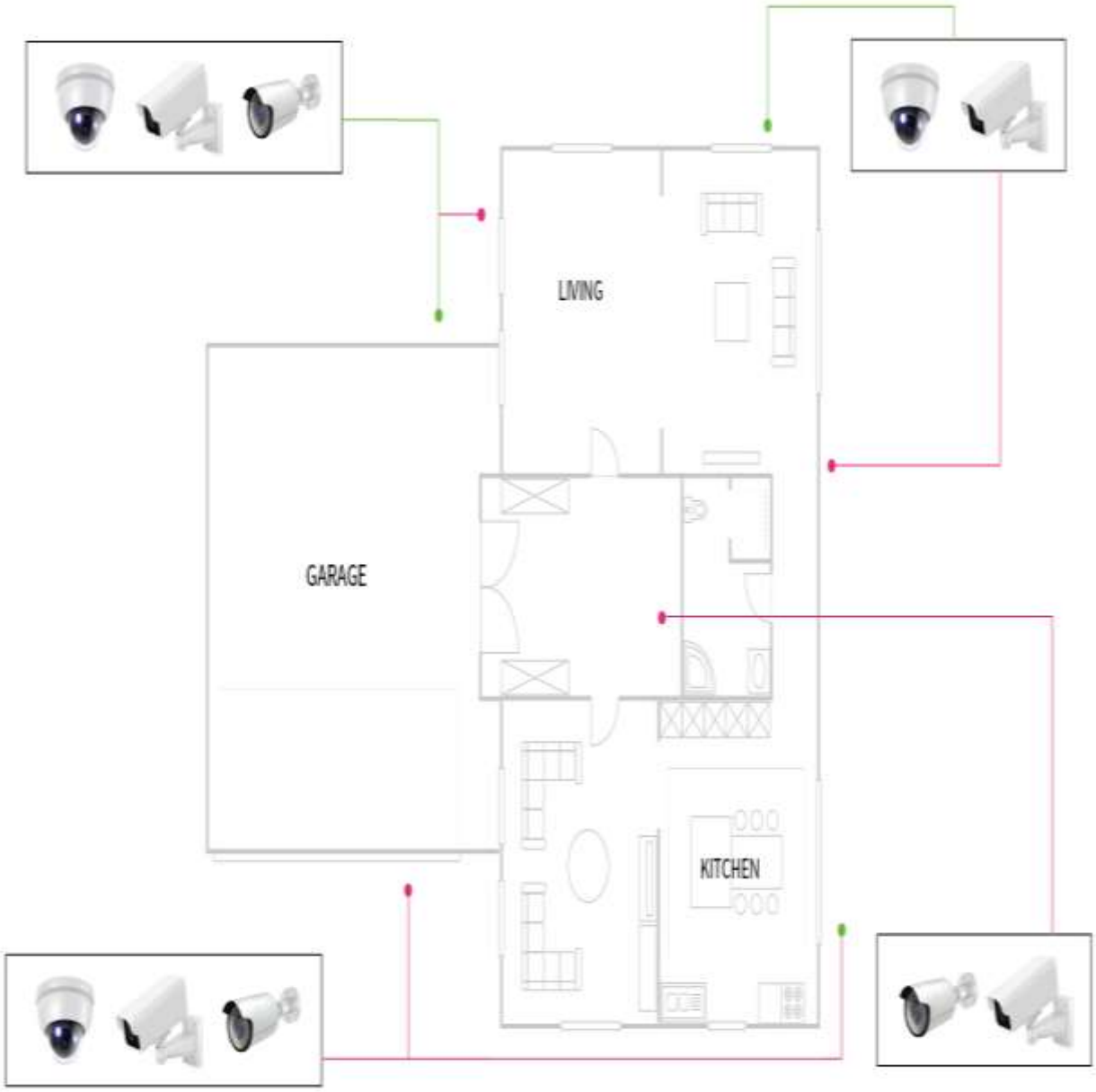
The trainee's manual is organized into Learning Outcomes, which is broken down into indicative content that includes both theoretical and practical activities. It provides detailed information on the key competencies required for each learning outcome, along with the objectives to be achieved.

As a trainee, you will start by addressing questions related to the activities, which are designed to foster critical thinking and guide you towards practical applications in the labor market. The manual also provides essential information, including learning hours, required materials, and key tasks to complete throughout the learning process.

All activities included in this training manual are designed to facilitate both individual and group work. After completing the activities, you will conduct a formative assessment, referred to as the end learning outcome assessment. Ensure that you thoroughly review the key readings and the 'Points to Remember' section.

MODULE CODE AND TITLE: NITCI301 CCTV SYSTEM INSTALLATION

Learning Outcome 1: Plan CCTV system installation
Learning Outcome 2: Deploy CCTV equipment
Learning Outcome 3: Configure CCTV System
Learning Outcome 4: Operate CCTV system
Learning Outcome 5: Test CCTV System
Learning Outcome 6: Maintain CCTV system


LIVING
GARAGE
KITCHEN

Indicative contents
1.1: Introduction to CCTV system
1.2: Identification of CCTV system installation requirements
1.3: Design of CCTV Diagram
1.4: Estimation of CCTV System Cost

Key Competencies for Learning Outcome 1: Plan CCTV system installation.

Knowledge	Skills	Attitudes
• Description of CCTV System. • Description of CCTV Camera types and components. • Description of CCTV System site survey and site survey types. • Identification of CCTV Diagram Design software.	• Conducting Site Survey • Selecting of CCTV system Components. • Selecting of site based on site survey. • Designing CCTV system Diagram using designing software. • Preparing CCTV system cost estimation of the required components.	• Being a Self-Motivated. • Being a Teamwork player. • Having a Curiosity. • Having a Self-confidence. • Being an innovator. • Be a Critical thinker.



Duration: 10 hrs

Learning outcome 1 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Describe properly CCTV system based on CCTV system requirements.
2. Identify properly CCTV Camera installation requirements based on site survey findings.
3. Design correctly CCTV Diagram based on CCTV system requirements.
4. Estimate cost properly as per CCTV system installation requirements.



Resources

Equipment	Tools	Materials
● Cameras ● camera Mount ● covers ● power cables ● Switch ● Monitors ● Storage devices ● DVR / NVR	● Simulation software	● Internet bundles ● cables ● Paper ● pencil

Indicative content 1.1: Introduction to CCTV system

Duration: 3 hrs

Theoretical Activity 1.1.1: Description of CCTV System

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of CCTV System.

i. What is a CCTV system?
ii. What is the CCTV System Purpose?
iii. How does the CCTV system work?
iv. Discuss the types of CCTV cameras based on their categories?
v. What are the components of the CCTV System?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class.

4: For more clarification, read the **key readings 1.1.1.**

5: In addition, ask questions where necessary.

Key readings 1.1.1.:

- **Description of CCTV System**

✓ **Definition of CCTV camera system**

A CCTV (Closed-Circuit Television) system is a security and surveillance technology that uses video cameras to transmit a signal to a specific, limited set of monitors or recording devices.

A CCTV system refers to all aspects and components that build up the system, such as the camera, wiring, video footage storage (via the cloud or digital video recorder), monitors, access control and general system management.

CCTV cameras are used for security monitoring and surveillance in any type of facility. CCTV systems transmit video footage over a single channel, essentially creating a closed circuit. This means CCTV footage is displayed on a limited number of personal monitors and screens, not publicly broadcasted.

CCTV camera (camera) is a unit containing an imaging device producing a video signal from an optical image.

✓ **CCTV System main purpose**

Surveillance: refers to the continuous observation of a specific area or multiple areas using video cameras. The primary goal of surveillance is to gather and record visual information for various purposes, including security, investigation, and

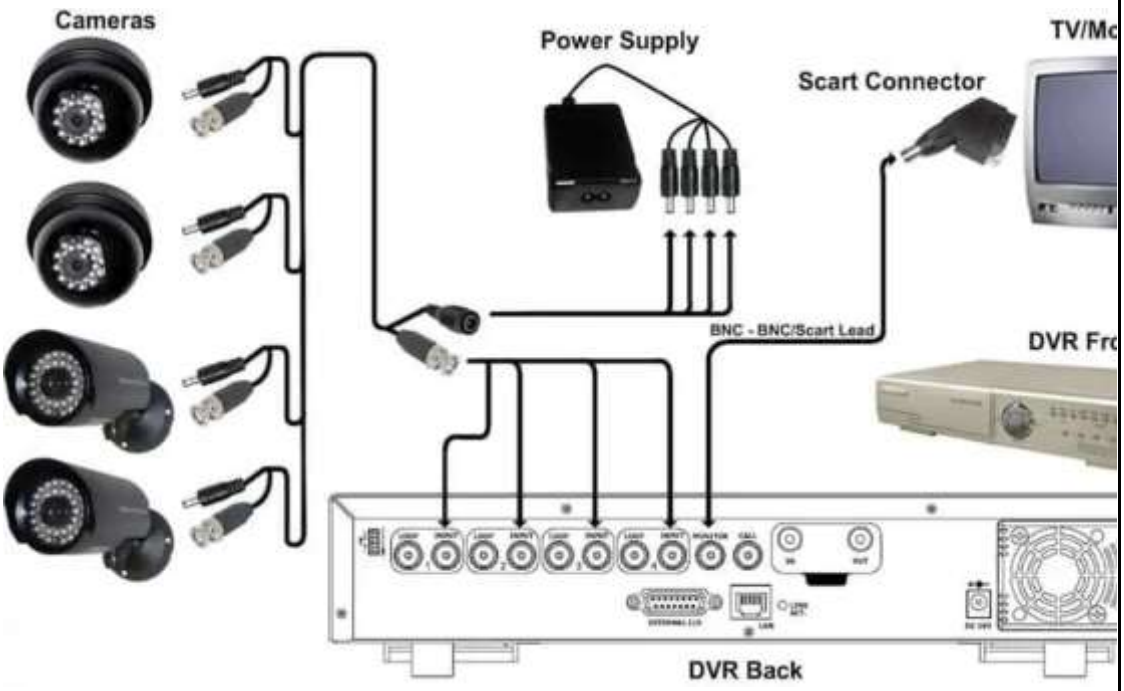
analysis. Video cameras are used to observe an area, connected to a recording device or IP Network and monitored in a control room.

- **Security:** refers to the measures and practices implemented to protect people, property, and assets from harm, theft, or damage. In the context of a CCTV system, security involves using surveillance tools and techniques to create a safe environment. The captured footage are used for controlling and evaluate environment.
- **Monitoring:** refers to the real-time observation and review of video footage captured by surveillance cameras. Monitoring ensures that any unusual or suspicious activity is detected and addressed promptly.

✓ **Working principles of CCTV system**

CCTV systems use cameras to capture video footage of a facility. This footage is sent to a recording device (a DVR or NVR), where it is stored for future reference. It is then sent from the cameras to a dedicated monitor, or even a smartphone or computer, which displays the footage in real time. CCTV security systems are connected with a coaxial cable, network cable, or wireless signal, depending on the system type.

CCTV works by the camera or cameras taking a constant sequence of images that are transmitted by cable or wirelessly (depending on the chosen system type) to the recording device to be displayed to the monitor, which enables an individual to see the sequence of images as video footage.



CCTV Systems: An Introduction to Security Video Surveillance (getsafeandsound.com)

✓ **Categories of CCTV Camera**

Categories of security cameras are categorized based on different criteria:

a. **Categories of CCTV System Based on Technology**

1. Analog CCTV Systems

Analog systems are the more traditional form of CCTV. These use coaxial cables to connect the camera and surveillance monitor or recording system. These systems tend to have a more difficult installation process, since they require more cabling than IP systems. Images in CCTV systems are captured in analog form and recorded on a DVR (Digital Video Recorder).

2. IP Systems

IP system is the modern systems using digital cameras and network cables. Images are captured in digital form and recorded on an NVR (Network Video Recorder) and can be accessed remotely via the internet.

IP systems, otherwise known as network cameras, are technically not "CCTV" in the same way as hardwired analog systems but for convenience they are often referred to as CCTV systems. IP-based systems run on digital IP and local area networks, they can run on a company's existing network, or can be set up on their own if data and bandwidth speeds are a concern. Unlike analog systems, IP CCTV cameras generally use ethernet cables, such as a Cat5E.

✓ **Analog Cameras versus IP/Digital Cameras**

Analog CCTV cameras offer high reliability according to their coax cables, but they generally shoot in much lower resolutions than their digital counterparts (though HD-over-Coax is a possibility).

IP security cameras are capable of much higher resolution up to 4k in many cases and thus provide better image quality. They are also usually compatible with many more features than analog based systems, such as video verification, video analytics, and advanced processing for performance in low-light recording situations.

IP systems tend to feature easier CCTV camera installation and less cabling than analog systems. IP cameras can also be scaled almost infinitely, allowing for far more flexibility and more powerful systems than analog cameras have traditionally allowed. For instance, while the network may be closed, many even allow for remote monitoring with a smartphone app.

Though analog systems lack the diverse features and performance capabilities of digital systems, they offer simplicity and a lower price tag, making them a better choice for some businesses. However, as most businesses and manufacturers

continue to transition to digital systems, IP-based systems are quickly becoming the more dominant form of surveillance.

- ✓ **Analogue CCTV cameras and Digital/ IP CCTV camera advantages and disadvantages**

Analogue CCTV cameras record images to a digital recorder which converts the video to a digital format.

Advantages Analogue CCTV cameras

- Low Cost
- Simplicity
- Lower Bandwidth Requirements
- More Design Options

Disadvantages Of Analogue CCTV cameras

- More Cabling
- Low Image Quality
- Less Coverage Area
- Positioning Limitations
- Port Limitations
- Low Wireless Capability (or Lack Thereof)
- No Encryption

IP camera, is a type of digital video camera that receives control data and sends image data via an IP network.

Advantages of (Internet Protocol) camera

- High Image Quality
- Large Coverage Area
- Fewer Cables Needed
- Positioning or Port Limitations
- Have Power over Ethernet (PoE) capabilities
- Wireless Capability
- Provide Encryption

Disadvantages of (Internet Protocol) camera

- Setup Complications
- Higher Initial Cost
- High Bandwidth Requirements
- Storage Requirements

b. **Categories of CCTV System Based on Usage:**

1. **Indoor CCTV Systems:** Designed for indoor use, typically in homes, offices, and commercial buildings.

2. **Outdoor CCTV Systems:** Designed for outdoor use, with weatherproof and vandal-resistant features. Used for monitoring external areas such as parking lots, entrances, and perimeters.

c. **Categories of CCTV System Based on Connectivity:**

1. **Wired CCTV Systems:** Cameras are connected to the recording device and power source using cables.
2. **Wireless CCTV Systems:** Cameras transmit data wirelessly to the recording device. Can be powered by batteries or wired to a power source.

d. **Categories of CCTV System Based on Features:**

1. **Day/Night CCTV Systems:** Equipped with infrared (IR) LEDs for night vision. Capable of recording in low light or complete darkness.
2. **PTZ (Pan-Tilt-Zoom) CCTV Systems:** Cameras can be remotely controlled to pan, tilt, and zoom. Useful for covering large areas and focusing on specific points of interest.
3. **HD (High Definition) CCTV Systems:** Provide high-resolution images for better clarity and detail. Typically available in 720p, 1080p, 4K, and higher resolutions.

✓ **Types of CCTV Camera**

There are various types of CCTV surveillance cameras, suitable for indoor and outdoor use in a range of building types and industries.

1. Bullet Cameras



Bullet cameras offer a sleek design and a flexible range of areas they can be placed, including on walls. The shape of a bullet camera allows it to accommodate a large lens, giving it a longer range than many other cameras. This makes it ideal for monitoring over long distances. Bullet cameras' good recording quality, motorized zoom capabilities, and infrared make them ideal for outdoor use and use in industries such as manufacturing, property management and farming.

2. Dome Cameras



Dome cameras are generally used indoors and mounted on ceilings. This allows them to survey large areas at a wide angle. Dome CCTV cameras are great for retail establishments, restaurants, hotels, and casinos, as they can completely monitor a large room. They are also well-suited to environments that get dirty easily, such as kitchens and warehouses. Dome cameras offer smaller, more discreet hardware than other camera types.

3. PTZ Cameras



PTZ (pan, tilt, zoom) cameras are shaped like dome cameras, but they allow users to remotely move the camera using an app, joystick, or computer program. This is ideal for panning certain areas and focusing in when the camera detects movement. With the ability to move and "look" around a room, PTZ cameras provide 360-degree coverage, and can do the job of several static cameras. PTZ cameras are great for protecting valuable objects, such as in retail settings.

4. Box Cameras



Box cameras are bulkier than many other camera types, which can be good for deterring crime by giving people an obvious sign they are being recorded. Box cameras are a modular option that let users configure the system for a range of different uses. This allows box cameras to be specifically equipped for outdoor use in rough weather.

5. **Thermal Cameras**



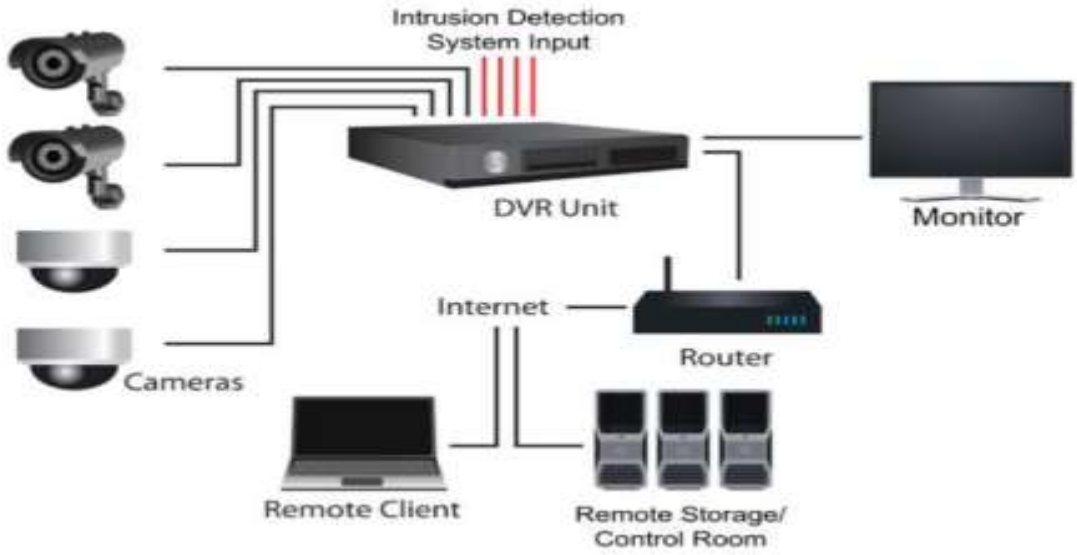
Thermal imaging cameras use heat mapping technology to detect people and objects. These are ideal for obtaining accurate security footage in environments with challenging lighting. Thermal CCTV cameras can also detect heat loss, gas leaks, and other repair issues in a facility. Thermal cameras are ideal for use in industrial buildings, defense, traffic monitoring, research and development facilities, and more.

6. **License Plate Reader Cameras**



License Plate Reader (LPR) cameras have high-resolution sensors that allow them to accurately capture license plate numbers day and night. Most security cameras cannot consistently capture license plates in every weather and lighting condition, let alone when vehicles are moving. LPRs are great for tolling, hotel overstay management, parking lots, and keeping track of who enters and exits any facility or community.

- **CCTV system components**



1. **Cameras:** is an electronic device that captures pictures, movies or other visual images either digitally or on film.
2. **Lenses:** Gathers light reflected from a subject and focuses the light on the image sensor
3. **Housings: Camera housings** are protective enclosures designed to safeguard cameras from various environmental factors, such as weather, vandalism, and dust. Different types of housings are used based on the installation environment and specific needs like Weatherproof Housings, Thermal Housings and Explosion-Proof Housings etc…

4. **Mounts:** Camera mounts are hardware components used to secure cameras in their designated positions. They provide stability and allow for precise positioning and adjustment of the cameras like Wall Mounts, Ceiling Mounts and Rack Mounts etc…
5. **Monitor:** is a device used for viewing recorded images and footage.
6. **Switch:** is a device which connects different devices in CCTV network and enabling them to exchange data
7. **Video recorders:** are the device where video recorded on the camera gets processed for storage & viewing. There are two types of video recorders: DVR (Digital Video Recorders) and NVR (Network Video Recorders).
8. **Data Storage**: is the storage device for a security camera system that is able to record, store and re-play videos non-stop from multiple feeds.
9. **Cables and Router:** With an analog system, a coaxial cable is required for transmitting video footage from the CCTV security cameras. This is one of the drawbacks of analog CCTV video surveillance, as the cable can be expensive and difficult to install, especially for larger camera networks, and those where cameras must be positioned in difficult locations.

✓ **CCTV System video storage**

CCTV video can be stored using various methods:

- **Local hard drive:** Store footage from cameras on local hard drives located in Digital Video Recorders (DVRs) or Network Video Recorders (NVRs).
- **Cloud Storage**: Stores video footage on remote servers, providing access from anywhere with an internet connection.
- **External Hard Drives**: Additional storage connected to DVRs/NVRs for expanded capacity.
- **SD Cards**: Used in standalone cameras for local storage of video footage.
- **Hybrid Systems**: Combine local and cloud storage for redundancy and flexibility.



Points to Remember

- A CCTV system refers to all aspects and components that build up the system, such as the camera, wiring, video footage storage, monitors, access control and general system management
- CCTV System main purposes are Surveillance, Security and Monitoring.
- The working principles of CCTV system is to capture video footage of a facility using cameras. This footage is sent to a recording device (a DVR or NVR), where it is stored for future reference.
- CCTV cameras are categorized based on Technology, Usage, Connectivity and Features.

- The Types of Cameras are Bullet Cameras, Dome Cameras, PTZ Cameras, Box Cameras, Thermal Cameras and License Plate Reader Cameras.
- The CCTV system components are Cameras, Lenses, Housings and mounts, Monitor, Switch, Video recorders (DVR, NVR), Data Storage, Cables and Router.
- CCTV Video Storage: Footage may be stored in a local Hard Drive (HDD) located in DVR or an NVR, it can be stored in SD Card or External Hard Drive, or it can be stored in the cloud.



Application of learning 1.1

A retail store owner wants to enhance the security of their premises. They decide to implement a CCTV system to monitor the store, prevent theft, and ensure the safety of employees and customers. As a trainee who learned the CCTV system basics, you are asked to explain to the retail store owner the CCTV system suitable and the working principles of CCTV system that is going to be installed and Explain the CCTV system components that will be needed for the installation.

Indicative content 1.2: Identification of CCTV system installation requirements.

Duration: 3 hrs

Theoretical Activity 1.2.1: Description of site survey

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of site survey.

i. What is a site survey?
ii. What are the types of physical site survey?
iii. Why is a site survey so important?
iv. How to Conduct a CCTV Camera Installation Site Survey for Maximum Security?
v. List three phases of conducting physical site survey.

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarifications, read the **key readings 1.2.1.**

5: In addition, ask questions where necessary.

Key readings 1.2.1.:

- **Description of site survey**

✓ **Introduction of Site survey**

Site survey is inspection of an area where work is proposed, to gather information for a design or an estimate to complete the initial tasks required for an outdoor activity. It can determine a precise location, access, best orientation for the site and the location of obstacles.

✓ **Types of physical site survey**

A site survey should address all aspects of specifying and building a CCTV system and it is an integral part of defining the requirements for discrete tasks and the role of each piece of equipment. Whenever possible, CCTV systems should be included in the planning and design stage of any new asset to ensure all necessary infrastructure requirements are adequately incorporated into the overall facility or asset design.

Physical site survey methodology is simply an examination of the physical environment in which CCTV camera will operate. The primary objectives for a physical site survey are to ensure that the location can accommodate a CCTV camera and to understand what challenges there may be.

Physical site survey includes two main types such as **Passive physical site survey** and **Active physical site survey methodology**

A. **Passive physical site survey methodology:** involve observation and documentation of the site without directly interacting with or altering it. These surveys provide an overview of the site's existing conditions and features. The most comprehensive survey type that covers the most important devices with its current performances and their characteristics.
B. **Active Physical site Surveys methodology** involve direct interaction with the environment and often include the use of tools and equipment to measure, test, or alter the site in some way. These surveys provide precise and detailed information about specific site conditions. This method is used to check environment, devices, equipment, according to the new technologies. It also enables you to set parameters that govern how the site survey is performed (such as the data rate).

   Considerations inherent in a physical site survey include the number of operators, local and remote operator consoles, layout, light levels, camera and lens selection and location, and power and data transmission.

- **What does a site survey include?**

  A site survey includes taking pictures, examining the materials and area where the sign will be installed, and verifying all measurements. Additionally, we re-examine all sign ordinances to make sure that your sign will fall within city code, and determine when a permit is needed, as well as securing these permits when necessary.
- **Why a site survey is so important?**

  A site survey is crucial in a CCTV system as it ensures optimal camera placement, coverage, and performance, ultimately enhancing the overall effectiveness of the surveillance system. By assessing the physical layout, lighting conditions, and potential obstacles, a site survey helps in selecting the appropriate types of cameras, lenses, and mounting solutions.

  It also identifies the best locations for cameras to cover critical areas and minimize blind spots. Additionally, a site survey helps in planning the cabling routes and power supply requirements, ensuring a smooth and efficient installation process. By addressing these factors beforehand, a site survey reduces the risk of future issues, enhances security, and maximizes the return on investment in the CCTV system.

  Site survey can help an organization even technician save valuable time, money and resources. As a result of the intelligence gathered through a site survey, you can make better decisions; reducing cost errors are made in the execution phase of your project. The information we need to know on site survey are:

- ✓ Gathering information about the design of installation
- ✓ Gathering information about the cost of the work
- ✓ Determination of precise location of installations.

- **Conduct a CCTV Camera Installation Site Survey for Maximum Security.**

  CCTV camera installation services are a great way to improve the security of your home or business. By installing CCTV cameras, you can deter crime, monitor your property, and keep an eye on your loved ones.

  There are many different CCTV camera installation services available, so it is important to do your research and find a reputable company that can meet your needs. When choosing a CCTV camera installation service, you should consider the following factors:

- ✓ **The type of cameras you want to install:** There are a variety of CCTV cameras available, including indoor and outdoor cameras, wired and wireless cameras, and analog and IP cameras.
- ✓ **The number of cameras you need**: The number of cameras you need will depend on the size of your property and the areas you want to monitor.
- ✓ **The features you want:** Some CCTV cameras have features such as night vision, motion detection, and facial recognition.
- ✓ **The cost of installation:** The cost of CCTV camera installation will vary depending on the type of cameras you choose, the number of cameras you need, and the features you want.

  Closed-circuit television (CCTV) is a powerful security tool that can be used to deter crime, identify suspects, and provide evidence in investigations. However, in order to be effective, CCTV systems must be properly installed and positioned. This is where a site survey comes in.

  A site survey is a process of evaluating a property to determine the best locations for CCTV cameras. The goal of a site survey is to identify all potential security risks and to ensure that the CCTV system will provide adequate coverage of the property.

- **A well-conducted site survey will help you to achieve the following:**

- Identify the most important areas to monitor.
- Determine the number and type of cameras needed.
- Choose the best camera locations.
- Consider the lighting conditions at the site.
- Evaluate the existing infrastructure.
- Estimate the cost of the project.

- **Conduct Physical site survey Phases**

For a CCTV site survey, there are many steps (phases) and considerations that a supplier needs to cover. These can be divided into three categories; pre-assessment, assessment and post assessment.

✓ **PRE-ASSESSMENT**

The aim of a pre-assessment is for a potential provider to have a better understanding of the business and its potential problems before visiting a site. Therefore, a pre-assessment could involve:

- Reviewing the business online to understand the brand.
- Ensuring what needs to be assessed is clear, such as buildings, grounds and car parks.
- Identifying key assets and ranking these in a critical order.
- Researching key pain points commonly faced within the industry.

✓ **ASSESSMENT**

Once on-site, the CCTV installer should conduct a thorough assessment to be able to best guide a business with the next steps. Consequently, a provider should look to address the following aspects during the survey:

- Indication of typical light levels.
- Surveillance zones.
- Requirements for remote monitoring and video motion detectors.
- Analytical requirements to detect advanced events.
- IT network assessment.
- How the CCTV system will operate including if it will be viewed 24/7 or called upon following an event?
- Assessment of the mains power supply.

✓ **POST-ASSESSMENT**

Once a site survey has been completed, the security supplier should produce a detailed report and accurate costings of the recommendations for the CCTV requirements so your business is fully informed before making a decision. Any recommendations provided to your business should include camera locations and position as well as taking into consideration scalability and future-proofing of your proposed installation.

You must have a valid reason for installing CCTV on your premises. You need to know the designated areas in which the recording will take place and subsequently notify your neighbours. A sign must be visible, stating that surveillance is taking place. Ensure your camera is positioned away from your neighbors property.



Practical Activity 1.2.2: Conduct site survey



Task:

1: Referring to previous activities (1.2.1) you are requested to perform the given task. The task should be done **individually**.

As a trainee who studied how to conduct the Site survey in CCTV system installation, you are asked to conduct a site survey in your computer lab. by locating where the CCTV system components (Camera, DVR and Screen) can be placed, locate Cable pathways and provide the distance between the components.

2: List out procedures to Conduct site survey given in task.

3: Referring to procedures provided on task 2, Perform the required tasks

4: Present your work to the trainer and whole class.

5: Read key reading (**1.2.2)** and ask clarifications where necessary.

6: Perform the task provided in **application of learning 1.2**



Key readings 1.2.2

- **Introduction**

  Before installing a surveillance system, It is recommended to do a site survey and document it for future needs and upgrades.

  Site surveys is inspection of an area where work is proposed, to gather information for a design or an estimate to complete the initial tasks required for an outdoor activity.

  Site survey: is a process of evaluating a property to determine the best locations for CCTV cameras. The goal of a site survey is to identify all potential security risks and to ensure that the CCTV system will provide adequate coverage of the property

- **Conduct a CCTV Camera Installation Site Survey for Maximum Security.**

  Conducting a CCTV camera installation site survey is crucial for ensuring maximum security and effectiveness of the surveillance system.

  The steps to conduct a comprehensive site survey:

1. **Define Objectives and Requirements:** Identify the objectives of the surveillance system, such as monitoring specific areas, preventing unauthorized access, or deterring criminal activity. Determine the requirements, including coverage area, camera placement, resolution, and recording capabilities.
2. **Survey the Site Layout:** Visit the site and assess the layout, architecture, and physical characteristics of the area to be monitored. Identify points of entry, exits, high-traffic areas, blind spots, and vulnerable zones that require surveillance

coverage.

3. **Assess Security Risks:** Evaluate potential security risks and threats, such as theft, vandalism, unauthorized access, or safety hazards. Consider environmental factors, lighting conditions, weather exposure, and other variables that may impact surveillance effectiveness.
4. **Identify Camera Locations:** Determine optimal camera locations based on the site layout, coverage requirements, and security objectives. Position cameras to cover critical areas, chokepoints, perimeters, and valuable assets while minimizing blind spots and obstructions.
5. **Consider Camera Types:** Select appropriate camera types and specifications based on the surveillance requirements and environmental conditions. Choose cameras with features such as high resolution, wide dynamic range, infrared night vision, vandal resistance, and weatherproofing as needed.
6. **Evaluate Lighting Conditions:** Assess ambient lighting conditions at different times of the day and night to ensure adequate illumination for surveillance. Consider installing additional lighting fixtures or infrared illuminators to enhance visibility in low-light environments.
7. **Check Power and Connectivity:** Verify the availability of power sources and network connectivity for camera installations. Ensure that sufficient power outlets and network infrastructure are accessible at camera locations or plan for alternative power and connectivity solutions if needed.
8. **Plan Cable Routes:** Plan cable routes for power and data cables from cameras to recording devices or network equipment. Consider cable management techniques to conceal and protect cables from tampering or damage.
9. **Evaluate Regulations and Compliance:** Review local regulations, laws, and privacy guidelines governing CCTV installations to ensure compliance with legal requirements. Obtain necessary permits or permissions if required for surveillance activities.
10. **Test Camera Placement:** Conduct on-site tests or simulations to verify the effectiveness of camera placements and coverage angles. Use temporary mounting solutions or mobile cameras to assess camera views and adjust positions as needed.
11. **Document Findings and Recommendations:** Document survey findings, including camera locations, coverage maps, technical specifications, and installation recommendations. Create a detailed site survey report to share with stakeholders and installation teams for reference.
12. **Review and Finalize Installation Plan:** Review the site survey report with project stakeholders, security personnel, and installation teams to finalize the installation plan. Address any concerns, modifications, or additional requirements before proceeding with the CCTV camera installation.

- **Additional tips for conducting a site survey.**

✓ Be sure to bring a camera with you on the walk through so that you can take pictures of potential camera locations.

✓ Talk to the people who work at the property to get their input on the security risks and the desired outcome of the survey.

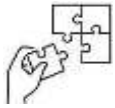
✓ Consider the budget when developing the plan for the CCTV system.

✓ Be sure to test the system thoroughly before it is put into operation.



Points to Remember

- Site surveys is inspection of an area where work is proposed, to gather information for a design or an estimate to complete the initial tasks required for an outdoor activity.
- Types of physical site survey are Passive physical site survey and Active physical site Survey.
- When choosing a CCTV camera installation service, you should consider the following factors: the type of cameras you want to install, the number of cameras you need, the features you want and the cost of installation.
- For a CCTV site survey, there are many steps and considerations that a supplier needs to cover. These can be divided into three categories: Pre-assessment, Assessment and Post assessment.
- During site survey, the important things to be considered are Site survey layout, measurements, components positioning and cables pathways allocation.
- The steps that must be followed for a comprehensive CCTV site survey are define Objectives and Goals, Gather Information, Risk Assessment, Legal and Regulatory Compliance, Identify Camera Locations, Camera Types and Specifications, Lighting Conditions, Power and Connectivity, Storage Requirements, Network Considerations, System Integration, Environmental Factors and Documentation.



Application of learning 1.2

In Kigali City, there is a newly constructed stadium called NEWB stadium, you are hired and as one of the technician team, you are asked to conduct site survey at NEWB stadium by allocating where the cameras and DVR/NVR must be placed and perform the measurements between the devices.

Indicative content 1.3: Identification of CCTV system installation requirements.

Duration: 3 hrs

Theoretical Activity 1.3.1: Description of CCTV Diagram Design

Tasks:

1: In small groups, you are requested to answer the following questions related to Description of CCTV Diagram Design.

1. What is a CCTV diagram?
2. What is CCTV diagram design?
3. What are the benefits of CCTV diagram design?
4. Provide the applications of CCTV Diagram.

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key **readings 1.3.1.**

5: In addition, ask questions where necessary.

Key readings 1.3.1.:

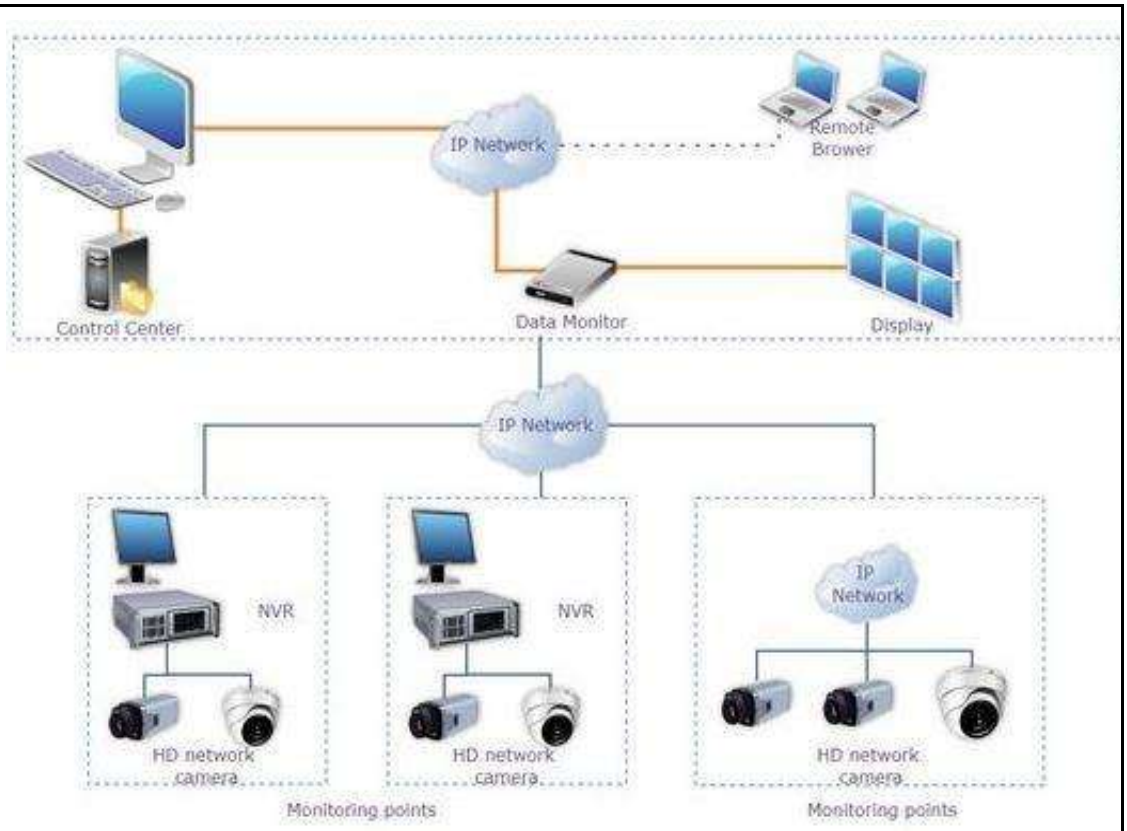
- **Design of CCTV Diagram**

✓ **Description of CCTV Diagram**

A CCTV (Closed-Circuit Television) diagram, also known as a CCTV system diagram or surveillance system diagram, is a visual representation that depicts the layout, components, connections, and configuration of a CCTV camera system. These diagrams are commonly used in the planning, design, installation, documentation, and maintenance of surveillance systems for security and monitoring purposes.

A CCTV diagram typically includes various elements such as cameras, monitors, DVRs (Digital Video Recorders) or NVRs (Network Video Recorders), switches, routers, cables, power supplies, and other relevant equipment. The diagram illustrates how these components are interconnected and arranged within the surveillance system, providing a comprehensive overview of its architecture and functionality.

CCTV diagrams can vary in complexity depending on the size and scope of the surveillance system. They may range from simple block diagrams showing basic connections between components to more detailed layouts depicting camera locations, viewing angles, cable routes, and control interfaces.



- **CCTV diagram design involves several key aspects**

1. **Component Representation**:
   Selecting appropriate symbols or shapes to represent the various components of the CCTV system, such as cameras, monitors, DVRs/NVRs, switches, routers, cables, and power supplies. Each component is typically represented by a specific icon or graphic element to facilitate easy identification.
2. **Layout and Arrangement**:
   Designing the layout and arrangement of components within the CCTV system diagram to accurately reflect their physical placement and connections. This includes positioning cameras at strategic locations, routing cables between devices, and organizing equipment within control rooms or monitoring centers.
3. **Connections and Wiring**:
   Illustrating the connections and wiring between components using lines, arrows, or connectors to indicate the flow of video signals, power supply lines, and data communication pathways. Clear labeling and annotations are added to indicate the type and purpose of each connection.
4. **Topology and Infrastructure**:
   Depicting the network topology and infrastructure of the CCTV system, including the configuration of LAN (Local Area Network) or WAN (Wide Area Network)

connections, Ethernet switches, routers, and wireless access points. Diagrams may also include details of server infrastructure, storage devices, and backup systems.

5. **Functional Representation**:
   Incorporating functional elements into the diagram to illustrate the operational aspects of the CCTV system, such as camera coverage areas, viewing angles, motion detection zones, recording schedules, and alarm triggers. This helps stakeholders understand how the system functions in real-world scenarios.
6. **Aesthetic Presentation**:
   Paying attention to the visual design and aesthetics of the CCTV diagram to ensure clarity, readability, and professionalism. This includes using consistent colors, fonts, line styles, and spacing to create a visually appealing and easy-to-understand representation of the surveillance system.
7. **Documentation and Annotations**:
   Adding text labels, annotations, and descriptions to provide context, explanations, and additional information about the components and connections depicted in the diagram. This helps clarify the purpose and functionality of each element for stakeholders and end-users.

- **The importance of designing a CCTV system Diagram**

Designing a CCTV (Closed-Circuit Television) system diagram is crucial for several reasons, as outlined below:

1. **Planning and Visualization**: A CCTV system diagram provides a visual representation of the surveillance system, allowing stakeholders to understand the layout, components, and connections involved. It helps in visualizing the placement of cameras, monitors, recording devices, and other equipment within the surveillance environment.
2. **Optimized Coverage**: By creating a CCTV diagram, designers can strategically position cameras to achieve optimal coverage of the monitored area. They can identify blind spots, determine camera angles, and ensure comprehensive surveillance coverage to maximize security effectiveness.
3. **Equipment Selection**: Designing a CCTV system diagram aids in selecting the appropriate equipment and components based on the surveillance requirements and environmental factors. It helps in choosing cameras with the right specifications, selecting suitable recording devices, and determining the type of networking infrastructure needed for connectivity.
4. **Resource Allocation**: The diagram helps in estimating the resources required for the CCTV system, including the number of cameras, recording capacity, storage requirements, and network bandwidth. It enables efficient resource allocation and budget planning to meet project objectives within constraints.
5. **Compliance and Regulations**: Designing a CCTV diagram ensures compliance with legal and regulatory requirements governing surveillance systems. It helps in

documenting camera placements, recording policies, privacy considerations, and other compliance-related aspects to adhere to industry standards and regulations.

6. **Integration with Existing Systems**: For existing surveillance systems or integrated security solutions, a CCTV diagram facilitates seamless integration with other systems such as access control, alarms, and analytics platforms. It ensures compatibility and interoperability between different security technologies.
7. **Troubleshooting and Maintenance**: A well-designed CCTV system diagram serves as a valuable reference for troubleshooting issues and performing system maintenance. It helps technicians identify equipment locations, cable routes, and connection points, simplifying troubleshooting tasks and minimizing downtime.
8. **Documentation and Training**: The diagram serves as documentation for the CCTV system's architecture and configuration, aiding in training for system operators, security personnel, and maintenance staff. It provides a clear overview of the system's setup and operation procedures for effective training and knowledge transfer.
9. **Future Expansion and Scalability**: By designing a scalable CCTV system diagram, designers can anticipate future expansion needs and accommodate growth requirements. It allows for easy scalability, addition of new cameras or devices, and upgrades to existing infrastructure without disrupting system operation.

- **Applications of CCTV Diagram Design**.

1. S**ecurity Planning for a Building:**

- **Identify Key Areas:** Mark entrances, exits, corridors, and high-risk areas like cash registers or safes.
- **Camera Placement:** Position cameras to cover these areas, ensuring overlapping fields of view to eliminate blind spots.
- **Diagram Creation:** Use software tools to create a detailed floor plan, marking camera locations and their fields of view.

2. **City Surveillance:**

- **Critical Zones:** Identify public spaces, intersections, and other high-traffic areas.
- **Network Layout:** Design a network of cameras that can be monitored centrally, ensuring coverage of key locations.
- **Diagram:** Map out camera locations on a city grid, indicating angles and areas covered by each camera.

3. **Industrial Site Monitoring:**

- **Hazardous Areas:** Focus on areas with heavy machinery, hazardous materials, or sensitive information.
- **Access Points:** Monitor access points to restricted areas.
- **Diagram:** Create a site map with cameras positioned to cover both general areas and specific high-risk points.

4. **Retail Store Surveillance:**

- **Customer and Employee Monitoring:** Place cameras to monitor customer activity, entrances, exits, and employee-only areas.
- **Loss Prevention:** Focus on areas prone to theft, such as aisles, checkout counters, and storage rooms.
- **Diagram:** Develop a store layout showing camera placements to ensure comprehensive coverage.

5. **School Security:**

- **Common Areas:** Include hallways, entrances, playgrounds, and cafeterias.
- **Sensitive Areas:** Cover areas like labs, server rooms, and storage areas.
- **Diagram:** Design a school map with strategically placed cameras for monitoring both general and sensitive areas.



Practical Activity 1.3.2: Use of CCTV Diagram design software



Task:

1: Referring to previous activity (1.3.1) you are requested to perform the given task. The task should be done individually.

As a trainee who learned diagram design in CCTV system installation, you are asked to draw the design of a CCTV diagram of a house with 4 rooms and 1 saloon (one camera for each room and 3 cameras in the saloon).

2: List out procedures to be used to perform the given tasks.

3: Referring to procedures and design provided on task 2, Perform the given tasks.

4: Present your work to the trainer and whole class

5: Read key reading 1.3.2 and ask clarification where necessary

6: Perform the task provided in application of learning 1.3



Key readings:

- **Use of CCTV Diagram Design software**

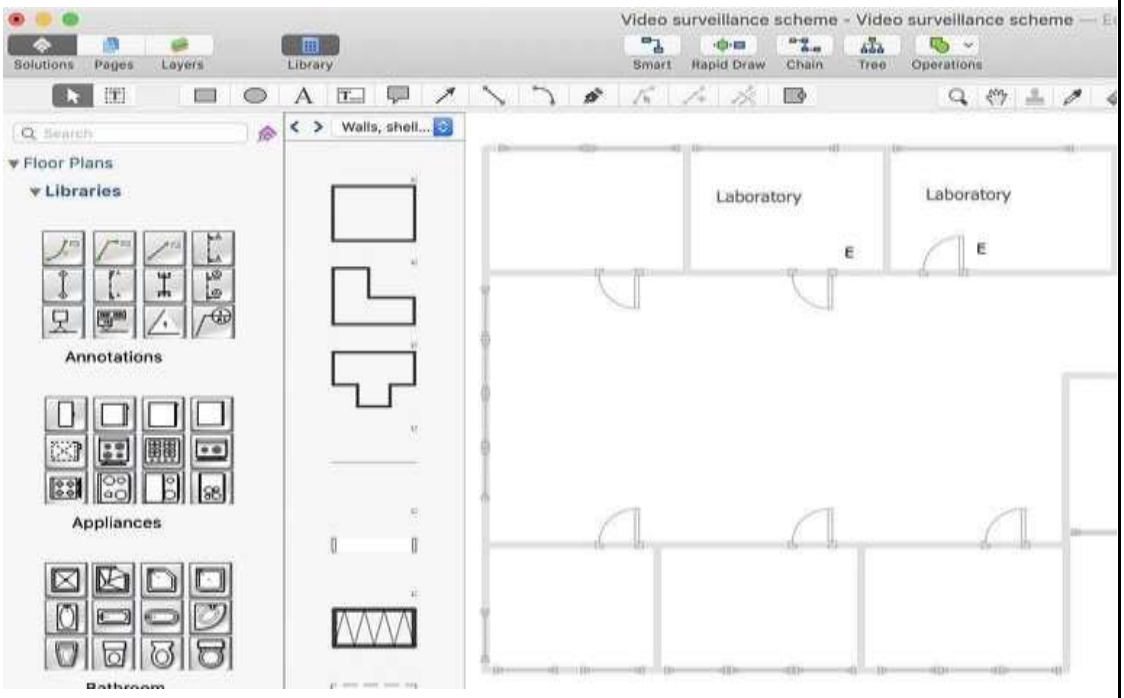
✓ **Selection of CCTV Diagram Design Software (ConceptDraw Diagram)**

ConceptDraw DIAGRAM is a powerful diagramming software that can be used to create various types of diagrams, including those depicting CCTV camera systems. Using ConceptDraw DIAGRAM with the Security and Access Plans Solution lets you create professional looking video surveillance CCTV system plans, security plans, and access schemes.

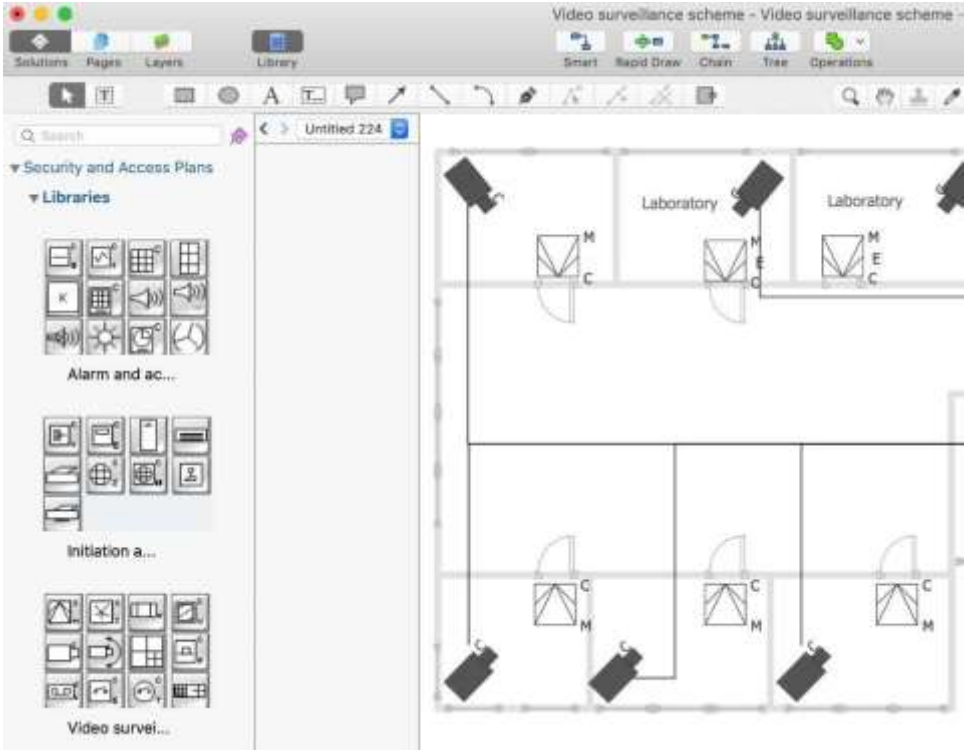
✓ **Steps of creating CCTV System Diagram using ConceptDraw Diagram**

1. Run ConceptDraw DIAGRAM.

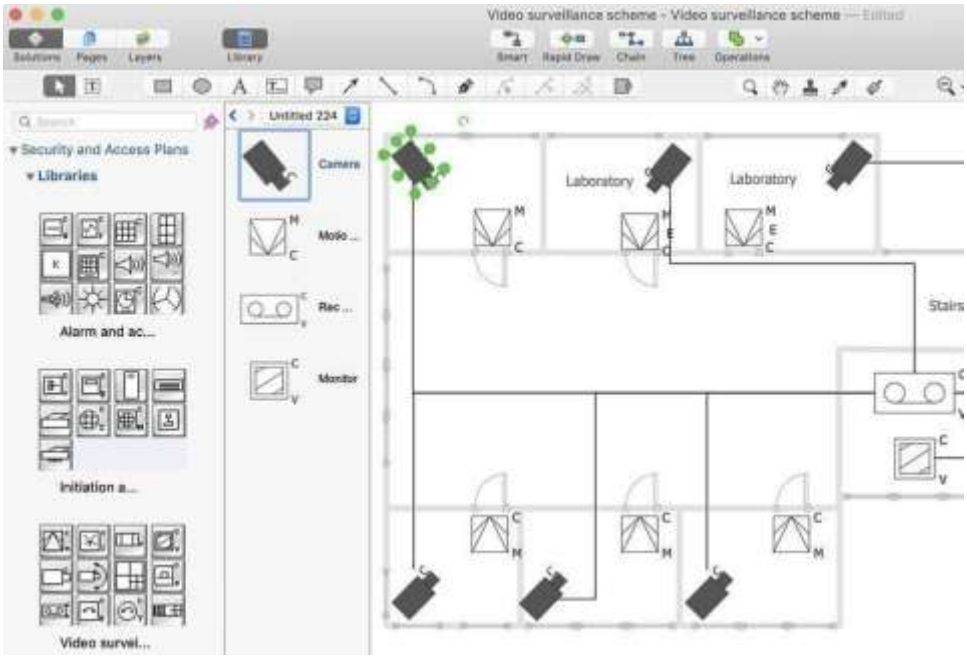
2. Use libraries supplied with the Floor Plans solution to depict the layout of the premises and other floor plan details. (See Pictures below)



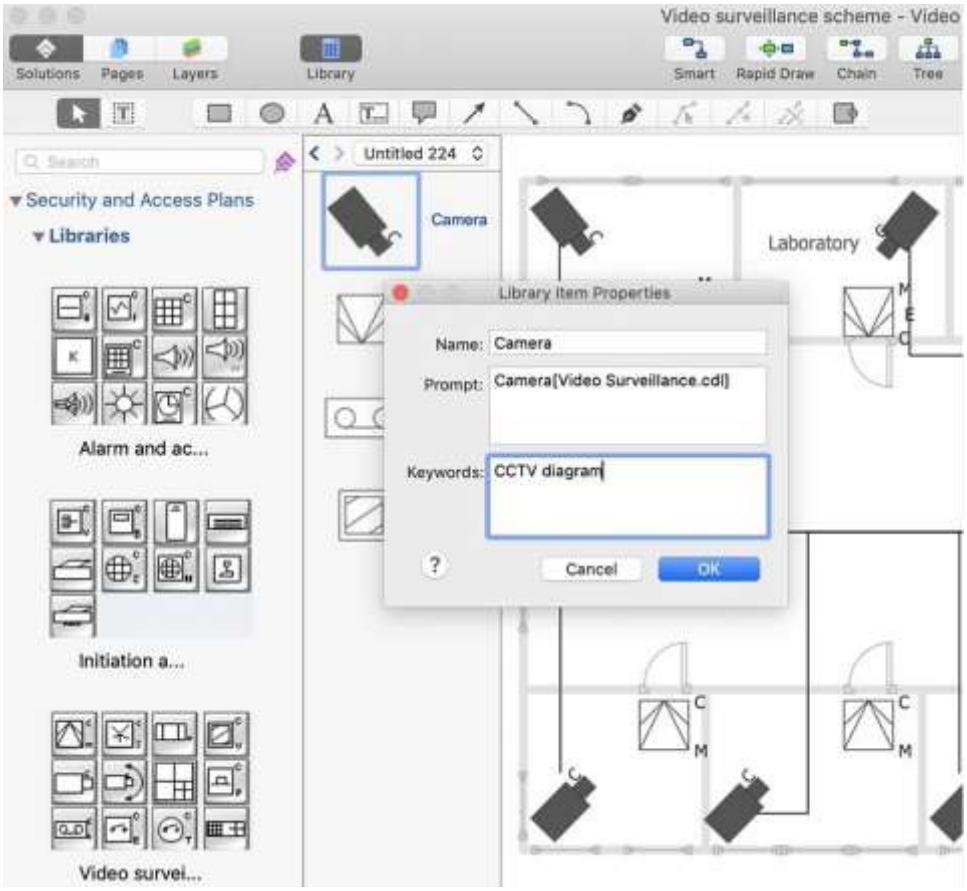
3. Open the Video surveillance library (Security and Access Plans solution) containing objects used for CCTV schemes.
4. The best way to get quick access to frequently used objects located in different libraries is to create your own custom library. To do this, open the **File** menu and select **Library**, then **New**. An empty library box will appear in the Library panel.



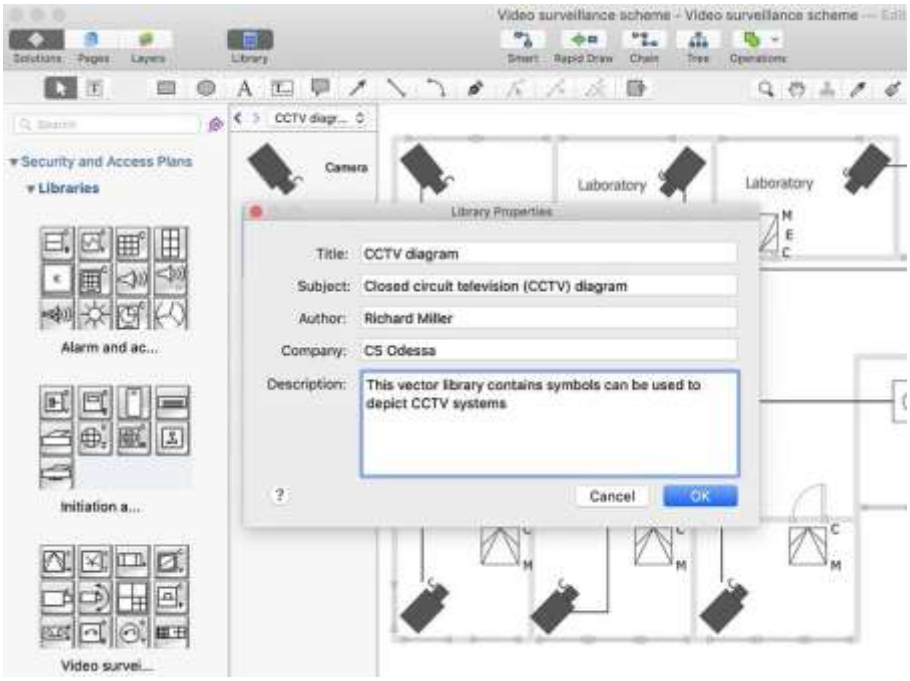
5. Drag and drop shapes from the drawing page to the new library.



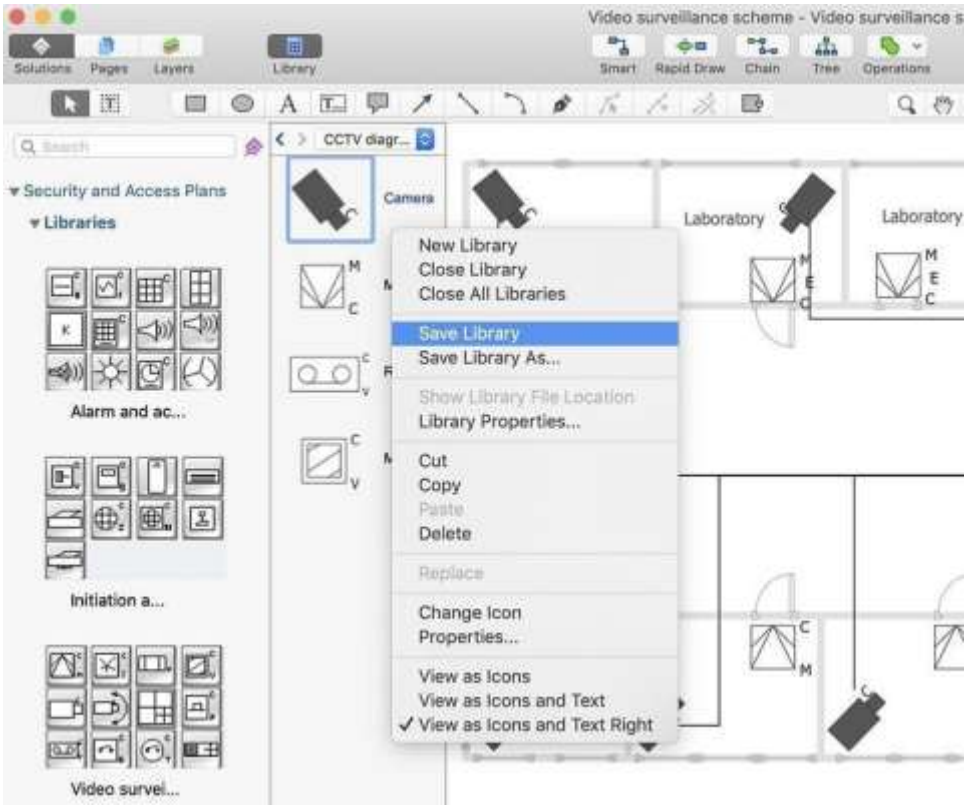
6. Set the shapes properties using context menu.



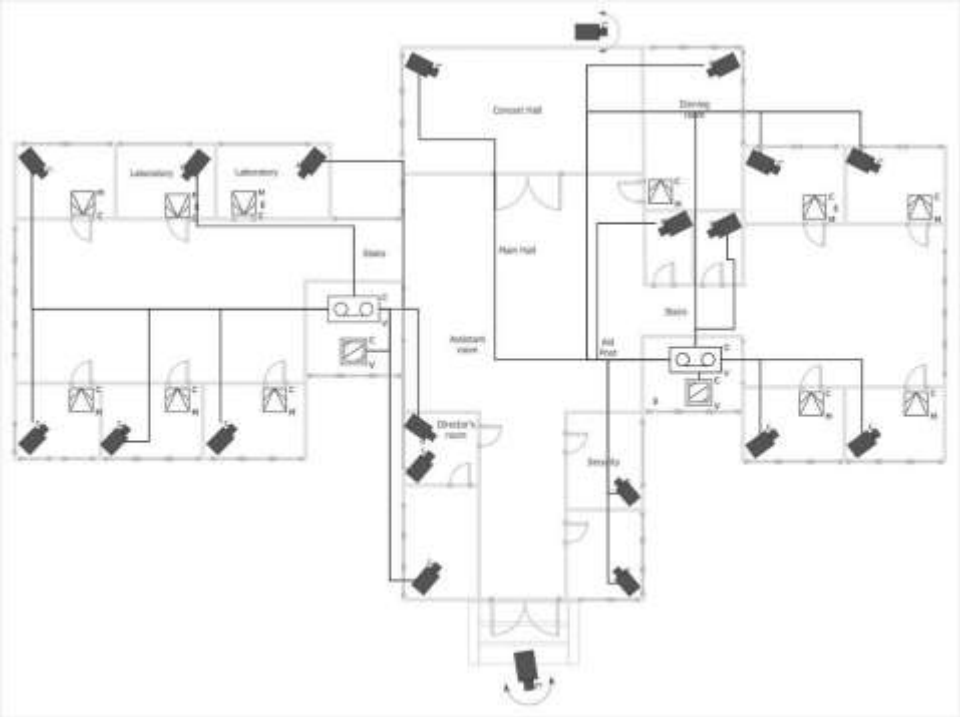
7. Set the library properties using context menu.



8. Save the library.



9. Use the new library to create a CCTV Schematic Diagram.



By following these steps, you may be able to create a full CCTV System Diagram.

Points to Remember

- CCTV system diagram is used to depict a system of video monitoring. The CCTV diagram provides video cameras placement strategy which capture and transmit videos to either a private network of monitors for real-time viewing, or to a video recorder for later reference. A CCTV network diagram shows the closed-circuit television and how it is connected to the camera and the network to record the live feed.
- The CCTV diagram provides several importance include Clarity and Communication, Component Placement, Resource Allocation, Installation and Maintenance, Vulnerability Identification.
- The applications of CCTV Diagram design are Security Planning for a Building, City Surveillance, Industrial Site Monitoring, Retail Store Surveillance and School Security.
- The Steps used to design CCTV Diagram Design software using Concept Draw Diagram software.
- The Steps used to design CCTV Diagram Design software using Concept Draw Diagram software.
    1. Install and Open ConceptDraw DIAGRAM.
    2. Create a New Document – Choose a blank page or a relevant template.
    3. Select CCTV Shapes and Symbols – From the library, add cameras, monitors, DVRs, NVRs, etc.

4. Place and Arrange Components – Position the components according to the CCTV layout.
5. Draw Connections – Connect cameras to monitors, NVR/DVR, and power sources.
6. Add Labels and Annotations – Clearly label each component for clarity.
7. Save and Export – Save your design and export it as needed (PDF, image, etc.).



Application of learning 1.3

In Kigali-Kicukiro, Experts Company in networking plans to install CCTV system in XYZ building. The network administrator has requested to present CCTV system diagram that must be used for the installation. As the network Administrator, you are asked to design a CCTV Schematic diagram of the XYZ building.

Indicative content 1.4: Estimation of CCTV system cost.

Duration: 1 hrs

Theoretical Activity 1.4.1: CCTV system cost estimation

Tasks:

1: In small groups, you are requested to answer the following questions related to the CCTV system cost estimation.

i. What is cost estimation?
ii. What are the item specifications of a CCTV system?
iii. How to estimate the cost of a CCTV system?

2: Provide the answers for the asked questions and write them on papers.
3: Present the findings/answers to the whole class
4: For more clarification, read the key **readings 1.4.1.**
5: In addition, ask questions where necessary.

Key readings 1.4.1.:

- **Estimation of CCTV System Cost**

✓ **Cost Estimation**

Cost estimation is the process of approximating the expenses, resources, and time required to complete a project, manufacture a product, deliver a service, or undertake any planned activity. It involves predicting the costs associated with various resources such as labour, materials, equipment, facilities, and any other expenses that might be involved.

✓ **Goal of cost estimation**

The goal of cost estimation is to provide an accurate forecast of the overall expenses that will be incurred throughout the lifecycle of a project or activity.

✓ **Key factors of CCTV system cost estimation**

Cost estimation for CCTV camera installation includes several key factors:

1. **Site Assessment**:

Conducting a thorough site assessment is the first step in cost estimation. This involves evaluating the physical layout, size, and environmental conditions of the installation site to determine the number and placement of cameras needed to provide adequate coverage.

2. **Camera Selection**:
   Selecting the appropriate CCTV cameras based on the surveillance requirements and application needs is essential. Consider factors such as resolution, field of view, low-light performance, weatherproofing, and special features (e.g., pan-tilt-zoom, infrared, vandal resistance) when choosing cameras.
3. **Additional Equipment**:
   Besides cameras, other equipment such as DVRs (Digital Video Recorders) or NVRs (Network Video Recorders), monitors, cables, power supplies, mounting hardware, and networking components may be required. Estimate the costs for each component based on the specifications and quantity needed.
4. **Installation Labor**:
   Labor costs for installation, configuration, and setup of the CCTV system are significant factors in cost estimation. Consider the complexity of the installation, the number of cameras, and the skill level required for installation technicians when estimating labor expenses.
5. **Infrastructure Requirements**:
   Assess the infrastructure requirements for the CCTV installation, including cabling (e.g., coaxial, Ethernet, fiber optic), conduit, junction boxes, and power outlets. Factor in the cost of materials and installation labor for infrastructure components.
6. **Integration with Existing Systems**:
   If integrating the CCTV system with existing security or IT systems, consider any additional costs associated with software integration, customization, compatibility testing, and training.
7. **Permits and Regulatory Compliance**:
   Determine if any permits or regulatory compliance requirements are necessary for the installation of CCTV cameras, such as local building permits, zoning regulations, or privacy laws. Budget for permit fees and any associated compliance costs.
8. **Maintenance and Support**:
   Estimate ongoing maintenance and support costs for the CCTV system, including software updates, equipment repairs or replacements, technical support services, and monitoring services if applicable.
9. **Contingency and Miscellaneous Costs**:
   Include a contingency budget for unexpected expenses or scope changes that may arise during the installation process. Additionally, budget for miscellaneous costs such as project management fees, insurance, and taxes.
10. **Vendor Quotations and Bids**:
    Obtain quotations and bids from reputable CCTV vendors, suppliers, or contractors to compare pricing and services. Use vendor proposals as a basis for cost estimation and negotiation.

11. **Total Cost Calculation**:
    Summarize the individual cost estimates for cameras, equipment, labor, permits, maintenance, and other expenses to calculate the total cost of the CCTV camera installation project.
12. **Review and Approval**:
    Review the cost estimation with stakeholders, decision-makers, or clients to ensure alignment with budgetary constraints and project requirements. Seek approval before proceeding with the installation project.

✓ **Items specifications**

In the context of CCTV camera installation, "items specification" refers to the detailed specifications and requirements for the various components and equipment involved in setting up a surveillance system. These specifications outline the specific characteristics, features, and performance criteria of each item to ensure compatibility, functionality, and reliability throughout the installation process.

Items and their specifications typically addressed in CCTV camera installation:

1. Camera Specification:

- Resolution: Specifies the camera's image resolution, often measured in megapixels (MP) or pixels (e.g., 1920x1080 for Full HD).
- Sensor Type: Specifies the type of image sensor used in the camera (e.g., CMOS, CCD).
- Lens Type: Specifies the lens focal length, aperture, and viewing angle required for the application (e.g., fixed lens, varifocal lens).
- Day/Night Functionality: Specifies whether the camera has day/night switching capability for low-light or nighttime surveillance.
- Ingress Protection (IP) Rating: Specifies the camera's level of protection against dust and water ingress (e.g., IP66 for outdoor use).
- Power Supply: Specifies the voltage and power requirements of the camera (e.g., 12V DC, PoE).
- Mounting Type: Specifies the mounting method compatible with the camera (e.g., wall mount, ceiling mount).
- Compression Format: Specifies the video compression format supported by the camera (e.g., H.264, H.265).

2. DVR/NVR Specification:

- Channel Capacity: Specifies the number of camera channels supported by the DVR/NVR (e.g., 4-channel, 8-channel, 16-channel).
- Recording Resolution: Specifies the maximum resolution supported for video recording (e.g., 1080p, 4K).

- Storage Capacity: Specifies the maximum storage capacity supported by the DVR/NVR for video recording (e.g., 1TB HDD, 4TB HDD).
- Compression Format: Specifies the video compression format used for recording and storage (e.g., H.264, H.265).
- Network Connectivity: Specifies the type of network connectivity supported by the DVR/NVR (e.g., Ethernet, Wi-Fi).
- Remote Access: Specifies whether the DVR/NVR supports remote viewing and playback via mobile devices or web browsers.
- Power Supply: Specifies the voltage and power requirements of the DVR/NVR (e.g., 12V DC, PoE).
- Operating System: Specifies the firmware or operating system running on the DVR/NVR (e.g., Linux-based, proprietary).

3. Cabling and Connectors Specification:

- Cable Type: Specifies the type of cables required for video transmission, power supply, and data connectivity (e.g., coaxial cable, Ethernet cable).
- Connector Type: Specifies the connectors used for terminating cables and connecting devices (e.g., BNC connectors, RJ45 connectors).
- Cable Length: Specifies the maximum allowable length of cables for reliable signal transmission and power delivery.
- Cable Quality: Specifies the quality and shielding properties of cables to minimize signal interference and degradation.

4. Accessories and Mounting Hardware Specification:

- Mounting Brackets: Specifies the type and size of mounting brackets required for camera installation (e.g., wall mount brackets, pole mount brackets).
- Power Adapters: Specifies the voltage, current, and connector type of power adapters required for camera and DVR/NVR power supply.
- Weatherproof Enclosures: Specifies weatherproof enclosures or housings required for outdoor camera installations.
- Surge Protectors: Specifies surge protectors or lightning arrestors required for protecting CCTV equipment from power surges and electrical damage.
- **Total Cost Formula**

  To calculate the total cost of a CCTV system, you can use a formula that considers all the components and associated expenses.

  Total Cost=∑(Camera Cost+Installation Cost+DVR/NVR Cost+Storage Cost+Monitor Cost+Cabling Cost+Maintenance Cost).

Points to Remember

- Cost Estimation is the process of approximating the expenses, resources, and time required to complete a project, manufacture a product, deliver a service, or undertake any planned activity.
- Items specification means a set of criteria to which a material must conform to be considered acceptable for its intended use. Specifications of CCTV items include the number of megapixels, aperture, focal length, sensor size, zoom type and methods of stabilisation and focusing system.
- Cost estimation is a process where project managers predict the amount of money they need to fund their projects. The process entails direct and indirect costs of the project. These costs may include utilities, materials, equipment, vendors, and employee compensation.
- The formula for finding total cost is simply **Number of Items * Item unit cost = total cost**.



Application of learning 1.4

A school administrator wants to enhance campus security by installing a CCTV system. They decide to implement this system to monitor the school premises, prevent vandalism, and ensure the safety of students and staff. As a trainee who has learned about estimating CCTV system costs, you are asked to provide the specifications of the CCTV system items to be installed, provide the cost of each specified item and calculate the total cost of the CCTV system installation.



Learning outcome 1 end assessment

Theoretical assessment

A. Choose the correct answer

1. **What is the primary purpose of a CCTV system?**
   A) To entertain customers
   B) To monitor and secure premises
   C) To provide internet access
   D) To enhance lighting in dark areas
2. **Which of the following is NOT a component of a CCTV system?**
   A) Cameras
   B) DVR/NVR
   C) Microphones
   D) Monitors
3. **Which type of camera is most commonly used in modern CCTV systems?**
   A) Analog CCTV Camera
   B) IP/Digital CCTV Camera
   C) Infrared Camera
   D) Film-based Camera
4. **What is the first step in installing a CCTV system?**
   A) Designing the CCTV Diagram
   B) Conducting a Site Survey
   C) Estimating the Cost
   D) Purchasing the Equipment
5. **What does DVR stand for in a CCTV system?**
   A) Digital Visual Recorder
   B) Dynamic Video Recorder
   C) Digital Video Recorder
   D) Data Verification Recorder
6. **Which of the following is a key difference between Analog and Digital/IP CCTV cameras?**
   A) Analog cameras provide higher resolution
   B) Digital/IP cameras can be accessed remotely over a network
   C) Analog cameras use Ethernet cables
   D) Digital/IP cameras are cheaper to install
7. **What is the importance of designing a CCTV Diagram?**
   A) It helps in choosing the color of the cameras
   B) It provides a visual representation of camera placement and coverage
   C) It ensures the cameras match the building's aesthetic
   D) It is required for legal compliance

8. **Which type of site survey involves physically visiting the location to assess the installation environment?**
   A) Virtual Site Survey
   B) Physical Site Survey
   C) Desktop Site Survey
   D) Technical Site Survey
9. **Which of the following software can be used to design a CCTV Diagram?**
   A) Microsoft Word
   B) ConceptDraw DIAGRAM
   C) Adobe Photoshop
   D) QuickBooks

10. **Which factor is NOT typically considered when estimating the cost of a CCTV system?**
    A) Camera resolution
    B) Cable length
    C) Camera installation height
    D) Storage capacity
11. **Why is conducting a site survey important before installing a CCTV system?**
    A) To determine the best spots for decoration
    B) To identify the necessary equipment and installation requirements
    C) To decide on the brand of cameras to purchase
    D) To prepare legal documentation for the installation
12. **Which of the following is typically included in the specification of CCTV system items?**
    A) Brand logo
    B) Model number and features
    C) Store location
    D) User preferences
13. **What is the main function of a CCTV camera in a security system?**
    A) To record audio in the environment
    B) To capture and transmit video footage
    C) To serve as a decoration
    D) To act as a fire alarm
14. **Which type of site survey is likely to be the most accurate in determining CCTV installation requirements?**
    A) Desktop Site Survey
    B) Virtual Site Survey
    C) Physical Site Survey
    D) Hypothetical Site Survey

15. Analog CCTV System consist of the following materials, EXCEPT:

a) Camera
b) Connector
c) Hard Disk for recording
d) NVR

B. Read carefully and answer to the following statement by TRUE if the statement is correct, or FALSE if the statement is not correct.

1. CCTV is a video system that consists of strategically placed video cameras around an area that records footage, and is then transmitted to a display monitor(s) for real-time viewing as well as footage playback.
2. CCTV surveillance installation is a system consisting of electronic or other devices designed constructed or adapted to monitor or record images on or in the vicinity of premises.
3. Interconnections means by which messages and/or signals are transmitted between CCTV system components.
4. CCTV works by the camera or cameras taking a constant sequence of images that are then transmitted by cable or wirelessly to the recording device and then on to the display monitor.
5. One of the main purpose of CCTV is to enhance customer experience and reduce security costs.
6. To view the video of IP camera, the DVR needs to be connected to a monitor or router to be broadcast through an internal network for remote access.
7. The quality of Image of IP camera is low compared to the Analog digital camera.
8. Initial cost of analog digital camera is higher than IP camera.
9. Specifications of CCTV items include the number of megapixels, aperture, focal length, sensor size, zoom type and methods of stabilization and focusing system.
10. The Active Surveys methodology is method used to check environment, devices, equipment, according to the new technologies.
11. Designing a CCTV system that blends into the background of your facility can provide non-invasive security footage that's always available to reference, should it ever be needed.

Practical assessment

In Musanze District's Cyuve Sector, a networking company is planning to install a CCTV system in the KAKA building. To successfully complete this project, the company is recruiting a skilled CCTV technician. You are eligible to take the practical exam, and the network administrator has requested that you conduct a site survey, present a detailed CCTV diagram of the building, and provide a comprehensive cost estimation for the installation. The system should include 12 cameras, strategically installed at the main entrances (2 cameras), hallways (4 cameras), stairwells (3 cameras), and parking areas (3 cameras) to ensure full coverage and security.



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Indicative contents
2.1 Selection of tools, material and equipment of CCTV installation
2.2 Conducting CCTV Trunking
2.3 Assembling CCTV equipment
2.4 Fixing and connecting CCTV controller devices and camera

Key Competencies for Learning Outcome 2: Deploy CCTV Equipment.

Knowledge	Skills	Attitudes
• Identification of tools, materials and equipment. • Description of CCTV trunking type. • Identification of assembling CCTV equipment. • Identification of assembling CCTV equipment. • Identification of controller devices and camera.	• Selecting of tools, materials and equipment. • Assembling process. • Perform CCTV trunking. • Connecting CCTV Controller devices.	• Being a Self-Motivator. • Being a teamwork player. • Being an innovator. • Having a creativity spirit.



Duration: 10 hrs

Learning outcome 2 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Select properly tool, materials and equipment based on CCTV installation requirements.
2. Conduct appropriately piping (Trunking) according to CCTV diagram.
3. Assemble properly CCTV equipment based on manufacturer's guidelines.
4. Fix properly CCTV equipment based on CCTV diagram.
5. Connect properly CCTV devices based on CCTV diagram.



Resources

Equipment	Tools	Materials
● Computer ● Monitor ● Power Supply ● Switch ● Storage devices ● Cameras ● Recording Devices (NVR/DVR)	● Tape measure ● Crimping tool ● Wire stripper ● Drilling machine ● Hammer ● Pliers ● Screw drivers	● Internet bundles ● Connectors ● Cable ties ● Cable trunks ● Pipes ● Screws ● Wall Mounts ● Power extension ● Labelling tags ● CCTV Power supply box

Indicative content 2.1: Selection of tools, material and equipment of CCTV installation.

Duration: 2 hrs

Theoretical Activity 2.1.1: Description of tools, materials and equipment

Tasks:

1: In small groups, you are requested to answer the following questions related to the description tools, materials and equipment.

i. Differentiate tools, equipments and materials?

ii. Give examples of tools, equipment and materials for a CCTV system.

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the **key readings 2.1.1**

5: In addition, ask questions where necessary.

Key readings 2.1.1.:

- **Tools**

  Tools are objects or gadgets used to lift out precise duties or operations. They are usually handheld and manipulated by means of hand to operate a variety of actions. Tools are frequently easy and portable, designed to be a useful resource in precise guide movements or processes.

  Examples: Cutting tools (Pliers), drilling tools, fixing tools(Screwdrivers), patching tools, Testing tools(Cable Testers).

- **Equipments**

  Equipments are pertained to large and greater complicated machinery, devices utilized for particular features or tasks.

  Examples: Computer, Monitor, Power Supply, Switch, Storage devices, Cameras, Recording Devices (NVR/DVR).

- **Materials**

  Materials refer to the components or elements used to create or assemble something. They are the uncooked components or sources that bear processing or transformation to be utilised in a precise context.

  Examples: Internet bundles, Connectors, Cable ties, Cable trunks, Pipes, Screws, Wall Mounts.

Practical Activity 2.1.2: Check tools, materials and equipment deployed.

Tasks:

1: Referring to previous activities (2.1.1) you are requested to perform the given task. The task should be done individually.

As a trainee who learned how to select tools, equipment and materials, you are asked to categorise tools, materials and equipment accordingly in the workshop.

2: List out the procedures to be used to perform the given tasks.

3: Referring to the procedures of selecting tools, materials and equipment provided in task2, perform the given tasks

4: Present your work to the trainer and whole class.

5: Read key readings 2.1.2 and ask clarification where necessary.

6: Perform the task provided in **application of learning 2.1**

Key readings 2.1.2

- **Tools**

  The following are the types of tools used in CCTV camera system installation:

  - **Cutting tools**

    A cutting tool is a pointed tool mounted in a machine tool and used for cutting materials like pliers.

  - **Drilling tools**

    Drilling tools, cylindrical end-cutting tool used to originate or enlarge circular holes in solid material. Usually, drills are rotated by a drilling machine and fed into stationary work, but on other types of machines a stationary drill may be fed into rotating work or drill and work may rotate in opposite directions.

  - **Fixing tools**

    These are the tools used to place CCTV components definitely and more or less permanently a wall like screwdrivers.

  - **Patching tools**

    It is used for inserting wire into insulation-displacement connectors on punch down blocks, patch panels, keystone modules, and surface mount boxes (also known as biscuit jacks).

  - **Testing tools**

    It is an installation and troubleshooting tester for digital/IP, HD coax and analogue security camera systems.

    Ex: Multimeter, Cable Tester

- **Equipment**

The following are the types of the equipment used in CCTV camera system installation:

- **Computer**

  A computer is an electronic device that manipulates information, or data. It has the ability to store, retrieve, and process data. You may already know that you can use a computer to type documents, send email, play games, and browse the Web.

- **Monitor**

  CCTV Monitor is one of the most important elements of CCTV systems that provide real-time feedback from CCTV cameras and security surveillance equipment. CCTV cameras digital signal needs to be converted into a visual picture and this can happen thanks to the specially designed monitors.

- **Power Supply**

  A CCTV power supply box, also known as a power distribution box, allows surveillance system installers to easily manage the power to multiple CCTV cameras at a central point (usually at the location of the DVR). This allows your camera installation to be neater.

- **Switch**

  A PoE switch for IP cameras is a device used to connect and power the IP cameras via Ethernet cables. The primary function of a PoE switch in IP camera system is to interconnect the cameras to an NVR (Network Video Recorder) and transmit the data (video and audio) to be recorded in such a device.

- **Storage devices**

  The current dominant storage technology is still magnetic **hard disk drives,** although solid state electronic storage in the form of flash drives, SD cards and solid state drives (SSD) are becoming more popular due to their affordability and capacity increase.

- **Cameras**

  This is a device for recording visual images in the form of photographs, film, or video signals. These cameras share the images across the internet, so CCTV footage can be easily accessed. Network cameras are ideal for both domestic and commercial purposes.

- **Recording Devices (NVR/DVR)**

  A DVR converts analogue footage into a digital format, while an NVR typically only works with digital footage. DVR systems process data at the recorder, while NVR systems encode and process data at the camera before transmitting it to the recorder for storage and remote viewing.

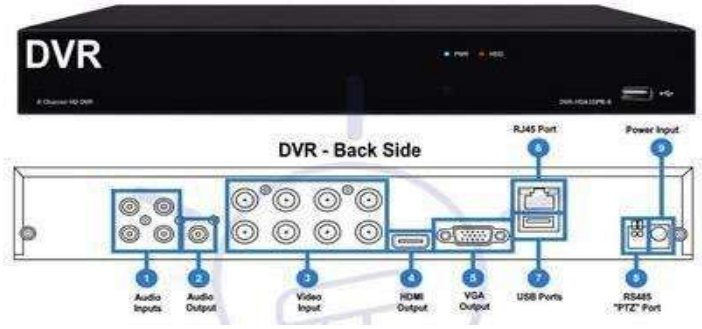
  Basically, both NVR and DVR systems are responsible for video recording. However, they differ in how they record the footage. The DVR system processes data at the recorder, while the NVR system encrypts and processes data at the camera before sending it to the recorder for storage and remote viewing. Both

systems handle data differently. Thus, they need different types of cameras. Most of the DVRs are used with analogue cameras, while NVRs are used with IP cameras. Most importantly, a DVR based surveillance system is a wired security system and NVR system can be a wired or wireless system.

The following table represents the basics comparison between NVR and DVR.

DVR	NVR
Already have analog or other coaxial based cameras installed	Already have CAT5e or CAT6 cabling in place and do not want to change it
Less complex setup and troubleshooting	Support resolutions of 8MP and 12MP
More affordable and easier to maintain	Use IP Cameras, Digital cameras.
Allow cameras from an accessible IP address to be recorded	Allow cameras from an accessible IP address to be recorded.
Records WiFi Security Cameras connected to the network	Records WiFi Security Cameras connected to the network.
	12MP Camera Support on some models
	Fisheye Camera Dewarp on some models
	Supports Two-way Talk Cameras
	Video Analytics

- **DVR**



Stands for "Digital Video Recorder." A DVR is basically a VCR that uses a hard drive instead of video tapes. It can be used to record, save, and play back television programs.

The DVR system is designed for recording video in a digital format. These security systems are priced lower than NVR systems, and this is one of the advantages of DVR systems, especially for small or domestic applications which do not encompass long-distance data transfer or remote viewing.

- **Analog Cameras:**
  As discussed before, the DVR system uses analog cameras. The camera is responsible for streaming an analog single to the recorder, which then processes the video data. Unlike NVRs, most DVR cameras are less expensive.

- **Coaxial Cables:**
  The coaxial cables connect the analog camera to the DVR system. The use of coaxial cable may not seem significant due to their limitations; however, ultimately data is transferred over this cable type. Coaxial cable does not support power supplying devices on the same line. Thus, another cable is needed to carry electrical signals, while the other cable carries data signals for video transmission. These cables are stiffer and wider than Ethernet cables, making installation a challenge. Also, audio is a limitation as the standard coaxial cable may not support audio transmission.
- **AD Encoder:**
  DVR recorders heavily rely on AD (Analog to Digital) encoders, which is responsible for processing the raw data streaming from the camera into viewable footage. Thus, every security camera system needs to be connected to both the recorder and a separate power source.
- **Benefits of DVR Systems**

  There are a few noticeable benefits of a DVR system, making them increasingly attractive.

❖ Minimal bandwidth usage
❖ Signal stability
❖ Low cost

○ **NVR**



The NVR full form is network video recorder. As the name suggests, NVR recorders record videos from the network directly using Cat5 or Cat6 Ethernet cables with RJ45 plugs. NVR systems record and store video footage directly from the network it lives on. These systems work with an advanced type of camera, called **IP**

cameras. IP cameras can actually capture and process video and audio data themselves. They do so using either an Ethernet cable or wirelessly via an existing Wi-Fi network.

NVR is a network-attached computer system that includes a software program for recording video in a digital format to storage devices such as a disk drive, USB flash drive, SD memory card, and many more. Popularly known as POE security camera systems, it is more flexible than DVR systems.

The following components are required to set up an NVR system:

- **IP Cameras:**
  NVR system uses IP cameras, which act as standalone image capturing devices. These cameras are capable of processing video data before sending it to the recorder. IP cameras are robust, and capable of recording and transmitting audio in addition to the image. Advanced hardware on these cameras improves intelligent video analytics such as license plate and facial recognition.
- **Ethernet Cables:**
  The NVR system makes use of standard Ethernet cables such as Cat5e and Cat6 to connect the camera to the recorder. They are easier to set up owing to their thin size and shape and cost-effectiveness and easy availability compared to coaxial cables.
- **Recorder:**
  The NVR recorder is only responsible for storing and viewing the footage. The system doesn't process video data.
- **Benefits of NVR System**

  Here are some important advantages of the NVR system:

❖ Better image quality
❖ More system flexibility
❖ Easy to install

- **Material**
  The following are the types of materials used in CCTV camera system installation:

➢ **Internet bundles**
  Internet service providers normally sell their data for browsing the Internet in specific bundles packages.
  Some security cameras can consume as little as 5 Kbps in "steady-state", and others as high as 6 Mbps and beyond. The average bandwidth consumption of an IP cloud camera is 1-2 Mbps

➢ **Connectors**
  Most surveillance DVRs and CCTV cameras have BNC female output. This connector allows you to connect one of those devices to a cable that also has a

female BNC end. BNC to RCA cable connector and RJ45 (CAT5) for NVR to POE Switch.

- **Cable ties**

  A cable tie is also often known as a zip tie and used to fasten or bind several electronic cables or wires together or to route cables and wires together in a manageable format. Most of the cable ties used are made from nylon, making them very flexible.

- **Cable trunks**

  Cable trunking is installed as a protective routing and covering system for electrical cables and wires which prevents accidental damage.

  The key use of cable electrical trunking systems is to keep the cabling tidy and hidden. The tough white plastic hollow is a safe and professional way to protect the electrical wiring installation. The cable electrical trunking wiring system provides an aesthetic look to the wiring, keeping it tidy and in one place.

- **Pipes**

  Used for electrical cable protection, installed underground or for road crossing. These pipes can be joined in long length for horizontal direct drilling (HDD), a trenchless method of installing underground pipes.

- **Screws**

  The **wire** seat kit comes with **screws** and ties. This **wire** organizer holder and the tie are made of nylon material.

- **Wall Mounts**

  A wall mount bracket is **a device used to secure a flat panel display (such as a TV or monitor) to a wall**. The most common use for a wall mount is to allow the user to position the display in a more comfortable viewing position.

- **Power extension**

  It is a length of flexible electrical power cable (flex) with a plug on one end and one or more sockets on the other end (usually of the same type as the plug).

- **Labelling tags**

  The key purpose of using cable labels is to make it as easy as possible to know the location of the other end of the cable. In addition, labelling a cable can tell you its length, type and so on and/or it can tell you what is connected to each end of it.

- **CCTV Power supply box**

  Security camera installers can use a CCTV power supply box, also known as a security power distribution unit, **to concentrate the power source of several cameras to a single location**. Wiring becomes neater and easier to manage when used with RG59U Siamese cable or video balloons.



Points to Remember

- Tools are usually handheld and manipulated by means of hand to operate a variety of actions. Examples: Cutting tools, drilling tools, fixing tools, Patching tools, Testing tools.
- Equipment pertains to large and greater complicated machinery, devices utilized for particular features or tasks. Examples: Computer, Monitor, Power Supply, Switch, Storage devices, Cameras, Recording Devices (NVR/DVR).
- Materials refer to the components or elements used to create or assemble something. Examples: Internet bundles, Connectors, Cable ties, Cable trunks, Pipes, Screws, Wall Mounts.
- Selection of tools.
- Selection of materials.
- Selection of equipments.



Application of learning 2.1

SAVANA Factory want to install CCTV system for better management of equipment and consumables in its environment.
You are hired as a CCTV System installer and assigned the following tasks:

- Select CCTV system tools, materials and equipment required to facilitate this installation

Indicative content 2.2: Conducting CCTV Trunking.

Duration: 3 hrs

Theoretical Activity 2.2.1: Description of CCTV Trunking

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of CCTV Trunking:

i. What are the CCTV trunking installation types?
ii. What are the CCTV trunking materials types?
iii. State the importance of cable trunking in CCTV camera installation.
iv. What is cable marking, laying and pulling?

2: Provide the answer for the asked questions and write them on papers.

3: Present the findings/answers to the whole class.

4: For more clarification, read the key readings 2.2.1.

5: In addition, ask questions where necessary.

Key readings 2.2.1.:

- **Description of installation types**

✓ **CCTV system installation types**

Visible: Visible cable installations involve running cables along the surfaces of walls, ceilings, or other structures. These cables are often secured using clips or conduits, making them easily accessible for maintenance but potentially more susceptible to tampering or environmental damage.

Built in: Built-in cable installations incorporate the cables within the structure of the building, such as inside walls or ceilings. This method hides the cables from view, providing a cleaner appearance and better protection against tampering and environmental factors.

Semi built in: Some parts of installations are visible while others are hidden.

Water proof also known as Submarine installation is a method used only when no other cable system can be used. This installation supplies circuits that must cross expanses of water or swampy terrain.

Underground: installation means the installation is used where it is necessary to provide a high degree of safety and mechanical protection, or where above-ground conductors would be unattractive.

- **CCTV Trunking Materials types**

Cable trunking is an enclosure which encases cables to protect them and provide space for other electrical equipment. There are different materials types of cable trunks.

a) Plastic trunking is an enclosure made by plastic especially PVC. This trunking has prevailed as a material for cable trunking, thanks to its robust characteristics. It is flame-resistant, largely withstands UV radiation and offers smooth surfaces which look great around walls or floor lines. Residential or office environments will often use plastic/PVC trunking.

b) Stainless steel (**Metal Trunking**) It is an enclosure made by metal especially stainless steel. You can choose aluminium trunking for walls or steel trunking if you want to run cables under the floor. If electrical cables need to be run underground, then flush floor trunking is a great option. For more industrial applications or areas of high footfall metal trunking rules.

c) Wooden trunking is a cable enclosure made by wood. These wooden cable trunking are eco-friendly and are non-conductors of electricity with absolute flame-retardant features.

- **Importance of Cables Trunking**

There are many benefits to installing cable trucking which include:

- ✓ **Cable protection:** The cables are enclosed reducing the risk of damage to the cables. Cable trunking systems reduces the damage as without cable, a trunking cable can be pulled, tangled, or crammed together in tight places causing strain on the cable. The hollow enclosure keeps the cables organised as well as safe from dust.
- ✓ **Improved design:** cable trunking is often used for aesthetics as without cable trunking, it would be an eyesore to have cables run loosely around your walls and floors. Cable trunking tidies up the layout of the wiring. Using cable trunking will smarten up the appearance of the workplace or home, removing the appearance associated with exposed CABLING.
- ✓ **Easier cable management:** Maintenance of cable trunking systems installation are easier than any other cabling system. Cable trunking and in wall cable conduit keep the cables in good condition. Cable trunking is usually a square or rectangular shape with hinges sections providing electricians easy access to cable if required which makes cable management so much easier.
- ✓ **Lower risk and injury:** Cable trunking wiring systems enhance safety. A variety of dangers are present from poorly organised wires such as tripping hazards, cables stocks, and even the risk of fire.
- ✓ **Cable trunking have a long life span:** Without cable trunking, cables will be easily damaged. Outdoor/external cable trunking protects against factors that would have a negative impact on the cable lifetime for example various weather

conditions. Installing wide trunking will keep the cables organised and separated and enables cool air within the cables for safe operation.

- **Cable marking, laying and pulling**

✓ **CCTV Cable Marking**

CCTV cable marking involves labeling cables to identify their purpose, destination, or type. This helps in organizing and managing the cables during installation and maintenance, ensuring easy identification and troubleshooting.

✓ **CCTV Cable Laying**

Cable laying refers to the process of placing cables along designated paths from the CCTV cameras to the recording and monitoring equipment. It involves planning the route, ensuring minimal interference, and protecting cables from damage.

✓ **CCTV Cable Pulling**

Cable pulling is the physical act of threading or pulling the cables through conduits, walls, ceilings, or underground paths to reach their final installation points. This requires careful handling to prevent damage to the cables and ensure proper connectivity.



Application of cable laying and pulling

- **Cable laying Methods**

The common methods of cable laying are:

a) Direct in the ground in trenches (underground cables).
b) In cable trenches in outdoors switchyards.
c) In cable trays or cable ducts

d) Fixed with clamps (usually at walls and ceilings).

e) In conduits

- **Cable laying arrangement**
  Multicore cables are laid in "flat formation" arrangement and single core cables may be laid in "trefoil" arrangement or in "flat formation" arrangement. For single core cables "trefoil" arrangement is the most advised method.

- **Segregation of cables**
  No matter what method is used for laying, cables must be segregate taking into account the voltage level and function, in order to avoid possible electromagnetic interferences that can disturb the networks and the signals cables are carrying on. Segregation means that cables with different voltage levels and/or different functions must not be laid at the same physical support and must be separated.
  Segregation of cables must be done in accordance with standards and regulations, as

- Power cables (by voltage level).
- Control cables.
- Communication cables.

- **Cable marking**
  Cables must be identified along cable runs, using cable markers or labels, showing the references of each cable. This identification shall be according to the following principles:

a) Every 150 m
b) Changes in direction
c) Both ends
   For underground cables splices must be marked and along the cable run wood or concrete markers shall be installed to identify cable run.

- **Cable pulling**
  Cable pulling if it is not done properly may cause damages to the outer sheath of the cables, this leading, sooner or later, to a fault in the cable.

  Pulling tension must be in accordance with the type of cable and the instructions of the manufacturer must be followed.

  Also maximum bending radius indicated by the manufacturers must not be exceed. For long runs, when installing cables in trenches and cable trays special equipment and tools must be used:

a) Cable pulling winch (mechanical drive)

b) Cable feeders for reels
c) Cable rollers
d) Tirfors

When installing cables in conduits approved lubricants of type compatible with cable jacket must be used to reduce pulling tension.

If more than one cable is to be installed in the same conduit they must be installed simultaneously. A “fish tape” shall be used.

- **Appropriate areas whereby CCTV cameras can provide added protection**

- **Retail Storefronts & Interior Areas:** To Deter Criminal Activities & Monitor Employee Productivity
- **Parking Garages:** Monitor & Protect Against Intrusions, Theft.
- **Warehouses/Loading Docks:** Review Shipping/Receiving Protocols & Deliveries
- **Casinos/Hotels:** Deter Theft & Other Malicious Activities
- **Restaurants/Bars:** Manage Front- & Back-of-House Activities
- **Office Environments:** Employee Activity & Productivity Measurement
- **Stadiums:** Monitor Crowd Control & Social-Distancing Protocols
- **Public Safety:** Assist Law Enforcement During Criminal Investigations & Traffics



Practical Activity 2.2.2: Perform CCTV Trunking installation



Task:

1: Referring to previous activities (2.2.1.) you are requested to perform the given task. The task should be done **individually.**

As a trainee who learned trunking in CCTV system installation, you are asked to make installation of CCTV system trunks that will be used to carry cables in the room provided.

2: List out the procedures to be used to perform the given tasks.

3: Referring to the procedures of CCTV camera trunks installation provided in task2, Perform the given task.

4: Present your work to the trainer and whole class.

5. Ask for more clarifications.

6: Perform the task provided in application of learning 2.2



Key readings 2.2.2

- **Perform CCTV Trunking installation process.**

✓ **Planning:**

- Assess the site and create a detailed plan for the trunking route.
- Determine the type and quantity of trunking required.

✓ **Preparation:**

- Gather all necessary tools and materials, including trunking, cables, mounting brackets, and screws.
- Mark the planned trunking route on walls, ceilings, or floors.

✓ **Trunking Installation:**

- Cut trunking to the required lengths using appropriate cutting tools.
- Secure the trunking to the walls or other surfaces using screws and brackets at regular intervals.

✓ **Cable laying and Pulling:**

- Feed or lay the CCTV cables through the installed trunking, ensuring they are properly supported and not stretched.
- Use cable ties or clips inside the trunking to keep cables organized and prevent tangling.

✓ **Connection and Testing:**

- Connect the cables to the CCTV cameras and the recording/monitoring equipment.
- Test the system to ensure all cameras are functioning correctly and the video feed is clear.

✓ **Finalization and labelling:**

- Close the trunking covers securely.
- Label the trunking and cables for easy identification during maintenance.
- Perform a final inspection to ensure everything is properly installed and tidy.



Points to Remember

- Cable trunking is an enclosure which encases cables to protect them and provide space for other electrical equipment.
- **CCTV installation types are** Visible cable cameras installations, Built in Cables cameras installations, Semi built in cable cameras installations, Water proof also known as Submarine installations **and** Underground camera installation
- **CCTV Trunking Materials types are Plastic trunking (enclosure made by plastic** especially PVC), **Stainless steel** (**Metal Trucking**), **Wooden trunking** (enclosure made by wood).

- CCTV cable marking involves labeling cables to identify their purpose, destination, or type. This helps in organizing and managing the cables during installation and maintenance, ensuring easy identification and troubleshooting.
- Cable laying refers to the process of placing cables along designated paths from the CCTV cameras to the recording and monitoring equipment. It involves planning the route, ensuring minimal interference, and protecting cables from damage.
- **CCTV** Cable pulling is the physical act of threading or pulling the cables through conduits, walls, ceilings, or underground paths to reach their final installation points. This requires careful handling to prevent damage to the cables and ensure proper connectivity.
- The installation process for CCTV cabling trunking involves several steps to ensure proper routing and protection of cables within the trunking system which are Planning and Preparation, Marking and Mounting trunks, Cutting and Configuring Trunking, Routing Cables, Securing and Connecting Cables, Completing Installation and perform Final Inspection and Cleanup.



Application of learning 2.2

XYZ complex market would like to install CCTV SYSTEM INSTALLATION in their buildings. You are one of the technician team that is going to install that CCTV System and you are assigned to perform the following tasks:
You are asked to install the CCTV trunking system using a visible installation type with PVC trunking.

Indicative content 2.3: Assembling CCTV equipment

Duration: 3 hrs

Theoretical Activity 2.3.1: CCTV system equipment assembling

Tasks:

1: In small groups, you are requested to answer the following questions related to the of description CCTV system equipment assembling:

i. What is rack?
ii. How to fix rack?
iii. What is camera assembling and mounting?
iv. What is DVR/NVR assembling and mounting?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key **readings 2.3.1.**

5: In addition, ask questions where necessary.

Key readings 2.3.1.:

- **Definition of Rack**

  A rack is a structure or piece of equipment that has multiple levels, slots, or holders designed for holding, displaying or storing various items. It is often made from materials like metal, wood, or plastic and can be used in a variety of settings including homes, businesses, laboratories, control rooms.

  A rack, in an cctv camera installation context, is a supporting framework that holds hardware modules. In this context, racks typically contain DVR or NVR, and other CCTV system equipment. Racks make it possible to contain a lot of equipment in a small physical footprint without requiring shelving.



The standard size for racks is nineteen inches across, although the ETSI (European Telecommunications Standards Institute) standard is 23". Nineteen inch racks are typically 73.5 inches high. For this purpose, the size of equipment designed to be rack-mounted is specified in terms of rack units (U), which equal 1.75 inches. Racks contain multiple mounting slots called bays, each designed to hold a hardware unit secured in place with screws.

- **Mount a Rack.**

  Rack Mounting refers to the process of installing equipment into a standard-sized rack or enclosure designed to hold multiple pieces of electronic or networking equipment.

  CCTV Camera assembling/mounting

✓ **Camera assembling:** Camera assembling involves putting together the different components of a CCTV camera to prepare it for installation like attaching lens with the body of camera.

✓ **Camera Mounting:** Camera mounting refers to the process of installing the assembled camera at a specific location to ensure optimal coverage and functionality.

✓ **Types of camera mounts:**

For the installation of network camera or camera housing & enclosure in video security projects, CCTV is offering a variety of mounting brackets like:

a) Wall mount bracket, which should be fixed on the walls
b) Pole mount bracket, which should be fixed on the poles,
c) Corner mount bracket which should be in the corner sides

- **DVR/NVR assembling/mounting**

✓ **DVR/NVR Assembling:** Assembling a DVR (Digital Video Recorder) or NVR (Network Video Recorder) involves preparing the device for operation by connecting its components and ensuring it is ready to record and manage video footage.

Some steps must be performed include:

- Unpack the DVR/NVR and check all included components (unit, power supply, cables, mounting hardware
- Insert Hard Drives: Connect and secure hard drives into the designated slots or bays.
- Connect Peripherals like Power Supply, Monitor, Mouse/Keyboard: Connect a mouse and/or keyboard for setup and operation.
- Connect Cameras: Connect CCTV cameras using coaxial cables or Connect IP cameras using Ethernet cables.
- Network Connection: Connect the DVR/NVR to your network router or switch using an Ethernet cable for remote access and communication.
- ✓ **DVR/NVR Mounting:** Mounting a DVR/NVR involves securely installing the device in a location that ensures stability, accessibility, and proper ventilation.



Practical Activity 2.3.2: Perform CCTV Trunking installation



Task:

1: Referring to previous activities (2.3.1) you are requested to perform the given task. The task should be done **individually**.

As a trainee who taught cable trunks installation in CCTV system installation, you are asked to assemble CCTV camera, rack and DVR/NVR in the place provided.

2: List out the procedures to be used to perform the given tasks.

3: Referring to the procedures of installing trunking in task2, Perform the given tasks,

4: Present your work to the trainer and whole class.

5. Perform the task provided in application of learning 2.3.2. for more clarifications

6: Perform the task provided in application of learning 2.3



Key readings 2.3.2.

- **Camera assembling and mounting**

The mounting process for CCTV cameras involves several steps to ensure proper installation and optimal positioning for effective surveillance coverage.

CCTV camera assembling and mounting process:

- ✓ Gather all necessary tools and equipment (screwdrivers, drill, screws, mounting bracket, etc.).

- ✓ Ensure you have the camera, power supply, and necessary cables (power, video, network).
- ✓ **Choose Location:** Select an appropriate location for the camera, considering optimal coverage and lighting conditions.
- ✓ **Mark Holes:** Use the mounting bracket as a template to mark the drilling holes on the surface.
- ✓ **Drill Holes:** Drill holes at the marked points.
- ✓ **Attach Bracket:** Secure the mounting bracket to the surface using screws and anchors, ensuring it is firmly fixed.
- ✓ **Assemble Camera:** Connect cameras with its mounts



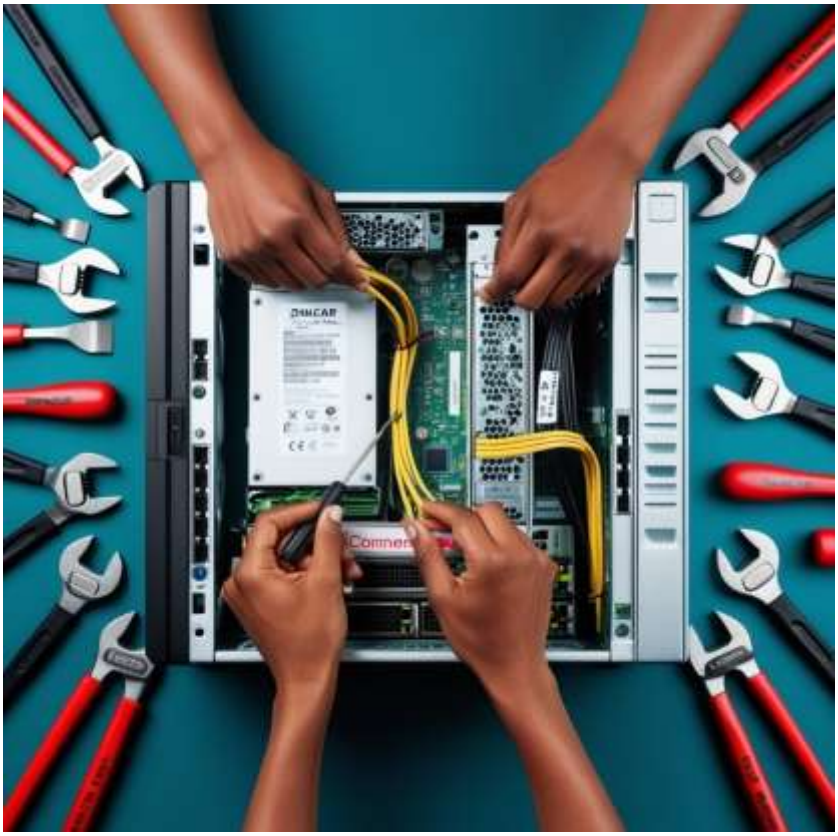
- ✓ **Run Cables:** Run the necessary cables (power, video, network) through the mounting bracket or wall, as required.
- ✓ **Connect Cables:** Connect the cables to the camera, ensuring proper connections (e.g., power cable to power port, network cable to Ethernet port).



- ✓ **Mount Camera:** Attach the camera to the mounting bracket, using screws or other provided fasteners.



- ✓ **Adjust Angle:** Adjust the camera angle and position to cover the desired area.
- ✓ **Power Up:** Connect the power supply to the camera and plug it into a power outlet.
- ✓ **Network Connection:** If using a network camera, connect the Ethernet cable to the network switch or router.
- ✓ **Access Camera:** Use a computer or monitor to access the camera's interface (often via a web browser or dedicated software).
- ✓ **Configure Settings:** Configure camera settings such as resolution, frame rate, and network settings.
- ✓ **Test Camera:** Check the camera feed to ensure it is capturing the desired area and the video quality is satisfactory.
- ✓ **Fine-Tune Position:** Make any necessary fine adjustments to the camera's position and angle.
- ✓ **Secure Connections:** Ensure all connections are secure and protected from environmental factors.
- ✓ **Label Cables:** Label the cables for easy identification during maintenance.

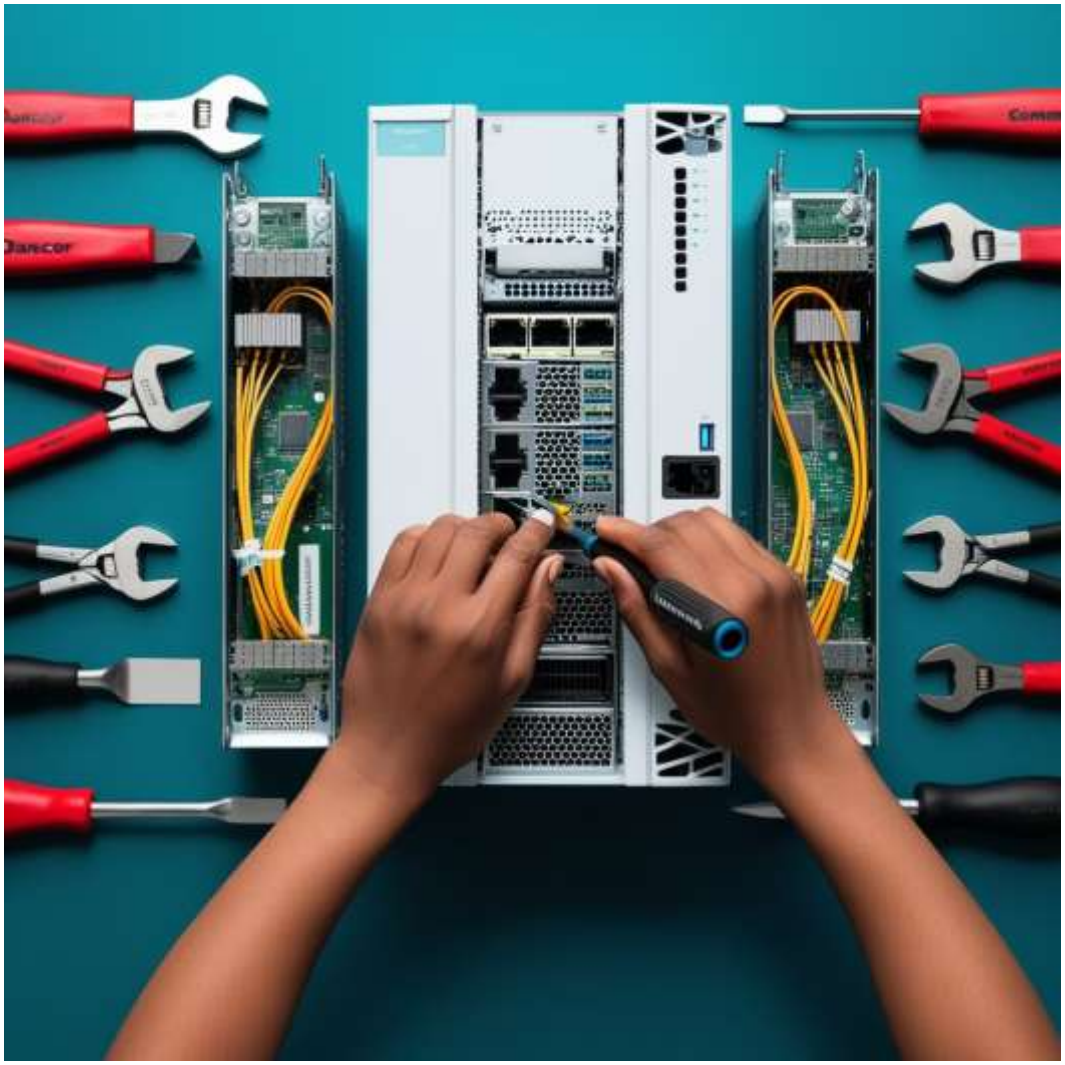


- **Rack Mount**

  Rack mounting process:

  - ✓ **Choose Location:** Select an appropriate location for the rack, ensuring it is stable and has sufficient space for ventilation and access.
  - ✓ **Gather Tools and Materials:** Ensure you have the necessary tools (screwdrivers, drills, level, measuring tape) and mounting hardware (screws, bolts, brackets).
  - ✓ **Follow Instructions:** Assemble the rack according to the manufacturer's instructions, ensuring all parts are securely fastened.
  - ✓ **Check Stability:** Ensure the rack is stable and level.
  - ✓ **Mark Holes:** Position the rack in its designated spot and mark the holes for drilling.
  - ✓ **Drill Holes:** Drill holes in the marked positions.
  - ✓ **Attach Rack:** Use bolts or screws to secure the rack to the floor or wall, ensuring it is firmly anchored.
  - ✓ **Position Rails/Shelves:** Install mounting rails or shelves inside the rack according to your equipment requirements.
  - ✓ **Secure Rails/Shelves:** Fasten the rails or shelves securely using the provided hardware.
  - ✓ **Attach Rack Ears/Brackets:** If the equipment has rack ears or brackets, attach them to the equipment.

- ✓ **Install Equipment:** Slide the equipment into the rack, aligning it with the mounting holes on the rails or shelves.
- ✓ **Secure Equipment:** Use screws or bolts to secure the equipment to the rack, ensuring it is firmly in place.
- ✓ **Route Cables:** Run cables through the rack, using cable management features such as cable ties, clips, or trays to keep them organized.
- ✓ **Label Cables:** Label the cables for easy identification and maintenance.
- ✓ **Connect Power:** Plug in the equipment to the power supply, using a rack-mounted power distribution unit (PDU) if available.
- ✓ **Network Connections:** Connect network cables to the appropriate ports on the equipment.
- ✓ **Check Stability:** Ensure all equipment is securely mounted and the rack is stable.
- ✓ **Verify Connections:** Check all connections to ensure they are secure and functioning properly.
- ✓ **Test Equipment:** Power on the equipment and verify it is operating correctly.



- **DVR/NVR Mounting**

Steps to Mount and Assemble a DVR/NVR

- ✓ **Choose Location:** Select a suitable location for mounting the DVR/NVR, considering accessibility, ventilation, and proximity to cameras and network connections.
- ✓ **Gather Tools and Materials:** Prepare necessary tools (screwdrivers, drill), mounting hardware (brackets, screws), and the DVR/NVR unit.
- ✓ **Determine Mounting Method:** Decide whether to mount the DVR/NVR in a rack (using rack ears) or on a wall/surface (using wall brackets).
- ✓ For rack mounting, align the DVR/NVR with the rack rails and mark the hole positions for rack ears.
- ✓ For wall mounting, mark the positions for wall brackets based on the DVR/NVR's mounting holes.
- ✓ Place the DVR/NVR onto the wall brackets or into the rack, and secure it with screws or bolts.
- ✓ Connect the DVR/NVR to a power source using the provided power adapter or cable.
- ✓ Connect the cameras to the DVR/NVR using coaxial cables (for analog systems) or Ethernet cables (for IP systems).
- ✓ Connect the DVR/NVR to the network router or switch using an Ethernet cable for remote access and communication.
- ✓ Connect a monitor to the DVR/NVR using the appropriate video output port (HDMI, VGA, etc.) to view the live feed and setup menus.



- **NVR configuration process:**

NVR can be set up with or without a Power over Ethernet (PoE) or Non-PoE switch.

How to Configure NVR for IP Camera with a PoE Switch?

You can find Ethernet switches for IP surveillance in different configurations. The configuration starts with a hardware setup. Here is how you can proceed with the hardware setup:

- ✓ Take the Ethernet cable from the router and connect it at the back of NVR. Ensure that NVR is connected to the Internet.
- ✓ Use Ethernet cables to connect the IP cameras to PoE NVR. As the NVR is enabled with the PoE technology, it can easily power the IP cameras.
- ✓ Take an HDMI cable or a VGA cable to connect NVR and TV or monitor. Ensure that you use the right input to connect the TV or the monitor— HDMI 1, HDMI 2, or VGA. When these steps are performed properly then NVR and IP camera connection can be easily made. Software setup is the next step in the configuration, and this can be easily done by following the instruction manual provided by the manufacturer.



- **How to Configure NVR for IP Camera without PoE Switch?**

As discussed before, NVR without PoE switch has no Ethernet ports at its back panel. It utilizes an external power adapter or an external PoE switch to power each IP camera. Like NVR with a PoE switch, the configuration of NVR for IP camera without a POE switch also begins with the hardware configuration. Here is how to proceed:

- ✓ Take an Ethernet cable to connect the LAN port on the external PoE switch and the router. Then connect the router with the non-PoE NVR.
- ✓ Use Ethernet cables to connect all IP cameras to the PoE switch RJ45

ports. The PoE switch will deliver power as well as support video transmission.

- ✓ Use an HDMI cable or VGA cable to connect the monitor and the NVR. Always use the right input for making the connection.

Recording will be enabled only when users add the cameras to NVR. This can be easily done by referring to the steps in the instruction manual.



Points to Remember

- A rack is a structure or piece of equipment that has multiple levels, slots, or holders designed for holding, displaying or storing various items.
- Rack Mounting refers to the process of installing equipment into a standard-sized rack or enclosure designed to hold multiple pieces of electronic or networking equipment.
- Steps to mount a rack are: Choose an appropriate location, assemble rack components, Fix the rack to the floor or wall, Place, fix and Install equipments, organize cables and finally Connect power and network cables securely and label them for easy maintenance.
- Camera assembling involves putting together the different components of a CCTV camera to prepare it for installation like attaching lens with the body of camera.
- Camera mounting refers to the process of installing the assembled camera at a specific location to ensure optimal coverage and functionality.
- Types of camera mounts are Wall mount bracket, Pole mount bracket and Corner mount bracket.
- DVR/NVR Assembling: Assembling a DVR (Digital Video Recorder) or NVR (Network Video Recorder) involves preparing the device for operation by connecting its components and ensuring it is ready to record and manage video footage.
- DVR/NVR Mounting: Mounting a DVR/NVR involves securely installing the device in a location that ensures stability, accessibility, and proper ventilation.
- CCTV camera assembling and mounting process includes Gathering all necessary tools and equipment (screwdrivers, drill, screws, mounting bracket, etc.), Ensure you have the camera, power supply, and necessary cables (power, video, network). Choose Location, Mark Holes, Drill Holes, Attach Bracket, Run Cables through the mounting bracket, Connect Cables, Mount Camera, Adjust Angle and Power Up.
- Rack mounting process involves choosing location, Gather Tools and Materials, Assemble the rack, Check Stability, Install Equipment.
- Steps to Mount and Assemble a DVR/NVR includes choosing Location, Gather Tools and Materials, Determine Mounting Method.



Application of learning 2.3

XYZ complex market wants to install CCTV systems in their buildings for better management of equipment in their environment, as a CCTV camera system installer, you are asked to install CCTV cameras by mounting the track, assemble and mount the camera and configure recording devices.

Indicative content 2.4: Assembling CCTV equipment

Duration: 2 hrs

Theoretical Activity 2.4.1: Fixing and connecting CCTV controller devices and camera

Tasks:

1: In small groups, you are requested to answer the following questions related to the Fixing and connecting CCTV controller devices and camera.

i. What are cable terminals?
ii. What is patching?
iii. What are the types of patching?
iv. What are the patch code types?
v. What are the patch panel management types?
vi. What is tagging?

2: Provide the answer for the asked questions and write them on papers.

3: Present the findings/answers to the whole class.

4: For more clarification, read the key readings 2.4.1.

5: In addition, ask questions where necessary.

Key readings 2.4.1.:

- **Terminating cables**

Cable Termination is the connection of the wire or fibre to a device, such as equipment, panels or a wall outlet, which allows for connecting the cable to other cables or devices.

- **Patching**

A patch panel is a piece of hardware with multiple ports that helps organize a group of cables. Each of these ports contains a wire that goes to a different location. Patch panels can be quite small, with just a few ports, or very large, with many hundreds of ports. They can also be set up for fibre optic cables, cat5 cables, RJ45 cables, and many others.

➢ **Advantages of Patch Panels**

Patch panels are an important piece of equipment in the data center. Using a patch panel comes with many advantages that make it well worth the investment. Some of the most significant benefits of using patch panels include:

-**Scalability**: After installing a patch panel, you can easily add new devices without having to run new cables end-to-end.

-**Reduces Cable Clutter**: Patch panels are typically located closer to the actual equipment. This makes it possible to use a shorter patch cable. From the panel,

a fibre optic or other high capacity connection runs to the next network or the internet.

-**Inexpensive**: Patch panels aren't considered "smart" devices in that they don't perform any function other than facilitating the passing of data. This means they are very inexpensive.

-**Lowering Cable Cost**: Having a patch panel allows you to use shorter cables, which cost less than longer ones. Typically, you can also use inexpensive cat-5 cables to get to the patch panel rather than costly fibre optics.

-**Easier Maintenance**: If you ever need to run a test cable, test a port, or perform any other common maintenance tasks, it can be done more quickly and is much easier than if each device had a cable run to its final destination.

- **Types of patch panels**

Basically, according to the cables connected, there are two types of networking patch panel: **Ethernet patch panel and fibre optic patch panel**. **Ethernet Patch Panel**

Ethernet patch panel, also called copper patch panel, provides interconnections for Ethernet applications in local area network. There are many types of Ethernet patch panels, among which Cat5e patch cable, Cat6 patch cable, and Cat6a patch cables are commonly used in network cabling. As for ports of the copper patch panel, various options also can be found, such as 24 ports and 48 ports. If you want to buy an Ethernet panel, you still have to consider its designs: shielded or unshielded, flat or angled, punch down or feed-through, and loaded or unloaded. With various options, you can always find a needed type.

- **Fibre Optic Patch Panel**

Fiber optic patch panel, the other branch of networking patch panel, is designed to organize fiber cables. Similar with Ethernet patch panels, the fiber optic patch panels also are varied in designs, such as adapter types, fiber types and fiber count. Conventionally, fiber optic patch panel is commonly used as rack mount fiber patch panel or wall mount patch panel, which also applied for the Ethernet patch panel.'

- **Patch code types**

A Patch Code consists of two wide bars (0.2 inches (5.1 mm) ± 0.01 inches (0.25 mm)) and two narrow bars (0.08 inches (2.0 mm)). The bars are separated by three narrow spaces, so the Patch Code symbols are a fixed length.

Patch Codes do not contain any encoded data. Instead, the pattern of the code tells scanning software which specific action to perform. There are six Patch Code patterns:

- Patch 1 — Wide, wide, narrow, narrow (WWNN). This is used for post-scan image control.
- Patch 2: Wide, narrow, narrow, wide (WNNW). This code assigns "image level 2" to the scanned document.
- Patch 3: Wide, narrow, wide, narrow (WNWN). This code assigns "image level 3" to the scanned document.

- Patch 4: Narrow, wide, wide, narrow (NWWN). Patch 4 is used for post-scan image control.
- Patch 6: Narrow, narrow, wide, wide (NNWW). This Patch Code is also used for image control.
- Patch T: Narrow, wide, narrow, wide (NWNW). Also known as the “transfer patch,” this code assigns a predefined image level to the next scanned document.

Patch Codes are used to increase the speed and accuracy of batch document processing. With the right barcode scanning software, it’s possible to scan large document volumes without the need to manually separate batches or worry that documents are perfectly aligned.

- **Patch panel management types**

Patch panel cable management is involved in two components: **fibre optic patch panels** and **Ethernet patch panels**. Different kinds of patch panel meet the demand to effectively manage high-density structured cabling in different applications.

Today’s data centers require a reliable, scalable, and manageable cabling infrastructure, and then the patch panel cable management solutions address these trends and facilitate the efficiency of high-density data center cabling. By the way, when purchasing the cabling infrastructure, there is no single solution that will meet all of the cable management needs.

- **Tagging**

The winning technology, Tag and Track (TNT), is a multi-camera tracking system that continuously tracks all persons or objects within an existing surveillance network. TNT can instantly locate any tagged person at any time along with the path they have taken throughout their journey in the network. An operator can tag an individual by a simple mouse click on that person in live or recorded video.

TNT automates the time consuming element of the day to day work of CCTV control room operators who manually trail suspects from camera to camera. It also allows operators to "follow" multiple tagged targets automatically, making staff more efficient.

The technology was initially developed by Kingston computer science specialists led by Dr. James Orwell, who is based in the University's Digital Imaging Research Centre. "Once operators notice potentially suspicious behavior, they can tag the person concerned by clicking on their image. This triggers the system to go back to the database and immediately create a full route for the person concerned using notes it has stored. The technology can provide predictions about individuals’ steps beforehand as well as find out where they move on to afterwards.

Practical Activity 2.4.2: Fixation and connection of CCTV device

Task:

1: Referring to previous activities (2.4.1) you are requested to perform the given task. The task should be done **individually**.

As a trainee who taught fixation and connection of CCTV devices, you are asked to fix and connect the CCTV devices given.

2: List the criteria to be used to select tools, materials and equipment.

3: Referring to procedures provided on task 2, Perform the given task.

4: Present your work to the trainer and whole class.

5: Read key reading 2.4.2 and ask clarification where necessary

6: Perform the task provided in application of learning 2.4

Key readings 2.4.2

- **Fixation and connection of CCTV Camera devices:**

✓ Ensure proper mounting of cameras for optimal coverage (Refer to Indicative Content 2.3).

✓ Use appropriate hardware to securely attach cameras to mounting brackets or surfaces.

✓ Align cameras correctly to capture desired areas without obstruction.

✓ Connect power cables and video/Data cables securely to cameras. For PoE (Power over Ethernet) cameras, connect the Ethernet cable which provides both power and data.

✓ Ensure cables are properly routed to prevent damage or interference.

✓ Test camera functionality after installation to verify proper connection and operation.

✓ Connect coaxial cable to the camera's video output and the DVR's input.

✓ For IP Cameras, Connect the cameras to the NVR using Ethernet cables. Ensure that the NVR is connected to a network switch or router for communication.

✓ Connect a monitor to the DVR/NVR using HDMI, VGA, or another compatible video output port for viewing live feeds and setup menus.

✓ Connect the DVR/NVR to the network using an Ethernet cable to enable remote access and management.

✓ Connect the DVR/NVR to a power source using the provided power adapter or cable.

✓ Power On: Turn on the cameras, DVR/NVR, and any associated equipment.

✓ Configuration: Access the system's interface to configure settings, such as camera

positions, recording schedules, and network settings.

- ✓ Testing: Verify that all cameras are functioning correctly and transmitting video to the DVR/NVR. Check connections and image quality.
- ✓ Cable Management: Organize and secure cables to prevent damage and ensure a neat installation.

- **Procedures for Terminating, Patching, and Tagging Cables:**

1. Terminating Cables

- ✓ **Preparation**
  - Gather necessary tools and materials such as wire strippers, crimping tools, and connectors.
  - Ensure you have the correct type of cables (e.g., coaxial for analog cameras, Ethernet for IP cameras).
- ✓ **Termination Process**
  - **Strip the Cable:**
    - **Coaxial Cables:** Remove the outer insulation to expose the shield and inner conductor. Strip the inner conductor to fit the connector.
    - **Ethernet Cables:** Remove the outer jacket, expose the twisted pairs, untwist them, and arrange according to the wiring standard (T568A or T568B).
  - **Attach Connectors:**
    - **Coaxial Cables:** Slide a BNC connector onto the stripped end and crimp it securely with a crimping tool.
    - **Ethernet Cables:** Insert the arranged wires into an RJ45 connector and use a crimping tool to secure the connector.
  - **Test Termination:** Use a cable tester to check for correct connections, ensuring no shorts, open circuits, or miswires.

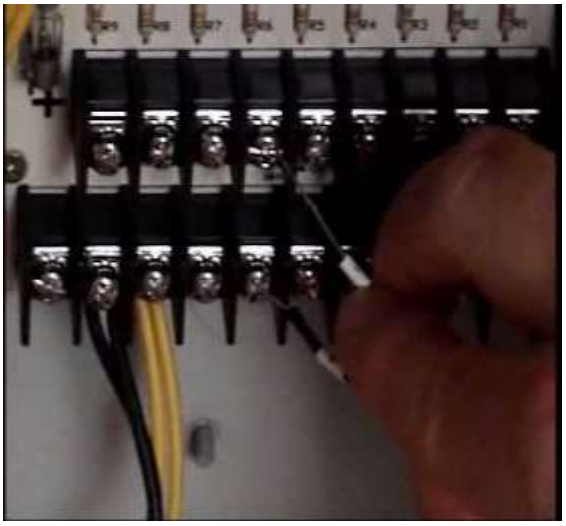




2. Patching:

- ✓ **Preparation**
  - Set up patch panels in a central location for organized connections.
  - Gather patch cords of appropriate lengths for connecting devices.
- ✓ **Patching Process**
  - **Organize and Route Cables:**
    - Route cables from CCTV cameras to the patch panel or switch.
    - Ensure cables are neatly arranged and do not obstruct airflow or create tangles.
  - **Patch Connections:**
    - **Patch Panels:** Insert each cable into the corresponding port on the patch panel. Use patch cords to connect the patch panel to the DVR/NVR and network switches.
    - **Direct Connections:** Connect cables directly between devices and switches or DVRs/NVRs.
  - **Secure Connections:** Ensure all connections are securely seated and properly aligned. Avoid loose or poorly connected cables.





3. Tagging:

- ✓ **Preparation**
  - Obtain durable labels or tags for cables and connectors.
  - Ensure labels are clear, easy to read, and resistant to wear.
- ✓ **Tagging Process**
  - **Label Cables:**
    - **At Both Ends:** Label each cable at both ends to identify its source and destination. Include details like camera number, location, and connection point (e.g., "Camera 1 - Front Entrance").
    - **Consistent Format:** Use a consistent labeling format for clarity.
  - **Label Connectors:**
    - **On Patch Panels and Switches:** Label connectors to identify which ports are used for which connections.
    - **Detailed Information:** Include details such as port numbers and device names.
  - **Documentation:** Keep updated documentation or diagrams of cable layouts, patch panel connections, and tagging information.





4. Final Checks:

- ✓ **Verify Terminations:** Ensure all cables are correctly terminated and connections are secure.
- ✓ **Test the System:** Perform functionality checks to confirm all devices are operational.
- ✓ **Review Labels and Documentation:** Ensure that all cables are properly tagged and that documentation accurately reflects the installation setup.



Points to Remember

- Cable terminals are connectors or junction points where CCTV cables are terminated or connected to other equipment. They provide the interface for connecting cables to devices such as cameras, DVRs/NVRs, or monitors.
- The purpose of Cable Terminals is to ensure a secure and stable connection between the CCTV system's various components, facilitating the transmission of video, power, and data signals.
- Patching refers to the process of connecting and routing cables to various ports or devices within a CCTV system. This is typically done using patch panels or patch cords.
- Purpose of Patching is to allows for flexible management and organization of cable connections, making it easier to configure, troubleshoot, and maintain the CCTV system.
- Types of Patching are Direct Patching, Panel Patching and Patch Cord Patching.
- Patch Cord Types are Ethernet Patch Cords, Coaxial Patch Cords and Fiber Optic Patch Cords.
- Patch Panel Management Types are Fixed Patch Panels, Sliding Patch Panels and Angled Patch Panels.
- Tagging involves labeling cables and connectors within a CCTV system to identify their purpose and destination.
- The purpose of tagging is to helps in organizing and managing the cabling system, making it easier to trace connections, perform maintenance, and troubleshoot issues. Tags typically include information like the cable's source, destination, and function.
- Fixation and connection of CCTV Camera devices steps.
- Termination cables Process involves Strip the Cable, Attach Connectors, Test Termination.
- Patching Process involves Organize and Route Cables, Patch Connections and Secure Connections.
- Tagging Process involves Label Cables and Label Connectors.



Application of learning 2.4

NKANDA Factory want to CCTV SYSTEM INSTALLATION in its factory storage building, as one of the CCTV system installer team in the industry, you are asked to terminating, patching and tagging cables.



Learning outcome 2 end assessment

Theoretical assessment

A. Select the letter corresponding to the correct answer.

1. Which of the following tools is used for creating holes in walls during a CCTV installation?

A) Cutting tools
B) Drilling tools
C) Fixing tools
D) Patching tools

2. Which equipment is necessary for storing video footage in a CCTV system?

A) Monitor
B) Power Supply
C) NVR/DVR
D) Switch

3. Which material is commonly used to organize and secure cables in a CCTV installation?

A) Screws
B) Cable ties
C) Labelling tags
D) Pipes

4. What is the purpose of using a rack mount in a CCTV system?

A) To store footage
B) To mount and organize equipment
C) To transmit video signals
D) To power the cameras

5. Which type of trunking is most suitable for outdoor installations exposed to weather?

A) Visible trunking
B) Built-in trunking
C) Semi built-in trunking
D) Waterproof trunking

6. Which material is NOT typically used for CCTV trunking?

A) Plastic
B) Stainless steel
C) Wooden
D) Glass

7. Which of the following is used to secure CCTV cables to walls or ceilings?

A) Cable trunks

B) Cable ties

C) Power extension

D) Labelling tags

8. What type of tool is used for connecting and securing network cables to patch panels?

A) Drilling tool

B) Fixing tool

C) Patching tool

D) Cutting tool

9. Which device is used to power multiple CCTV cameras from a single source?

A) Power extension

B) CCTV Power supply box

C) NVR/DVR

D) Monitor

10. In a CCTV installation, what is the primary function of patch panels?

A) To mount cameras

B) To record video footage

C) To organize and connect network cables

D) To provide power to cameras

11. Which type of cable laying arrangement involves separating cables to avoid interference?

A) General method

B) Cable laying arrangement

C) Segregation of cables

D) Cable bundling

12. Which of the following is NOT a testing tool used in CCTV installation?

A) Cable tester

B) Multimeter

C) Screwdriver

D) Network tester

13. What is the main advantage of using stainless steel trunking over plastic trunking?

A) Lower cost

B) Better flexibility

C) Higher durability and strength

D) Easier installation

14. Which of the following is used to label and identify cables in a CCTV system?

A) Cable ties
B) Screws
C) Labelling tags
D) Wall mounts

15. Which type of trunking is typically embedded within walls for a cleaner installation?

A) Visible trunking
B) Built-in trunking
C) Waterproof trunking
D) Semi built-in trunking

16. …….. used to place CCTV components definitely and more or less permanently a wall.

A) Drilling tool
B) Cutting tool
C) Fixing tool
D) Patching tool

17. …….. a device used to connect and power the IP cameras via Ethernet cables.

A) Computer
B) Monitor
C) Power supply
D) Switch

18. The important advantages of the NVR system are the following, except:

A) Better image quality
B) More system flexibility
C) Easy to install
D) Minimal bandwidth usage

19. Which of the followings are the materials of CCTV installation? (select three)

A) Cable sites
B) Computer
C) Connectors
D) Internet bundles
E) Monitor
F) Switch

20. The following are the Appropriate areas whereby CCTV cameras can provide added protection, except:

A) Bathroom
B) Office Environments
C) Parking Garages
D) Retail Storefronts & Interior Areas
E) Warehouses/Loading Docks

B. Answer the following statement by True if the statement is correct, otherwise False.

1. In Cable trunking system, the lid is kept in place by turnbuckles.
2. Lighting trunking is a trunking material.
3. In cable lying arrangement, Multicore cables are laid in “flat formation” arrangement and single core cables may be laid in “trefoil” arrangement or in “flat formation” arrangement.
4. Segregation of cables and cable marking are advisable while cable laying and pulling.
5. Built-in devices or features are included in something as a part of it, rather than being separate. True
6. Scalability, inexpensive, are the advantages of patching.

Practical assessment

A networking company get the job to install CCTV systems in a building of 4 rooms, as the company technician, you are tasked to perform the following tasks:

Task:
You are asked to Fix and Assemble the following CCTV Equipment and materials in provided area.
- 1 Rack mount
- Wall trunking
- 4 Camera
- 1 Recording devices (NVR/DVR)



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CCTV Technician
TV
DVR

Indicative contents
3.1. Management of Administration parameters settings
3.2. Management of CCTV user accounts
3.3. Configure CCTV IP addresses
3.4. Apply setting of CCTV recordings
3.5. Document the CCTV functionalities

Key Competencies for Learning Outcome 3: Configure CCTV System

Knowledge	Skills	Attitudes
● Description of default password. ● Identification of recording storage. ● Identification of CCTV user account management. ● Description of dynamic and static IP. ● Identify of CCTV Diagram Design software. ● Description of CCTV recording.	● Change default password. ● set recording and storage.	● Be a Teamwork player. ● Critical thinking. ● Have a Creativity spirit. ● Be an Innovator.



Duration: 10 hrs

Learning outcome 2 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Manage correctly the administration parameters on CCTV camera configuration
2. Set correctly CCTV account to the user privileges based on CCTV system configuration.
3. Configure IP addresses correctly base on the type of CCTV camera.
4. Set correctly recording settings on CCTV System
5. Document properly CCTV System based on CCTV functionalities



Resources

Equipment	Tools	Materials
● Computer ● Monitor ● Power Supply ● Switch ● Storage devices ● Cameras ● Recording Devices (NVR/DVR)	● Web browser	● Internet bundles

Indicative content 3.1: Management of Administration parameters settings

Duration: 2 hrs

Theoretical Activity 3.1.1: Description for configure CCTV System administration parameters settings

Tasks:

1: In small groups, you are requested to answer the following questions related to the Description of configure CCTV System administration settings

i. What is username and password?
ii. What is the default password?
iii. What is recording storage?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class.

4: For more clarification, read the key readings 3.1.1

5: In addition, ask questions where necessary.

Key readings 3.1.1.:

- **Description for configure CCTV System**
  - ✓ **Username and Password.**

A username and password are credentials used for authentication, particularly in computer systems, websites, and online services, to verify the identity of a user before granting access to a system or resource. Here's what they entail:

✓ **Username**

A username is a unique identifier chosen by a user to represent their identity within a system or service. It is often a string of characters, such as letters, numbers, or symbols, that is associated with a specific user account. Usernames are used to log in to a system or service and differentiate between different users accessing the same system.

✓ **Password**

A password is a secret string of characters chosen by a user to authenticate their identity and gain access to a system or service associated with their username. Passwords are intended to be known only to the user and are used as a form of authentication to prevent unauthorized access to sensitive information or resources. Passwords are typically kept confidential and should be sufficiently complex and unique to protect against unauthorized access.

When a user attempts to log in to a system or service using their username and password, the system verifies the entered credentials against stored records to authenticate the user's identity. If the provided credentials match those on record, access is granted, and the user

is allowed to proceed. If the credentials do not match or are invalid, access is denied, and the user is typically prompted to re-enter their credentials or take appropriate action to recover or reset their password.

A default password is a standard preconfigured password for a device.

Default passwords for CCTV cameras vary depending on the manufacturer and model. However, it's important to note that default passwords are often set by manufacturers for initial setup and should be changed immediately after installation to enhance security and prevent unauthorized access.

If you have a CCTV camera and you're looking for the default password, you can typically find it in the product documentation provided by the manufacturer. This documentation may include user manuals, quick start guides, or online resources available on the manufacturer's website.

Recording storage: Recording storage in CCTV (Closed-Circuit Television) cameras refers to the capacity and method of storing video footage captured by the camera for later retrieval and analysis.

There are several methods of recording storage in CCTV cameras:

- ✓ **Local Storage**

Many CCTV cameras are equipped with onboard storage options, such as SD cards or internal memory, allowing them to record video locally without the need for external storage devices. Local storage is convenient and suitable for standalone camera setups or small-scale surveillance systems.

- ✓ **DVR (Digital Video Recorder)**

DVRs are dedicated devices used to record and store video footage from multiple CCTV cameras. They typically feature multiple channels for connecting cameras and offer various storage capacities, ranging from built-in hard disk drives (HDDs) to expandable storage options. DVRs provide centralized storage and management of recorded footage and often offer features such as scheduled recording, motion detection, and remote access.

- ✓ **NVR (Network Video Recorder)**

NVRs are similar to DVRs but are designed to work with IP (Internet Protocol) cameras, which transmit video over a network. NVRs capture and store video footage from IP cameras, providing centralized storage and management of recorded footage. They typically offer high scalability and flexibility, allowing for the integration of a large number of cameras and the use of advanced features such as video analytics and cloud storage.

- ✓ **Cloud Storage**

Some CCTV cameras and recording systems offer cloud storage options, allowing video footage to be uploaded and stored securely in the cloud. Cloud storage offers benefits such as remote access to footage from anywhere with an internet connection, automatic backups, and scalability. However, it may require a subscription fee and depend on the reliability and security of the cloud service provider.

- ✓ **Hybrid Solutions**

Hybrid solutions combine elements of local storage, DVRs, NVRs, and cloud storage to provide flexible and scalable recording options. These solutions may involve a combination of onsite and offsite storage, allowing users to leverage the benefits of different storage methods based on their specific requirements and preferences.
The choice of recording storage method depends on factors such as the scale of the surveillance system, budget, storage requirements, remote access needs, and security considerations.



Practical Activity 3.1.2: Perform administration parameters settings



Task:

1: Referring to previous activities (3.1.1) you are requested to perform the given task. The task should be done **individually**.

As a trainee who taught how to change password and setting CCTV recording on storage, you are asked to change the default password of a CCTV camera system to "12345" and set the camera to record.

2: List out procedures of configuration to be used to perform the given tasks (3.1.1).

3: Referring to procedures provided on task 2, Perform the given task.

4: Present your work to the trainer and whole class.

5: Read key reading 3.1.2 and ask clarifications where necessary.

6: Perform the task provided in application of learning 3.1



Key readings 3.1.2

- **Configuration of CCTV System**

✓ **Change default password**

Changing the default password on a CCTV camera is essential for security purposes as it helps prevent unauthorized access to the camera and its footage.

Here are the general steps to change the default password on a CCTV camera:

1. **Log In to the Camera:**

- Once the camera's web interface loads, you will be prompted to log in.
- Enter the default username and password for the camera. This information is often provided in the camera's user manual or on a sticker attached to the camera.
- Click or tap the Login button to proceed.

2. **Access the Camera Settings:**

- Once logged in, navigate to the camera settings or configuration page. The location of this page may vary depending on the camera model and firmware version.

- Look for an option or menu item related to user management, account settings, or security settings.

3. **Change the Password:**

- Locate the option to change the password for the camera's administrator or user account.
- Enter the current/default password in the appropriate field.
- Enter a new, strong password in the New Password field. A strong password typically includes a combination of uppercase and lowercase letters, numbers, and special characters. Ensure the password is memorable but difficult for others to guess.
- Retype the new password in the Confirm Password field to verify it.

4. **Save the Changes:**

- After entering the new password, look for a Save, Apply, or OK button to apply the changes.
- Click or tap the button to save the new password settings.

5. **Logout and Test:**

- Logout of the camera's web interface to ensure the changes take effect.
- Close the web browser or navigate away from the camera's web interface.
- Attempt to log back in to the camera using the new password to verify that the password change was successful.

✓ **Configuration of recording storage**

When all the camera images are showing, you are then ready to set up the DVR/NVR.

The DVR setup wizard will automatically show when you switch on the recorder and you will see the wizard on the monitor and use a mouse connected to the DVR to navigate the system.

Setting up recording storage on a CCTV camera involves configuring where and how the video footage captured by the camera will be stored. Here are the general steps to set up recording storage on a CCTV camera:

1. **Choose Storage Location**:

Determine where you want to store the recorded footage. Common storage options include:

a) **Onboard SD card:** Some CCTV cameras have built-in slots for SD cards, allowing you to record directly to the card.

b) **Network-attached storage (NAS):** Connect the camera to a NAS device on the network to store footage centrally.

c) **Digital Video Recorder (DVR) or Network Video Recorder (NVR):** Connect the camera to a DVR or NVR to record footage to its internal storage or external hard drives.

2. **Prepare the Storage Device**:

If using an SD card, ensure it is properly inserted into the camera and formatted according to the camera's specifications.

If using a NAS, DVR, or NVR, ensure the device is powered on, connected to the network, and configured properly to receive and store footage from the camera.

3. **Access Camera Settings**:
   Connect to the CCTV camera's web interface using a web browser on a computer connected to the same network as the camera.
   Log in to the camera using the administrator credentials.
4. **Navigate to Recording Settings**:
   Once logged in, navigate to the recording settings section of the camera's configuration menu. The exact location of this setting may vary depending on the camera model and firmware version.
5. **Configure Recording Schedule**:
   Set up a recording schedule based on your requirements. Options typically include continuous recording, scheduled recording (e.g., recording during specific hours of the day), or motion-triggered recording (recording only when motion is detected).
6. **Select Storage Location**:
   Choose the storage location where you want the footage to be saved. This could be the onboard SD card, NAS device, DVR, or NVR.
7. **Adjust Recording Settings**:
   Customize recording settings such as video resolution, frame rate, and compression settings according to your preferences and storage limitations.
8. **Save and Apply Settings**:
   After configuring the recording settings, save the changes and apply them to the camera.
9. **Test Recording**:
   Verify that the camera is recording footage correctly by performing a test recording and reviewing the saved footage to ensure it meets your expectations.
10. **Monitor Storage Space**:
    Regularly monitor the available storage space on the recording device to ensure that it does not fill up and overwrite older footage prematurely. Consider setting up alerts or notifications for low storage space if supported by the recording device.

By following these steps, you can effectively set up recording storage on your CCTV camera and ensure that video footage is captured and stored according to your requirements.

Points to Remember

- **Username and Password:** Credentials used to authenticate users and grant access to systems or accounts, Username identifies the user, while the password verifies their identity.
- **Default Password is a** Pre-set password assigned by manufacturers for initial access to devices or systems.
- **Recording Storage:** Storage space allocated for storing recorded footage captured by CCTV cameras. Recording storage can be in the form of local storage devices (such as hard drives or Network Attached Storage) or cloud-based storage services.
- Steps to Change default password.
- Steps to Set recording storage.

Application of learning 3.1

TERA Company Ltd would like to recruit a CCTV system controller for better management of their equipment. You are one recruited to the post and you have to perform the following task:
To change the default password of the DVR to “MyBusiness23” and set where the recording footage must be stored.

Indicative content 3.2: Management of CCTV user account

Duration: 2 hrs

Theoretical Activity 3.2.1: Describe CCTV user account.

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of CCTV user account.

i. What is user account?
ii. What is factor authentication?
iii. Differentiate the two types of factor authentication?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key readings **3.2.1.**

5: In addition, ask questions where necessary.

Key readings 3.2.1.:

- **Description of User account**

✓ **What is user account?**

A user account in a CCTV (Closed-Circuit Television) camera refers to a configured set of credentials that allow individuals to access and manage the camera's settings, configuration, and recorded footage. User accounts are used to control access to the camera's web interface or management software and can have varying levels of permissions and privileges.

Key aspects of user accounts in CCTV cameras:

1. **Username:** A unique identifier chosen by the user to represent their identity within the camera's system. Usernames are typically alphanumeric strings and are used to log in to the camera.
2. **Password:** A secret string of characters chosen by the user to authenticate their identity and gain access to the camera's settings and features. Passwords are intended to be known only to the user and are used in conjunction with the username during the login process.
3. **Access Levels:** User accounts may have different access levels or roles, each with its own set of permissions and privileges. Common access levels include:

- **Administrator:** Users with administrative privileges have full access to all camera settings and features, including configuration, recording, and playback.

- **Operator:** Users with operator privileges have limited access to camera settings and features, typically restricted to viewing live video feeds, playback of recorded footage, and basic configuration options.
- **Viewer:** Users with viewer privileges have restricted access and are typically limited to viewing live video feeds without the ability to modify camera settings.

4. **Permissions:** User accounts may be assigned specific permissions that dictate what actions they can perform within the camera's interface. Permissions may include the ability to configure recording schedules, adjust camera settings, export recorded footage, and manage user accounts.
5. **Security:** User accounts play a crucial role in the security of a CCTV camera system. It's essential to choose strong and unique passwords for user accounts to prevent unauthorized access. Additionally, user accounts should be regularly reviewed and updated, and unused or outdated accounts should be disabled or removed to reduce the risk of security breaches.

   In general, user accounts provide a mechanism for controlling access to CCTV cameras and ensuring that only authorized individuals can view and manage the camera's settings and recorded footage. Proper management of user accounts helps maintain the security and integrity of the CCTV camera system.

✓ **Factor authentication in CCTV**

Two-factor authentication (2FA) in the context of CCTV cameras is a security mechanism that requires users to provide two different forms of identification before gaining access to the camera's settings or recorded footage. It adds an extra layer of security beyond just a username and password combination. Here's how two-factor authentication typically works with CCTV cameras:

1. **Username and Password:** The user first enters their username and password as usual to log in to the CCTV camera's web interface or management software.
2. **Second Authentication Factor:** After entering the username and password, the user is prompted to provide a second form of identification. This second factor can take various forms, such as:

- One-time passcode (OTP) sent via SMS or email to a registered mobile phone or email address.
- Biometric authentication, such as fingerprint or facial recognition.
- Physical token, such as a smart card or security key, that the user must present or insert into a reader.
- Time-based authentication code generated by an authenticator app installed on a mobile device.

3. **Authentication Verification:** The user enters the second authentication factor (e.g., OTP, biometric data, physical token) to complete the authentication process.

4. **Access Granted:** If both the username/password combination and the second authentication factor are verified successfully, access to the CCTV camera's settings or recorded footage is granted.

   Two-factor authentication adds an additional layer of security by requiring something the user knows (password) and something they have (second authentication factor) to authenticate their identity. This helps mitigate the risk of unauthorized access in case the username and password are compromised.

   Implementing two-factor authentication on CCTV cameras enhances the security of the camera system, especially in scenarios where access to sensitive areas or critical infrastructure is involved. It helps prevent unauthorized access, data breaches, and tampering with camera settings or recorded footage.

✓ **Difference between the two types of factor authentication**

Two-factor authentication (2FA) typically involves the use of two different types of factors to authenticate a user's identity. These factors fall into three main categories: something you know, something you have, and something you are.

Differentiation between the two types of factor authentication based on these categories:

1. **Traditional Two-Factor Authentication (2FA):**

- Something You Know: This factor involves information that only the user should know, such as a password, PIN, or passphrase.
- Something You Have: This factor involves a physical item that the user possesses, such as a smartphone, security token, or smart card.

2. **Modern Two-Factor Authentication (2FA):**

- Something You Know: Similar to traditional 2FA, this factor involves information known only to the user, such as a password or PIN.
- Something You Have: This factor involves possession of a physical item, such as a smartphone, security key, or authentication token.
- Something You Are: This factor involves biometric characteristics unique to the user, such as fingerprints, facial features, or iris patterns.

More detailed differentiation between the two types of factor authentication:

a. **Traditional 2FA:**

Pros:

Relatively easy to implement and use.

Provides an additional layer of security beyond just a password**.**

Cons:

Vulnerable to phishing attacks and social engineering if users are tricked into revealing their passwords.

Passwords can be forgotten or stolen, compromising security.

b. **Modern 2FA:**

Pros:

Offers stronger security compared to traditional 2FA by incorporating biometric factors.

Resistant to phishing attacks and social engineering, as biometric data cannot be easily replicated or stolen.

Provides a more user-friendly experience by eliminating the need to remember complex passwords.

Cons:

Requires specialized hardware or software for biometric authentication, which may be costly to implement.

Biometric data may raise privacy concerns if not properly handled or secured.

Some biometric authentication methods may have limitations in accuracy or reliability.

In summary, both traditional and modern 2FA methods offer enhanced security compared to single-factor authentication (e.g., password-only authentication). While traditional 2FA relies on a combination of something you know and something you have, modern 2FA incorporates additional biometric factors for increased security and convenience.

A user account is comprised of a username, password and any information related to the user.

- ✓ **Authentication factors**
- **Knowledge factor authentication:** It requires users to provide one piece of verifiable information to authenticate
- **Two-factor authentication:** users are required to provide two pieces of verifiable information to authenticate.
- ✓ **Identification of user account**

A user account is comprised of **a username, password and any information related to the user**.

When you open box of any camera, DVR, NVR or Any other IP Based device, first thing comes in your mind about is the default IP address and password.

The following table represents almost all addresses and username and passwords.

Sn.	CCTV Company Name	Default Username	Default Password	Default IP Address
1.	Hikvision	admin	12345	192.0.0.64
2.	TVT	admin	123456	192.168.226.1
3.	Sony	admin	admin	192.168.0.100

4.	Samsung	root	4321 or admin	192.168.1.200
5.	Samsung	admin	4321 or 111111	192.168.1.200
6.	FLIR	admin	flirad min	192.168.250.116
7.	Avigilon	admin	admin	no default/DHCP

Knowledge factor authentication

It requires users to provide one piece of verifiable information to authenticate.

1. Users provide the required information, which could be a password, a PIN, or fingerprints.
2. The online resource compares the information provided with the authentication information it has stored in the system.
3. If the authentication information provided matches the information in the system, users are granted access. If it doesn't match, users are denied access.

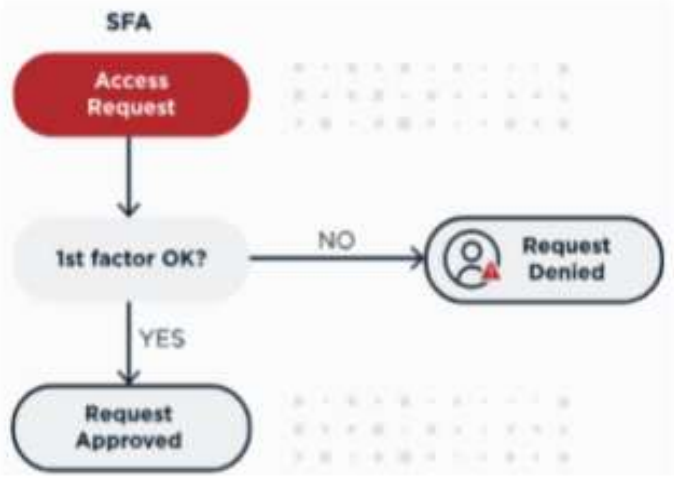
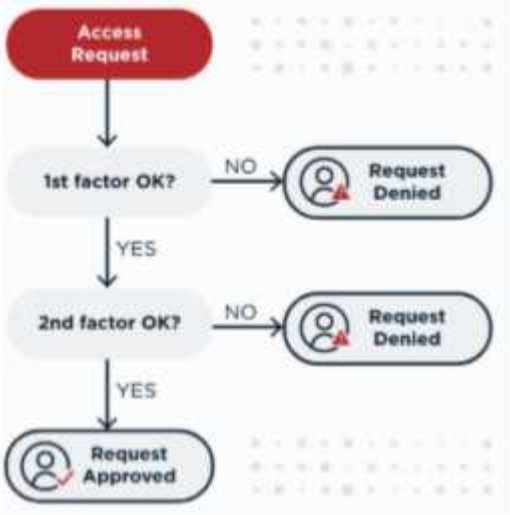


Figure 1Knowledge factor authentication

✓ **Two factor authentication**

With two factor authentication (2FA), users are required to provide two pieces of verifiable information to authenticate. 2FA was designed to add an additional layer of security to sensitive information. Primary credentials and passwords are often forgotten or compromised, so 2FA can be used to help ensure that sensitive information is secure.



✓ **How does 2FA works?**

1. Users provide the first piece of required information, which could be a password or PIN.
2. The online resource compares the information provided with the authentication information it has stored in the system.
3. If the authentication information provided matches the information in the system, users are asked to provide the second piece of required information, which could be a one-time passcode (OTP), or a fingerprint.
4. If the authentication information provided matches the information in the system, users are granted access. If it doesn't match, users are denied access.



Practical Activity 3.2.2: Perform new user account creation



Task:

1: Referring to previous activities (3.2.1) you are requested to perform the given task. The task should be done **individually**.

As a trainee who taught how to create users accounts, you are asked to create different CCTV user accounts (Admin and guest).

2: List out procedures used to create users accounts.

3: Referring to the procedures in task2, perform the given tasks

4: Present your work to the trainer and whole class.

5: Read key readings 3.2.2 and ask for more clarifications where necessary.

6: Perform the task provided in application of learning 3.2



Key readings 3.2.1

- **Steps to create new user account**

  Creating a new user account in a CCTV camera typically involves accessing the camera's web interface or management software and configuring the user account settings.

  General steps to create a new user account in a CCTV camera:

  1. Log in to the Camera:

  Enter the administrator username and password when prompted to log in to the camera's web interface. These are typically the default credentials unless they have been changed previously.

  2. Navigate to User Management Settings:

  Once logged in, navigate to the user management or account settings section of the camera's configuration menu. The exact location of this setting may vary depending on the camera model and firmware version.

  Look for options such as "User Management," "Account Settings," or "User Configuration."

  3. Create a New User Account:

  In the user management settings, look for an option to add a new user account. This is usually labeled as "Add User," "New User," or something similar.

  Click on the "Add User" or equivalent button to begin creating a new user account.

  4. Enter User Details:

  In the new user account form, enter the details for the new user account, including:

a) **Username:** Choose a unique username for the new user account.
b) **Password:** Set a password for the new user account. Ensure it meets any password complexity requirements specified by the camera.
c) **Access Level:** Specify the access level or role for the new user account (e.g., administrator, operator, viewer).
d) **Permissions:** Set any specific permissions or privileges for the new user account, such as the ability to configure camera settings, view live video feeds, or access recorded footage.

  5. Save User Account Settings:

  After entering the user details and configuring the permissions, save the new user

account settings. Look for a "Save," "Apply," or "OK" button within the user management settings, and click on it to save the changes.



Points to Remember

- **User Account (for CCTV Camera):** Individualized credentials used to access CCTV camera systems, typically includes a username and password for authentication.
- **Factor Authentication (for CCTV Camera):** Security measure requiring multiple forms of identification for access, Adds layers of verification beyond just a password to enhance security and Commonly includes something the user knows (password) and something the user has (e.g., a mobile device).
- **Different Types of Factor Authentication (for CCTV Camera):** Single-Factor Authentication, Multi-Factor Authentication (MFA).
- **Steps to create new user account:** Device management, Account management, Add menu, Set a user type, username and password, Select the permission you want the user to have, Save change.



Application of learning 3.2

XYZ company located in Nyarugenge want to monitor the employees in their daily activities. As one of the CCTV system controller team, you are asked to use the existing installed CCTV system and perform the below task.

Task:

- Create two different user accounts (One as guest and another as Admin).
- Assign Password for each to those accounts.

Indicative content 3.3: Configure CCTV IP addresses

Duration: 2 hrs

Theoretical Activity 3.3.1: Description of CCTV IP addresses

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of CCTV IP addresses.

i. Differentiate static IP to dynamic IP?

2: Provide the answers for the asked question and write them on papers.

3: Present the findings/answers to the whole class.

4: For more clarification, read the key readings 3.3.1.

5: In addition, ask questions where necessary.

Key readings 3.3.1.:

- **What is an IP address?**

  An IP address, or Internet Protocol address, is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.

- **IP addresses serve two primary purposes:**

1. **Identification**: IP addresses uniquely identify each device on a network, allowing them to communicate with each other. Think of it as similar to a postal address for devices on the internet.
2. **Routing**: IP addresses are used by routers and other networking devices to route data packets to their intended destinations across networks.

- **Versions of IP addresses**

  There are two versions of IP addresses currently in use:

1. **IPv4 (Internet Protocol version 4)**:

- IPv4 addresses are 32-bit numerical addresses written in the form of four decimal numbers separated by periods (e.g., 192.168.0.1).
- The total number of available IPv4 addresses is limited, leading to the adoption of IPv6 to accommodate the growing number of devices connected to the internet.

2. **IPv6 (Internet Protocol version 6)**:

- IPv6 addresses are 128-bit numerical addresses written in hexadecimal format and separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- IPv6 provides a much larger address space than IPv4, allowing for an almost unlimited number of unique addresses.

- **Assigning IP address**

  IP addresses are typically assigned dynamically or statically:

1. **Dynamic IP Address**:

- Dynamic IP addresses are assigned to devices temporarily by a DHCP (Dynamic Host Configuration Protocol) server. These addresses may change over time as devices join or leave the network.

2. **Static IP Address**:

- Static IP addresses are manually configured for a device and do not change unless modified by an administrator. These addresses are typically used for servers, routers, and other network infrastructure devices.



Practical Activity 3.3.2: Perform CCTV IP address configuration



Task:

1: Referring to previous activities **(3.3.1**) you are requested to perform the given task. The task should be done **individually.**

As a trainee who taught how to configure cameras IP Addresses, you are asked to create and configure cameras static IP addresses in a network that you have.

2: List out procedures to perform the given tasks.

3: Referring to procedures of creating and configure IP address provided on task 2, Perform the given tasks

4: Present your work to the trainer and whole class.

5: Read key reading 3.3**.2** and ask clarifications where necessary.

6: Perform the task provided in application of learning 3.3



Key readings 3.3.2

- **CCTV IP address configuration**

  Configuring the IP address in a CCTV camera involves accessing the camera's settings through its web interface or management software and manually setting the IP address, subnet mask, gateway, and other network parameters.

  The general steps to configure the IP address in a CCTV camera:

1. **Connect to the Camera**: Ensure that the CCTV camera is powered on and connected to the network using an Ethernet cable. Connect your computer to the same network as the camera.

2. **Determine Camera's Default IP Address**: Refer to the camera's user manual or documentation to find its default IP address. This information is typically provided by the manufacturer and may be printed on a label attached to the camera.
3. **Access the Camera's Web Interface**: Open a web browser on your computer and enter the default IP address of the CCTV camera in the address bar. Press Enter to access the camera's web interface.
4. **Log in to the Camera**: When prompted, enter the username and password to log in to the camera's web interface. These credentials are usually provided by the manufacturer and may be the default username and password.
5. **Navigate to Network Settings**: Once logged in, navigate to the network settings or configuration section of the camera's web interface. The exact location of this setting may vary depending on the camera model and firmware version.
6. **Set IP Address**:

✓ In the network settings, locate the option to set the IP address. This may be labeled as "Network Settings," "TCP/IP Settings," or something similar.

✓ Enter the desired IP address for the camera, along with the subnet mask and gateway. Ensure that the IP address is unique and compatible with your network configuration.

7. **Apply Changes**: After entering the network settings, save the changes by clicking on the "Apply" or "Save" button within the camera's web interface.
8. **Restart the Camera**: Some cameras may require a reboot for the changes to take effect. Follow any prompts to restart the camera after applying the network settings.
9. **Verify Connectivity**: Once the camera has restarted, verify that it is accessible on the network using the new IP address. Open a web browser and enter the new IP address to access the camera's web interface.
10. **Test Functionality**: Test the camera's functionality by viewing live video feeds, adjusting settings, and verifying network connectivity.



Points to Remember

- **Static IP Address:** Remains constant and does not change over time, Ideal for devices that require permanent, fixed network addresses, such as CCTV cameras, Provides stability and consistency for accessing the CCTV camera remotely.
- **Dynamic IP Address:** Assigned dynamically by a DHCP server and can change periodically. Suitable for devices that do not require a permanent network address and can adapt to changing network configurations.
- CCTV IP Address Configuration

- Static IP configuration
- Dynamic IP configuration



Application of learning 3.3

A company named KANYANGE needs to install CCTV systems in its office for better management of employees and their tasks. As a one of the CCTV installer team of the company, you are asked to install CCTV cameras in the office of that company.

Task:

During installation, set the dynamic IP address in this CCTV system.

Indicative content 3.4: Apply setting of CCTV recordings

Duration: 2 hrs

Theoretical Activity 3.3.1: Description of setting CCTV recordings.

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of setting CCTV recordings.

i. Describe the video format.
ii. Which reasons may cause to delete recorded videos?
iii. How to make backup of deleted videos?
iv. Describe the recording mode types.

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key readings 3.4.1

5: In addition, ask questions where necessary.

Key readings 3.4.1.:

- **Describe the video format.**

The video format used in CCTV (Closed-Circuit Television) cameras depends on various factors, including the camera model, manufacturer, and specifications. However, the most common video formats used in CCTV cameras are:

1. **Analog CCTV Cameras**: Analog CCTV cameras typically use analog video formats such as PAL (Phase Alternating Line) or NTSC (National Television System Committee). These formats define the resolution, frame rate, and encoding standards for analog video signals transmitted by the camera.
2. **Digital CCTV Cameras**: Digital CCTV cameras, also known as IP cameras, use digital video formats to encode and transmit video data over networks. The most common digital video formats used in IP cameras include:

1. H.264: This is one of the most widely used video compression standards for IP cameras. It offers efficient compression while maintaining good video quality.
2. H.265 (HEVC): High-Efficiency Video Coding (HEVC) is a newer compression standard that provides better compression efficiency than H.264, allowing for reduced bandwidth and storage requirements.

3. MJPEG (Motion JPEG): Motion JPEG is a video compression format that compresses each frame of video separately as a JPEG image. It offers high-quality images but may result in larger file sizes compared to H.264 and H.265.
4. MPEG-4: MPEG-4 is another video compression standard used in some IP cameras. It offers efficient compression and good video quality, but it may not be as widely supported as H.264 and H.265.
3. **Resolution and Frame Rate**: CCTV cameras may support various resolutions and frame rates, depending on the camera's specifications and intended use. Common resolutions include standard definition (SD), high definition (HD), and ultra-high definition (UHD or 4K). Frame rates typically range from 24 frames per second (fps) to 30 fps for smooth video playback.
4. **Container Formats**: In addition to video compression formats, CCTV cameras may use container formats such as AVI (Audio Video Interleave), MP4 (MPEG-4 Part 14), or MOV (QuickTime Movie) to encapsulate the compressed video data along with audio and metadata.

   In general, the video format used in CCTV cameras plays a significant role in determining the quality, efficiency, and compatibility of the recorded video footage.

   ✓ **Delete CCTV footage?**

   There are several reasons why your CCTV footage might get deleted. For instance, the old footage may get overwritten with fresh data after a month by default. It may also depend on the CCTV storage capacity - lack of space can lead to deletion of old data. Or you may also lose recorded footage if someone manually deletes it - either accidentally or deliberately.

   In addition to this, you may also lose your CCTV footage due to any of the following reasons: Failure of DVR or hard drive, Fire or water damage, Corrupted or damaged SD cards, Virus or malware attacks.

✓ **Recover deleted CCTV Footage?**

Now that you are aware of why your CCTV footage gets deleted, let's move on to how you can try to get it back. Below are three effective methods that you can try out.

✓ **Recover The Deleted Footage from Backups**

If you have a backup storage system set up for your CCTV camera, getting back the deleted files won't be too difficult. You can get back the required footage by following the steps mentioned below.

- Locate the CCTV footage backup folder and open it.
- Search for the erased files and transfer them back to the original storage disk or SD
card.

✓ **Use Data Recovery Software**

In case you don't have a backup system in place, you would need to use third-party data
recovery software to retrieve your deleted or damaged files. It's best to stick to reputable
software for the task to ensure data privacy and device security. To recover your deleted
footage using this method, you need to follow the below-mentioned steps.

- Download and install the data recovery software on your computer.
- Run the program to scan for recoverable files.
- Locate the required files from the list and transfer them to your desired storage location

✓ **Recover The Deleted CCTV Camera Footage Using CMD**

It is also possible to retrieve your deleted CCTV footage using Control Prompt Command or CMD.

For this method, you need to follow the below steps:

- Open CMD or Command Prompt.
- Right-click on the start menu and select 'Run'. Next, type 'cmd' and press the enter
key.
- In the command prompt, type chkdsk (space) Drive Letter /f. The '/f' parameter helps
resolve any detected errors in the disk. Wait till the command finishes its actions.
- Next, use the parameters mentioned below to fix issues and restore the data.

−h: To show hidden attributes

−r: For locating bad sectors and recovering read-only information

−s: For commanding the system to look for a specific path location

✓ **Backup frequency**

The Backup Plan period is **the time when the DVR will backup the video footage**. The Record Period is the video that you want to backup. Some businesses want to archive video records for various reasons, and the schedule backup solution is an excellent option. If you need to have it offsite, it can still work because the software uses an address and login criteria to find the machine and access it for backup.

−First, you will need to download the Windows only program

−Run the program and login using admin as the username and admin as the password. After you login, you will see the main Schedule Backup window where you

can start to configure your backup.
−Start by clicking on the Config button and select Device Manage so you can add your
DVR or NVR to the software.
−You need to add your IP address, login credentials, and give it a good name. If you have more than one DVR or NVR to backup, it is a good idea to name each device by
the location so you can track them easier. If your computer is not at the site of the DVR or NVR, keep in mind that it will be using the upstream of the Internet connect to
send high quality video to the backup computer. Also consider that you will fill up your computer's hard drive very fast with high quality video, so you need to consider
have vast amounts of storage space to encompass large volumes of data.
−If you select Config then Option, you can configure the default save directories by selecting up to five devices as well as how much hard drive space will be used until it
moves onto the next device. The amount used will be the amount of available space
on the drive until it reaches the minimum free capacity amount. You can select the default option of Overwrite or Stop for when the drives are full.
−Our next step is to create a backup plan by clicking the Config button, and selecting
Backup Plan. When you see the backup plan box, click the Add button.
−This is where you will create your first backup plan. If you want to backup everything,
you would need to click the Multiple-channel button and select all cameras.

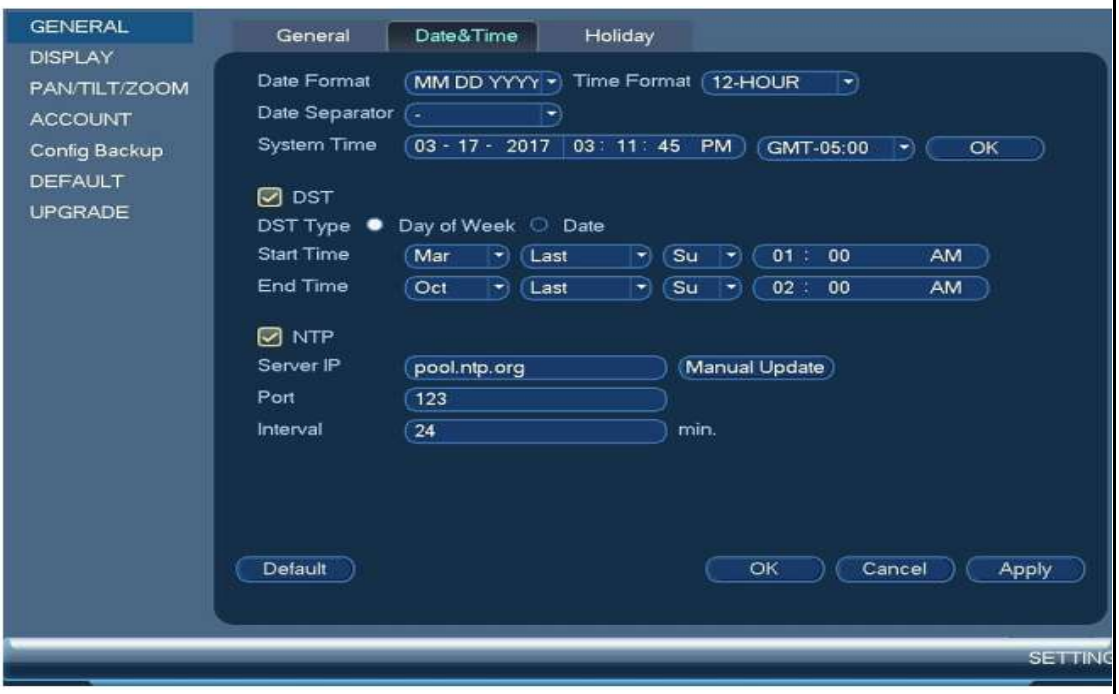
- **Set time and date**

  The time on your DVR / NVR can be configured manually, or automatically by enabling **Network Time Protocol (NTP).** If the recorder is showing the incorrect time, or the time is resetting, adjust your time configuration settings by enabling / disabling NTP, but, your system must have a constant Internet connection to use NTP.

  To enable / disable NTP:

  1. Right-click on the Live View to access the Quick Menu.
  2. Click Main Menu and then click the Setting icon
  3. Click Settings > General.
  4. Click the Date & Time tab. Shown on the figure below.
  5. Check / uncheck the **NTP** checkbox to enable / disable Network Time Protocol. Enable

NTP so that the DVR / NVR time is always correct. If the time is incorrect it could cause
the time stamp on a video to be incorrect.
6. Check / uncheck the **DST** checkbox to enable / disable automatic Daylight Savings
Time updates if applicable to your location.
7. If NTP is enabled, set the Time zone to match your location and click **Manual Update** to update the time.
8. Click **Apply** to save the settings, and then click **OK** to exit.



- **Set recording mode**

  ✓Recording modes

  –**Continuous**. This is our recommended recording method. The DVR records all the time, when the hard drive fills up it goes back to the start and records over the earliest footage on a rolling basis.

  –**Motion sensing.** The DVR monitors the picture coming in from the camera. If it sees movement then it starts recording. You can choose which areas of the screen the DVR looks for movement in, you don't have to use the whole screen as a trigger area.

  We don't recommend using an only motion-triggered recording. There is a risk you will miss vital footage and what happened in between the gaps?

  Our DVRs allow you to use both methods side by side.

  –**Alarm triggered.** The DVR starts recording when triggered by an external sensor such as a PIR (passive infra-red) motion sensor or magnetic contact "reed switch". This method is reliable in terms of not generating false triggers but involves additional wiring and sensor hardware.

–**Scheduled**. You can decide what type of recording takes place at various times throughout the day and on particular days of the week. This helps reduce the amount of space taken up on the hard drive but the downside is if an event occurs when the DVR is scheduled not to record you will miss it.

✓Setup recording mode

–Using the mouse, I click on the home button in the lower left of the screen, then select setup, advanced. When the main setup screen opens, click on the record button. Then click record again. The recording screen will be displayed and this is where you can configure the recording rules for each camera connected to your DVR. Each camera can have a unique configuration for normal, event, and panic type recording. Users
can set the active recording mode of each camera to normal (which means continuous), event (which is triggered on motion or alarm), normal + event (which does both), by schedule, or off.

–Users can also configure the resolution, quality, and frame rate for each camera and each recording type for that camera.



Practical Activity 3.4.2: Setting up time and date



Task:

1: Referring to previous activities (3.4.2) you are requested to perform the given task. The task should be done **individually**.

As a trainee who taught how to set time/date record schedule and recording mode, you are asked to schedule (set) the recording time/date on the CCTV system and provide the recording mode.

2: List out procedures of Setting up time and date the given tasks (3.4.2).

3: Referring to procedures and formulas provided on task 2, Perform the given tasks.

4: Present your work to the trainer and whole class.

5: In your trainee manual, read key reading of practical activity 3.4



Key readings 3.4.1

- **Setting up time and date in CCTV camera**

  Setting up the time and date in a CCTV (Closed-Circuit Television) camera is essential for accurate timestamping of video footage.

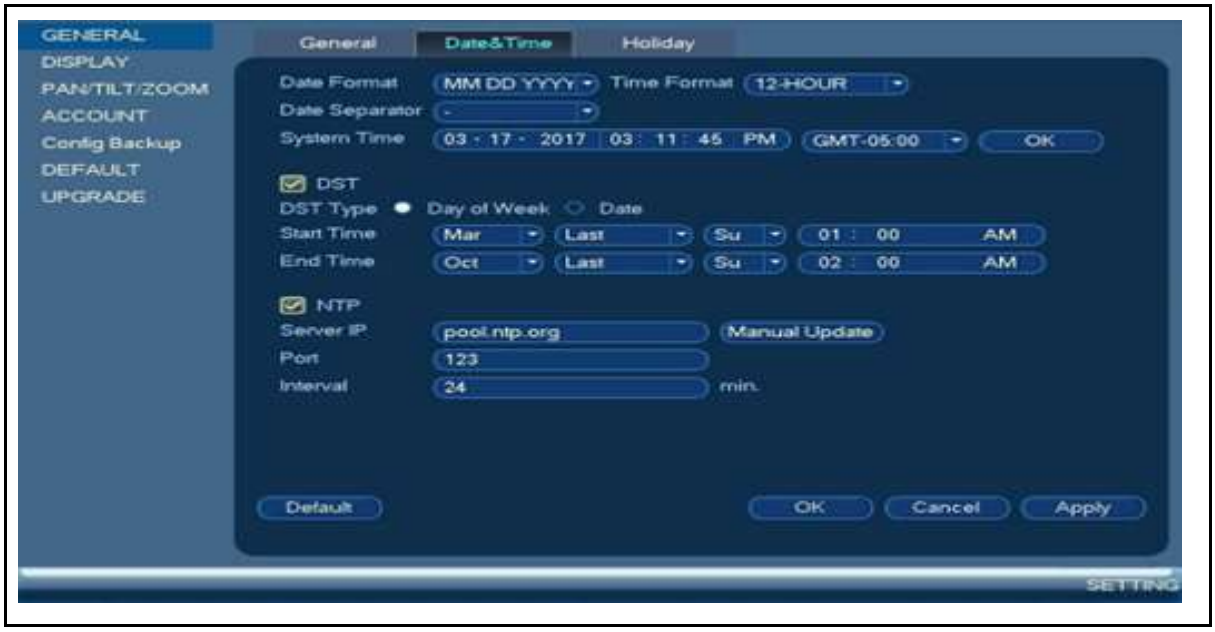
  The general steps to set up the time and date in a CCTV camera:

1. **Log in to the Camera**: Enter the username and password when prompted to log in to the camera's web interface. Use the administrator credentials or any other user account with permission to change settings.
2. **Navigate to Date and Time Settings**: Once logged in, navigate to the date and time settings section of the camera's configuration menu. The exact location of this setting may vary depending on the camera model and firmware version.
   Look for options such as "Date & Time," "Time Settings," or "System Settings."
3. **Set Time Zone**: Choose the appropriate time zone for your location from the available options in the camera's settings. This ensures that the time displayed by the camera is accurate for your geographical region.
4. **Set Date and Time**: Enter the current date and time in the corresponding fields provided in the camera's settings. Some cameras may provide options to set the date and time manually, while others may offer synchronization with a network time server (NTP server) for automatic time updates.
   If setting the date and time manually, ensure that the information entered is accurate and matches the current date and time.
5. **Apply Changes**:

- After entering the date and time settings, save the changes by clicking on the "Apply," "Save," or "OK" button within the camera's web interface.
- If the camera supports synchronization with an NTP server, enable this option to ensure that the date and time remain accurate and automatically updated.

6. **Verify Settings**: After applying the changes, verify that the date and time settings are displayed correctly in the camera's web interface. Check the timestamp on live video feeds or recorded footage to ensure that the date and time are accurate.
7. **Logout and Test**:

- Once the date and time settings are configured, log out of the camera's web interface to ensure that the changes take effect.
- Test the camera's functionality by viewing live video feeds, adjusting settings, and verifying that the date and time are displayed correctly.


GENERAL
DISPLAY
PAN/TILT/ZOOM
ACCOUNT
Config Backup
DEFAULT
UPGRADE
General
Date&Time
Holiday
Date Format
MM DD YYYY
Time Format
12-HOUR
Date Separator
System Time
03 - 17 - 2017 03 : 11 : 45 PM
GMT-05:00
OK
DST
DST Type
Day of Week
Date
Start Time
Mar
Last
Su
01 : 00
AM
End Time
Oct
Last
Su
02 : 00
AM
NTP
Server IP
pool.ntp.org
Manual Update
Port
123
Interval
24
min.
Default
OK
Cancel
Apply
SETTING



Points to Remember

- **Video Format** Refers to the encoding and storage method of video data, Common formats include H.264, H.265, MPEG-4, and MJPEG.
- **Reasons for Deleting Recorded Videos includes** Limited storage capacity on the recording device, Videos no longer needed for surveillance purposes etc….
- **Making Backup of Deleted Videos:** Utilize backup software or tools provided by the CCTV system, Transfer videos to external storage like USB drives or network-attached storage (NAS).
- **Recording Mode Types:** Continuous Recording, Motion-Activated Recording, Scheduled Recording and Event-Triggered Recording.
- Setting up Time/Date and Recording Mode
- Choose recording mode: Continuous Recording, Motion-Activated Recording and Scheduled Recording
- Adjust settings to optimize storage and coverage.
- Test recording modes for proper functionality and surveillance coverage.



Application of learning 3.4

A company named KIRWA needs to install CCTV systems in its office for better management of employees and their tasks. As one of the CCTV system installer team of the company, you are asked to install CCTV cameras in the office of that company, and set time and date on recording devices (DVR/NVR).

Indicative content 3.5: Document the CCTV functionalities

Duration: 2 hrs

Theoretical Activity 3.5.1: Description for documentation of CCTV functionalities

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of documentation of CCTV functionalities

i. What is documentation?
ii. What must be in documentation?
iii. What are CCTV system requirements?
iv. What are CCTV system settings?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key readings 3.5.1

5: In addition, ask questions where necessary.

Key readings 3.5.1.:

- **Introduction to documentation**

  Documentation: paperwork (or other media) prepared during the design, installation and hand over of the CCTV system, recording details of the CCTV system, including paperwork (or other media) related to maintenance (where applicable).

  Documentation is any communicable material that is used to describe, explain or instruct regarding some attributes of an object, system or procedure, such as its parts, assembly, installation, maintenance, and use. As a form of knowledge management and knowledge organization, documentation can be provided on paper, online, or on digital or analog media, such as audio tape or CDs.

  Documentation shall include all the information necessary to enable the client or specifier to ensure the CCTV system is appropriate for the application. The information provided in the documentation shall include the following:

  Client details: The name, address and the trading name (if different from the name of the client) and any other information necessary to clearly identify the client

  Contractor details: The name, address and trading name of the contractor (if a trading name is used) shall be included along with any other information necessary for the client to identify and/or contact the contractor. Headed company stationery with contractor details is acceptable in this regard.

Supervised area details: The name and address of the supervised area shall be included if different from the address of the client. This shall also include a description of the supervised area and an indication of what the area is used for.

Schedule of equipment: A schedule of the type and location of operational equipment (in words and/or diagrammatic form) shall be included.

Maintenance: The system design proposal shall include recommendations for the scheduled maintenance of the CCTV system or individual components including details of the frequency of any maintenance visits and a list of the work to be carried out during each visit.

Good CCTV documentation provides more than just where the cameras are located, but what networking devices they are connected to and how much bandwidth each segment is using, how and where cameras being displayed and by whom, what permissions does each User have. Maintenance becomes difficult and inconsistent without good up to date documentation.

CCTV documentation should be able to answer the following questions:

−Where is each camera located?

−What area is each camera look at, FoV (Field of View)

−Record settings for cameras that are used for specific purposes like Face Recognition,

AI perimeter protection, People counting, Plate Capture, Age recognition etc.

−What is the model and camera type, and how is it mounted?

−What IP address of each camera, network switch and Client PC

−The encoding settings each camera need to be set to, this is very important because it

relates to how much retention (recording history) your system will have, and the network bandwidth (equipment) that each segment requires, and is critical to know

when you wish to add more cameras, or higher resolution cameras.

✓CCTV system requirements

Depending on the type of camera you choose, you may only need some of the following advice. To cover everything, you may need to know, we've included instructions for installing all three CCTV camera types.

CCTV installation requirements are:

- Gather your gear
- Choose the camera's location
- Attach the camera
- Run the wiring
- Operate the system

Also you can consider the followings for CCTV system requirement, as:

- **Tools, materials and equipment**

Closed-circuit television (CCTV) equipment and tools are used in a range of industries, from security to surveillance. CCTV is a system that uses video cameras to transmit a signal to a specific place, on a restricted set of monitors. The purpose of CCTV is to monitor and record activity in a given area.

CCTV equipment and tools typically consist of cameras, monitors, DVRs (digital video recorders), and software for managing the system.

CCTV tools are classified in ***CCTV*** Installation ***Tools***, ***CCTV*** Crimp ***Tools***, Hand Held ***CCTV*** Testers, Wire Stripping ***Tools***, Quick Connect, BNC ***Tools***, RJ45 Crimp ***tool, etc***

CCTV materials are including coax cable, bnc connectors, network cabling, ***camera*** mounts, monitor mounts, etc

- **CCTV diagram**

CCTV (Closed-circuit television) diagram is used to depict a system of video monitoring. The CCTV diagram provides video cameras placement strategy.

A CCTV network diagram shows the closed-circuit television and how it is connected to the camera and the network to record the live feed.

- **Cost**

Cost is the amount of money needed to buy, install, do, or make something, or an amount spent for something.

Some of the factors that determine the price of a camera are listed below.

−The Quality & performance of the cameras CMOS sensor & digital signal processor

−The power of the Cameras CPU; AI functionality requires a powerful CPU

−Types, & quality of analytics supported by the camera

−The resolution supported by the camera, typically 2MP(1080p⁄FHD), 4MP(QHD), 5/6MP and 4K(UHD)

−Type and quality of the lens, a really good lens can cost more than the camera itself

−The cameras IK vandal resistance rating (ability to withstand impacts)

−The cameras IP ingress protection rating (ability to withstand water and dirt ingress)

−Other features like inbuilt microphone, inbuilt speaker, additional LEDs, IRs, alarm I/O

−PTZ (manoeuvrable) cameras usually cost more than an equivalently featured fixed camera.

Points to Remember

- **Documentation (CCTV Camera Installation):** Refers to the process of creating written records and guidelines.
- **Contents of Documentation include** Diagrams of camera placements and wiring, Inventory of equipments (cameras, cables, and recording devices), Configuration of IP addresses, usernames, and passwords, Maintenance schedules etc…
- **CCTV System Requirements include** Adequate coverage of the monitored area, Sufficient lighting conditions for optimal video quality, Stable power supply, Network connectivity, recording and monitoring devices.
- **CCTV System Settings like** camera parameters such as resolution, frame rate, and motion detection sensitivity, Network settings including IP addresses, Recording settings, User access controls and authentication

Application of learning 3.5

A company named NDKIZ Ltd needs to install CCTV systems in its office for better management of employees and their tasks. As one of the CCTV system installer team of the company, you are asked to install CCTV cameras in the office of that company, and then set the CCTV system settings and provide all required documents for installation.

Learning outcome 3 end assessment

Theoretical assessment

A. Read carefully the following statement, and answer by TRUE if the statement is correct, otherwise FALSE.

1. A user account is comprised of a username, password and any information related to the user. …….
2. Continuous, motion setting, and Alarm triggered are the modes of recording.
3. Dynamic IP address configuration works when your router doesn't support DynDNS service (Dynamic DNS).
4. Failure of DVR or hard drive can be a reason to lose your CCTV footage……
5. In most cases, after 15 days or a month depending upon the storage available in the DVR/NVR, old data is kept with fresh data by default, and thus old data is kept for long time. ……..

6. static IP addresses allows you to apply an IP address range to the listed cameras, or apply a specific IP address to a camera. …….
7. The DVR viewer software allows you to view your security cameras over the Internet. ……..
8. The time on your DVR / NVR can be configured manually, or automatically by disabling Network Time Protocol (NTP). …….
9. There are three basic video format for CCTV Camera, namely MPEG2, MPEG4, MJPEG……
10. With knowledge factor authentication (2FA), users are required to provide two pieces of verifiable information to authenticate. ……..
11. With two factor authentication, users are required to provide one piece of verifiable information to authenticate.

B. Choose the correct answer

1. What is the first step you should take when managing the administration parameters of a new CCTV system?

A) Set the recording storage
B) Change the default password
C) Configure IP addresses
D) Create user accounts

2. Which of the following is an important security measure when setting up CCTV user accounts?

A) Using generic usernames for all users
B) Setting up two-factor authentication
C) Allowing all users to share the same password
D) Disabling user accounts

3. What is the difference between Static IP and Dynamic IP in CCTV systems?

A) Static IP changes automatically, while Dynamic IP stays the same
B) Static IP remains constant, while Dynamic IP can change over time
C) Static IP is used for wireless cameras only
D) Dynamic IP is used for analog cameras only

4. Which video format is commonly used for CCTV recordings?

A) JPEG
B) MP4
C) GIF
D) PNG

5. What is the purpose of the overwrite capability setting in CCTV recordings?

A) To prevent accidental deletion of old footage
B) To automatically delete the oldest footage when storage is full
C) To ensure all footage is permanently saved
D) To increase the resolution of the recordings

6. When configuring a CCTV system, why is it important to set the correct time and date?

A) To enhance the video quality

B) To ensure accurate timestamping of recorded events

C) To improve the camera's field of view

D) To save storage space

7. Which of the following should be documented in the CCTV system documentation?

A) The color of the cables

B) The password used for the admin account

C) The CCTV diagram and system settings

D) The location of all physical keys to the building

8. What does setting the backup frequency in CCTV recording settings do?

A) It sets how often recordings are played back

B) It determines how frequently the system creates backups of the footage

C) It controls the speed of the camera

D) It adjusts the quality of the recorded video

9. Which method of authentication is based on something the user knows, like a password?

A) Biometric authentication

B) Knowledge factor authentication

C) Two-factor authentication

D) Token-based authentication

10. What is a key consideration when creating user accounts in a CCTV system?

A) Assigning the same privileges to all users

B) Identifying user roles and assigning appropriate access levels

C) Limiting the number of user accounts to two

D) Disabling password protection for ease of access

11. Why is it important to document the cost of a CCTV system during installation?

A) To estimate future maintenance expenses

B) To ensure the system fits within the budget

C) To provide an accurate value for insurance purposes

D) All of the above

12. What does the term 'Two-Factor Authentication' mean in the context of CCTV user accounts?

A) Using two passwords for each account

B) Requiring a password and a secondary form of verification, like a text message code

C) Creating two separate accounts for each user

D) None of the above

13. Which setting would you configure to ensure that your CCTV system retains recordings for a set period before deleting them?

A) Video format

B) Recording mode

C) Overwrite capability

D) Backup frequency

14. What is the benefit of assigning a Static IP address to a CCTV camera?

A) It reduces the cost of the system

B) It ensures the camera has a permanent, unchanging address

C) It allows the camera to move between networks easily

D) It enables the camera to function without an internet connection

15. What is an essential part of the documentation process when installing a CCTV system?

A) Recording the names of all visitors

B) Describing the tools, materials, and equipment used

C) Hiding the locations of the cameras

D) Encrypting all the documentation

Practical assessment

You are hired by the OZONE company located in NYAMASHEKE district Western Province, to install the CCTV camera system in their department and you asked to do the following activities:

- Change default password of DVR/NVR to MyPass@1!
- Set date and time
- Set recording features to frames per second (FPS)
- Format the hard disk of DVR/NVR.
- Configure CCTV static/dynamic IP addresses.



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Indicative contents
4.1. Monitor live footage of capture events
4.2. Maintain CCTV system control room
4.3. Perform Export/Playback of required information
4.4. Interpretation of CCTV system logs
4.5. Generate and interpret incident report

Key Competencies for Learning Outcome 4: Operate CCTV system.

Knowledge	Skills	Attitudes
● Description of live footage. ● Identification of type of monitors for CCTV. ● Description of control room. ● Description of Environmental condition. ● Identification of Define Export and Playback. ● Interpretation of CCTV system logs.	● Preform viewing live footage for CCTV. ● Perform control room operator. ● Performing steps of Export and Playback for required information.	● Be a Critical thinker ● Be a Teamwork player, ● Have a Creativity spirit, ● Be an Innovator



Duration: 20 hrs

Learning outcome 4 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Monitor effectively live footages based on captured events.
2. Maintain appropriately control room based on CCTV system condition.
3. Perform correctly export/playback based on required information.
4. Interpret properly CCTV system logs based on CCTV system status.
5. Generate correctly incident report based on CCTV system issue.



Resources

Equipment	Tools	Materials
● Computer ● Monitor ● Power Supply ● Switch ● Storage devices ● Cameras ● Recording Devices (NVR/DVR)	● CCTV Monitoring Software	● Internet bundles ● Power extension ● CCTV Power supply box

Indicative content 4.1: Monitor live footage of capture events

Duration: 4 hrs

Theoretical Activity 4.1.1: Description of live footage

Tasks:

1: In small groups, you are requested to answer the following questions related to the Monitor live footage of capture events.

i. What is live footage?

ii. What are the types of monitors for CCTV system?

2: Provide the answer for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarifications, read the key readings 4.1.1

5: In addition, ask questions where necessary.

Key readings 4.1.1.:

- **Definition of live footage**

Live footage: is a section of film or video that is been shot or recorded in real time. **Live monitoring** means a person viewing images live in real time as they are being captured and recorded by a camera.

Live video surveillance (also called remote video monitoring) starts with a series of cameras that are watched over by security professionals. These cameras are monitored at all times, providing 24/7 protection. However, a remote monitoring system includes more than just cameras.

- **Type of monitors for CCTV**

✓ **LCD Monitor: Liquid Crystal Display (LCD) monitor** is a flat-panel display that uses liquid crystal technology to produce images

✓ **LED Monitor: a Light Emitting Diode (LED) monitor** is an advanced type of flat panel display that uses Light Emitting Diodes for illumination. Compared to standard LCDs, an LED panel display is thinner and utilizes less power than LCD monitors

✓ **OLED Monitor: An organic light-emitting diode (OLED) monitor** is a type of flat panel display that produces its own light. OLED monitors gives you several advantages over traditional LCD monitors, including thinner panels and the use of less energy.

✓ **Plasma Monitor:** Plasma monitors are flat-screen monitors that use phosphors gas to provide color. Because the picture is produced by gases instead of light bulbs or

other heat sources, they are exceptionally thin and therefore can be mounted on walls.

- ✓ **CRT Monitor:** An old-fashioned computer monitor, or CRT (cathode ray tube) display, is one of the main types of computer monitors.
  They are large and bulky monitors that come equipped with a bulky box that connects to the back of them.
- ✓ **Touch Screen Monitors:** A touch screen monitor is a type of computer monitor that uses a touch-sensitive panel built into the front of the unit.
  The built-in touch screen allows you to touch on the screen to open various websites, photos, or applications effortlessly. **Unlike a computer monitor, a CCTV monitor is manufactured and designed to be on and working 24-7 all year long.**

When looking for a new CCTV monitor, the followings are the things to be considered:

- ✓ Size
- ✓ Power consumption
- ✓ Heat
- ✓ Connectivity
- ✓ Resolution



Practical Activity 4.1.2: Perform viewing live footage for CCTV



Task:

1: Referring to previous activity (4.1.1) you are requested to perform the given task. The task should be done **individually**.

As a CCTV Camera installer, you are requested to set live view footage for installed CCTV camera system.

2: List out procedures to be used to perform the given tasks (4.1.2).

3: Referring to procedures of viewing live footage provided on task 2, perform the given tasks

4: Present your work to the trainer and whole class

5: Read key readings 4.1.2 and ask clarification where necessary

6: Perform the task provided in application of learning 4.1



Key readings 4.1.2

- **Watch live CCTV footage on the security cameras.**
  Watching live CCTV footage on the security cameras typically involves accessing the camera's live stream through a web browser, mobile app, or dedicated monitoring software.
  The general steps to watch live CCTV footage on your internet security cameras:

1. **Connect to the Camera**: Ensure that your computer, smartphone, or tablet is connected to the internet via Wi-Fi or a cellular network. Make sure the camera is also connected to the internet and powered on.
2. **Access the Camera's Interface**: Open a web browser on your computer or launch the mobile app provided by the camera manufacturer on your smartphone or tablet.
3. **Login to the Camera**: Enter the camera's IP address or domain name in the address bar of the web browser, or launch the mobile app and log in with your username and password. This information is usually provided by the camera's documentation or setup instructions.
4. **Navigate to Live View**: Once logged in, navigate to the live view section of the camera's interface. This may be labeled as "Live View," "Live Feed," or something similar.
5. **Select the Camera**: If you have multiple cameras connected to your system, select the camera you want to view from the list of available cameras.
6. **Start Live Stream**: Click on the "Play" or "Live" button to start streaming live footage from the selected camera. The live video feed should appear on the screen, displaying real-time images captured by the camera.
7. **Adjust Settings (Optional)**: Depending on the camera's capabilities and the interface you are using, you may have options to adjust settings such as video resolution, frame rate, brightness, contrast, and zoom level to optimize the viewing experience.
8. **Monitor Live Footage**: Monitor the live footage from the CCTV camera in real-time. Use any available controls or features to pan, tilt, zoom, or adjust the viewing angle of the camera if supported.
9. **End Live View**: When finished watching live footage, click on the "Stop" or "Exit" button to end the live view session. This will stop the live video stream and return you to the main menu or dashboard of the camera's interface.
10. **Logout (Optional)**: If using a web browser to access the camera's interface, log out of the system to secure your account and prevent unauthorized access. This step may be optional, depending on the interface and your security preferences.



Points to Remember

- **Live Footage:** Real-time video feeds from CCTV cameras.
- **Types of Monitors for CCTV System:** Dedicated CCTV Monitors, LCD/LED Monitors and PC Monitors.



Application of learning 4.1

XYZ School systems in their workshops for better management of equipment and consumables. As a CCTV SYSTEM INSTALLER hired by the school, you are asked to install CCTV cameras in the workshops of and you will also need display devices known as monitors to display live footage.

Indicative content 4.2: Maintain CCTV system control room

Duration: 4 hrs

Theoretical Activity 4.2.1: Description of CCTV control room.

Tasks:

1: In small groups, you are requested to answer the following questions related to the of description of CCTV control room

i. What is control room?
ii. What are the functions of control room?
iii. Describe the control room operator?
iv. What must be the control room conditions?

2: Provide the answer for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key readings 4.2.1

5: In addition, ask questions where necessary.

Key readings 4.2.1.:

- **Description of control room**

✓ **Definition of control room**

A CCTV (Closed-Circuit Television) control room, often called a surveillance room, serves as the central hub in a security system where all CCTV footage is monitored and reviewed. Located in a secure building area, it provides a systemized space for the operators to manage security systems and processes efficiently. The control room has various data retrieval, analysis, and processing devices.

A control room: is the secure workplace for organizations in any type of industry, located in an adapted facility where the technology provides centralized monitoring, control and command for CCTV system.

- **Function of the control room**

- Improvement of security
- Incident detection through continuous monitoring
- Analysis of activity and assisted mitigation of further incidents

- **Control room operator**

Control Room Operator: is a professional person who works in the control rooms of large plants, in particular power plants, where they monitor all of the operations of the plants to ensure that everything is working properly.

A CCTV operator works in a central control room, monitoring and controlling a bank of screens. These are linked to surveillance cameras placed in public places,

buildings and places such as construction sites, to detect and prevent crime, and protect people

The Role of a Control Room Operator

- ✓ Monitor Operations.
- ✓ Monitor Emissions.
- ✓ Control Room Industries.
- ✓ Resolve Issues.
- ✓ Communicate with Others.
- ✓ Improve Safety.
- ✓ Record and Understand Data.
- ✓ Train Others.

- **Environmental condition**

  Environmental conditioning: is the modification of the environment of one or more organisms by their activities, including reaction and co-action (liberation of oxygen, for example, by water plants in an aquarium).

  The following must be considered for better setting control room environment condition:

- ✓ Control room arrangement
- ✓ Air condition system
- ✓ Electricity
- ✓ o Equipment status



Points to Remember

- **Control Room is** Centralized location equipped to monitor and manage operations, often in security or surveillance contexts.
- **Functions of Control Room include** Monitoring CCTV cameras for security surveillance, Responding to alarms and alerts, Coordinating emergency responses, Controlling access to secured areas and Recording/storing surveillance footage.
- **Control Room Operator:** Personnel responsible for monitoring activities within the control room like Observes CCTV feeds, responds to incidents, and communicates with relevant personnel.
- **Control Room Conditions:** Comfortable working environment conducive to concentration.



Application of learning 4.2

XYZ complex market would like to install CCTV SYSTEM INSTALLATION in their buildings. You are one of the technician team that is going to install that CCTV System and you are assigned to perform the following tasks:
You are asked to organize and arrange control room and provide equipment status.

Indicative content 4.3: Perform Export/Playback of required information

Duration: 4 hrs

Theoretical Activity 4.3.1: Description of Export/Playback for required

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of Export/Playback for required information

i. What is live Export?

ii. What is playback?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key readings 4.3.1.

5: In addition, ask questions where necessary.

Key readings 4.3.1.:

- **Define Export and Playback**

  "Live Export" in the context of CCTV (Closed-Circuit Television) cameras refers to the capability of exporting or saving live video footage in real-time as it is being captured by the camera. This feature allows users to instantly save or archive important events or incidents as they occur, without having to wait for playback or post-event analysis.

  Here's how "Live Export" typically works:

1. **Real-Time Recording:** The CCTV camera continuously captures live video footage of its surroundings, recording events as they happen in real-time.
2. **Exporting Live Footage:** When an event of interest occurs or when users want to save live footage, they can initiate the "Live Export" feature through the camera's interface or management software.
3. **Selection and Duration:** Users can specify the portion of the live footage they want to export and the duration they want to capture. This may involve selecting a specific timeframe or marking the beginning and end of the footage to be exported.
4. **Export Options:** Depending on the camera's capabilities and the software being used, users may have various export options, such as choosing the file format (e.g., AVI, MP4), video resolution, compression settings, and destination for saving the exported footage.

5. **Real-Time Saving:** Once the export settings are configured, the camera begins saving the live footage in real-time, creating a video file that captures the specified duration of the event.
6. **Notification and Confirmation:** Some systems may provide notifications or alerts to indicate that the live export process has started and provide confirmation once the export is complete.
7. **Accessing Exported Footage:** Once the export process is finished, users can access the exported video file through the camera's interface or management software. They can then review, analyze, or share the footage as needed for security, surveillance, or evidentiary purposes.

   "Live Export" is particularly useful in situations where immediate action or documentation is required, such as capturing evidence of security incidents, monitoring critical operations in real-time, or archiving important events as they unfold.

- **Video playback.**

  Video Playback is the process of reviewing recorded video footage captured by the camera's surveillance system. It allows users to view past events, incidents, or activities that have been recorded and stored by the camera or associated recording device. Video playback is a fundamental feature of CCTV systems and is essential for security, surveillance, and investigative purposes.

  CCTVvideo playback entails:

1. **Accessing Recorded Footage:** Users can access recorded footage through the camera's interface, dedicated monitoring software, or a connected video recording device such as a DVR (Digital Video Recorder) or NVR (Network Video Recorder). They typically navigate to a specific section or menu within the interface or software dedicated to video playback.
2. **Selecting Date and Time:** Users specify the date and time range for the footage they want to review. They may navigate through a calendar or input the desired date and time manually to locate the relevant recording.
3. **Playback Controls:** Once the desired footage is selected, users can control the playback using various controls provided by the interface or software. These controls typically include options to play, pause, stop, rewind, fast-forward, and adjust the playback speed of the video footage.
4. **Navigation and Scrubbing:** Users can navigate through the recorded footage by scrubbing through the timeline or using navigation buttons to move forward or backward in time. This allows them to quickly locate specific events or incidents within the recorded video.
5. **Playback Settings:** Depending on the capabilities of the CCTV system, users may have options to adjust playback settings such as video resolution, aspect ratio, audio playback, and synchronization with other camera feeds or sensor data.

6. **Event Tagging and Bookmarking:** Some CCTV systems allow users to tag or bookmark specific events or timestamps within the recorded footage for easy reference later. This feature is particularly useful for identifying and categorizing important incidents or activities.
7. **Exporting and Sharing:** Users may have the option to export or save selected portions of the recorded footage for further analysis, documentation, or sharing with law enforcement, security personnel, or other stakeholders.



Practical Activity 4.3.2: Performing video exporting and Playback

Task:

1: Referring to previous activities (4.3.1) you are requested to perform the given task. The task should be done **individually**.

As a CCTV Camera installer, you are requested to Performing video exporting and Playback for installed CCTV camera system.

2: List out procedures to perform exporting and playback on the given tasks.

3: Referring to procedures provided on task 2, Perform the given tasks

4: Present your work to the trainer and whole class.

5: Read key reading 4.3.2 and ask clarification where necessary

6: Perform the task provided in application of learning 4.3

Key readings 4.3.2

- **Performing Video exporting**

  Exporting video footage from a CCTV (Closed-Circuit Television) camera involves

retrieving recorded video data and saving it in a compatible file format for external use, such as archiving, analysis, or sharing with authorities.

The general steps to export video from a CCTV camera:

1. **Access Video Playback Interface:** Log in to the CCTV camera's interface, monitoring software, or connected video recording device (e.g., DVR, NVR). Navigate to the section or menu dedicated to video playback or recorded footage.
2. **Select Date and Time Range:** Specify the date and time range for the video footage you want to export. Use the calendar or input fields to select the start and end times of the desired recording.
3. **Locate Desired Footage:** Browse through the recorded footage to locate the specific events, incidents, or activities you wish to export. Use playback controls to navigate through the timeline and scrub through the video.
4. **Initiate Export Process:** Once the desired footage is located, look for an option to initiate the export process. This option may be labeled as "Export," "Save," "Download," or something similar.

   Click on the export option to begin the process of exporting the selected video footage.
5. **Specify Export Settings:** Depending on the CCTV system and software used, you may be prompted to specify export settings such as file format, video resolution, compression settings, and destination folder for saving the exported video file.

   Choose the appropriate settings based on your requirements and the capabilities of your system.
6. **Start Export:** After configuring the export settings, confirm your selections and start the export process. The CCTV system will begin exporting the selected video footage according to the specified settings.
7. **Monitor Export Progress:** Monitor the progress of the export process to ensure that it completes successfully. Some systems may provide a progress bar or indicator to show the status of the export.
8. **Verify Exported File:** Once the export process is complete, verify that the exported video file is saved in the designated location. Check the file to ensure that it contains the desired footage and plays back correctly.
9. **Save and Organize Exported Files:** Save the exported video file to a secure location on your computer or storage device. Organize the files according to date, time, or event to facilitate easy retrieval and access in the future.
10. **Optional:** Share or Archive Exported Footage; Depending on your needs, you may choose to share the exported footage with relevant parties such as law enforcement, security personnel, or stakeholders. Alternatively, you can archive the footage for future reference or investigation purposes.

- **Perform CCTV video Playback**

  Video playback in a CCTV (Closed-Circuit Television) camera allows users to review

recorded footage of past events or activities captured by the camera's surveillance system.

The general steps to perform video playback in a CCTV camera:

1. **Access Playback Interface**: Log in to the CCTV camera's web interface, dedicated monitoring software, or connected video recording device such as a DVR (Digital Video Recorder) or NVR (Network Video Recorder). Navigate to the section or menu dedicated to video playback or recorded footage.
2. **Select Date and Time Range**: Specify the date and time range for the video footage you want to review. This may involve using a calendar interface to select the desired date and entering the start and end times of the recording.
3. **Locate Desired Footage**: Browse through the recorded footage to locate the specific events, incidents, or activities you wish to review. Use playback controls to navigate through the timeline and scrub through the video.
4. **Playback Controls**: Once the desired footage is located, use playback controls to control the playback of the video. Common playback controls include play, pause, stop, rewind, fast-forward, and adjust playback speed.

   Some systems may also provide options to jump to specific timestamps or events within the recorded footage.
5. **Adjust Playback Settings**: Depending on the capabilities of the CCTV system, you may have options to adjust playback settings such as video resolution, aspect ratio, audio playback, and synchronization with other camera feeds or sensor data.
6. **Review Footage**: Review the recorded footage in real-time or at your preferred playback speed. Use playback controls to pause, rewind, or fast-forward through the video to analyze specific events or incidents in detail.
7. **Event Tagging and Bookmarking**: Some CCTV systems allow users to tag or bookmark specific events or timestamps within the recorded footage for easy reference later. This feature is useful for identifying and categorizing important incidents or activities.
8. **Export or Save Footage (Optional)**: If you need to save or export specific portions of the recorded footage for further analysis or documentation, look for options to export or save the footage. Follow the appropriate steps to export the footage in your desired format and resolution.
9. **End Playback**: Once you have finished reviewing the recorded footage, exit the playback interface or return to the main menu of the CCTV system.

Points to Remember

- **Video export**: refers to saving a copy of the current open document, database, image or video into a file format required by a different application. Applications may export to a variety of popular formats.
- **Video Playback:** is the playing back of computer or video content to screens that appear on camera.
- **Export Playback**: Involves selecting footage, setting export parameters, and saving the video file.
- **Playback**: Involves selecting the desired camera and time, and using playback controls to view the recorded footage.
- **Steps for Exporting Playback** include Access the CCTV System, Log In, Select the Playback Feature, Choose the Camera and Date, Preview the Footage, Set Export Parameters, Select Export Destination, Start the Export Process and Verify the Exported File.
- **Steps for Playback** include Access the CCTV System, Log In, Navigate to Playback, Select the Camera, Specify the Date and Time, Start Playback, Use Playback Controls and finally Zoom and Search (Optional).

Application of learning 4.3

KUKU Company Ltd located in Western Province, Rubavu District, recruit you and assign you to install CCTV cameras in the workshops of the GARE Market and you will also need to export and playback the stored footage to show to the room controller how the system works.

Indicative content 4.4: Interpretation of CCTV system logs

Duration: 4 hrs

Theoretical Activity 4.4.1: Description of CCTV system logs

Tasks:

1: In small groups, you are requested to answer the following questions related to the of description of CCTV system logs.

i. What is logs in CCTV System?

ii. Describe system logs metrics in CCTV system.

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class.

4: For more clarification, read the key readings 4.4.1

5: In addition, ask questions where necessary.

Key readings 4.4.1.:

- **Definition of logs**

  Logs typically refer to recorded data or events that are stored by the camera or associated recording device. These logs provide a chronological record of various activities, events, and system operations related to the CCTV system.

  More detailed explanations of logs:

1. **Event Logs:** CCTV cameras and recording devices often maintain event logs that record significant occurrences or actions detected by the system. These events may include motion detection, alarms triggered by sensors (e.g., door/window sensors), camera tampering, video loss, network connectivity issues, and system errors.
   Each entry in the event log typically includes details such as the date, time, type of event, camera identifier (if multiple cameras are connected), and any additional information relevant to the event.
2. **System Logs:** System logs capture information about the operational status and performance of the CCTV system itself. This includes logging system startup/shutdown events, configuration changes, firmware updates, system errors, disk usage, network connectivity status, and other system-level activities.
   System logs provide valuable diagnostic information for troubleshooting issues, monitoring system health, and ensuring the reliability and integrity of the CCTV system.

3. **Audit Logs:** Audit logs track user interactions and administrative actions performed on the CCTV system. This includes logging user logins, logouts, access attempts, permission changes, and other administrative activities.
   Audit logs help administrators monitor user activity, track changes made to system settings, and maintain accountability by recording who accessed the system and what actions they performed.
4. **Archived Logs:** CCTV systems may also support the archiving of logs for long-term storage and historical reference. Archived logs allow users to review past events, system activities, and user interactions over extended periods.
   Archiving logs is crucial for compliance with regulatory requirements, internal audits, incident investigations, and legal purposes.
   In general, logs in a CCTV camera serve as a detailed record of events, system operations, and user activities, providing valuable insights into the functioning and security of the surveillance system.

- **Description of system logs metrics**
  While **logs** are about a specific event, **metrics** are a measurement at a point in time for the system.
  System logs in a CCTV (Closed-Circuit Television) camera capture various metrics and parameters related to the operational status, performance, and configuration of the CCTV system. These metrics provide valuable insights into the functioning and health of the surveillance infrastructure.

  Description of common system logs metrics:

1. **System Startup/Shutdown Events**: Logs system startup and shutdown events, indicating when the CCTV system was powered on or off. These events provide insight into system uptime and availability.
2. **Configuration Changes**: Records changes made to the configuration settings of the CCTV system, such as camera settings, recording parameters, network settings, and user permissions. Configuration change logs help track modifications to the system setup and ensure compliance with security policies.
3. **Firmware Updates**: Tracks firmware updates applied to the CCTV cameras, recording details such as the date, time, and version of the firmware installed. Firmware update logs help ensure that cameras are running the latest software versions with bug fixes, performance improvements, and security patches.
4. **System Errors and Warnings**: Logs errors, warnings, and diagnostic messages generated by the CCTV system, indicating issues or abnormalities encountered during operation. System error logs provide insights into hardware failures, software bugs, network connectivity issues, and other operational problems that require attention.

5. **Disk Usage and Storage Capacity**: Monitors disk usage and storage capacity of the recording device (e.g., DVR, NVR), tracking metrics such as available storage space, used space, and disk utilization percentage. Disk usage logs help administrators manage storage resources efficiently and prevent storage capacity issues that could affect video recording.
6. **Network Connectivity Status**: Records information about the network connectivity status of the CCTV system, including network configuration details, IP addresses, subnet masks, gateway addresses, DNS settings, and connection status (e.g., connected, disconnected).
7. **Camera Health and Performance**: Captures metrics related to the health and performance of individual CCTV cameras, such as camera uptime, video signal quality, motion detection events, camera tampering alerts, and camera temperature. Camera health logs help identify issues with specific cameras and ensure continuous surveillance coverage.
8. **Authentication and Access Control**: Logs user authentication events, access attempts, login failures, and administrative actions performed on the CCTV system. Authentication logs provide an audit trail of user activity and help enforce access control policies.
9. **System Resource Utilization**: Tracks the utilization of system resources such as CPU usage, memory usage, and network bandwidth. Resource utilization logs help monitor system performance and identify potential bottlenecks or resource constraints.
10. **Timestamps and Event Sequencing**: Includes timestamps for each logged event, indicating the date and time when the event occurred. Event sequencing ensures that logs are organized chronologically, facilitating analysis and troubleshooting.



Points to Remember

- **Logs** are files that record events, warnings and errors as they occur within a software environment. Most logs include contextual information, such as the time an event occurred and which user or endpoint was associated with it. **The CCTV log** displays a paged grid containing all scheduled, current and completed recordings. The list is sorted based on start time.
- **Description of system logs metrics:** While **logs** are about a specific event, **metrics** are a measurement at a point in time for the system. **A log** is an event that happened and **a metric** is a measurement of the health of a system.

Application of learning 4.4

NKANDA Factory has installed CCTV SYSTEM in its factory storage building, as one of the CCTV system installer team in the industry, you are asked access and review system logs and metrics after provide the report showing those logs to the factory owner.

Indicative content 4.5: Generate and interpret incident report

Duration: 4 hrs

Theoretical Activity 4.5.1: Description incident report

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of incident report.

i. What is an incident report?
ii. What are the benefits of reporting?
iii. Describe the types of incidents to be reported.
iv. Identify the key elements of incident report.

2: Provide the answers for the asked questions and write them on papers.
3: Present the findings/answers to the whole class
4: For more clarification, read the key readings 4.5.1
5: In addition, ask questions where necessary.

Key readings 4.5.1.:

- **Generate and interpret incident report**

  An incident report: is a tool that documents any event that may or may not have caused injuries to a person or damage to a company asset.

- **The benefits of reporting**

✓ Increased communication
✓ Increased productivity
✓ Increased accuracy
✓ Increased timeliness
✓ Increased information
✓ Improved decision making

- **Four (4) types of incidents you should report:**

1. **Sentinel events**: these are unexpected occurrences that resulted in serious physical or psychological injury or death (e.g. slips, trips and falls, natural disasters, vehicle accidents, disease outbreak, etc.).

✓ Worker injury incident
✓ Environmental incident
✓ Property damage incident
✓ Vehicle incident

- ✓ Fire incident

  2. Near misses: these are situations where the people involved had no injuries but could have been potentially harmed by the risks detected.

  3. Adverse events: related to medicine, vaccines, and medical devices. These events occur when an act of commission or omission harmed a patient rather than from the existing disease or condition.

  4. No harm events: these are incidents that need to be communicated across an organization to raise awareness of any harm that may happen.

- **Key elements to be included in incident report**
- ✓ **Accurate:** All data must be clear and specific.
- ✓ **Factual:** An incident report should be objective and supported by facts.
- ✓ **Complete:** Ensure that all essential questions (what, where, when, why, and how) are covered in the incident report.
- ✓ **Graphic:** Photos, diagrams, and illustrations should be included as supporting evidence.
- ✓ **Valid:** Upon completion, those who are involved in the incident (e.g. victim, witnesses, manager, reporter, etc.) should sign off to testify and validate all the information that was mentioned in the incident report.
- **Incident report format**

  1. Introduction

  The first part of the incident report form covers the who, what, when, and where of the incident: 94 − Include the names of all the people involved in the incident. If names are not available, you can instead cite the person's role in relation to the incident e.g. the customer, the guest. − Summarize the incident itself in no more than three sentences e.g. a printer caught fire causing minimal damage to a room − Include the time and date of occurrence. An estimate works if the exact time could not be noted.

  − Note the location of the incident and be exact if possible e.g. the Mercato Conference Room, 19th floor Building A.

2. Body

This is where you lay out all of the incident's details in a comprehensive manner. Talk about the incident from start to finish, ensuring details are laid out in chronological order to avoid confusion. Make sure to include, who, what, when, and where mentioned in the introduction when they are mentioned.

3. Conclusion

Was the incident resolved? How? If the incident was not resolved, explain why, and provide the steps that need to be taken in order to resolve it.

4. **Sign off**

   Include the full name and signature of the incident report writer for accountability and record-keeping.



Points to Remember

- **Incident Report is a** Document detailing events or occurrences that deviate from normal operations or expectations.
- **Benefits of Reporting:** Facilitates identification of security vulnerabilities or system failures, Helps in implementing corrective measures to prevent future incidents.
- **Types of Incidents to be Reported are** Security breaches or unauthorized access, Equipment malfunctions or failures, Physical intrusions or unauthorized entries, Suspicious activities or behavior and Any other events affecting safety or security.
- **Key Elements of Incident Report:** Date, time, and location of the incident, Description of the incident, individuals involved, Actions taken.



Application of learning 4.5

NKANDA Factory wants to has installed CCTV SYSTEM in its factory storage building, as one of the CCTV system installer team in the industry, you are asked to generate and interpret report on the incidents occurred during CCTV system operations.



Learning outcome 4 end assessment

Theoretical assessment

A. Choose the correct answer

1. What is the primary function of a CCTV control room?

A) To store recorded footage

B) To monitor and manage live footage from CCTV cameras

C) To repair CCTV equipment

D) To install new CCTV cameras

2. Which type of monitor is typically used for viewing live footage in a CCTV system?

A) CRT Monitors

B) LED Monitors

C) Plasma Screens

D) Projectors

3. What does "Export" refer to in the context of CCTV systems?

A) Sending live footage to another monitor

B) Transferring recorded footage to external storage

C) Removing old footage from the system

D) Streaming live footage to mobile devices

4. Which environmental condition is crucial for maintaining an effective CCTV control room?

A) Ambient lighting

B) Air conditioning

C) Background music

D) Carpet flooring

5. What are CCTV system logs primarily used for?

A) Tracking the health and status of CCTV equipment

B) Storing live footage for future review

C) Recording user actions and system events

D) Analyzing footage for suspicious activities

6. What is the role of a control room operator in a CCTV system?

A) Installing cameras

B) Managing and monitoring live footage and alarms

C) Repairing equipment

D) Writing software for the CCTV system

7. What does the term "Playback" refer to in CCTV operations?

A) Reviewing recorded footage

B) Playing live footage on a different monitor

C) Backing up the system data
D) Configuring camera settings

8. What is a critical aspect of generating an incident report from CCTV footage?

A) Using high-quality monitors
B) Interpreting the system logs
C) Ensuring the footage is not edited
D) Ensuring the footage is from the correct time frame and location

9. Why is air conditioning important in a CCTV control room?

A) To keep the room comfortable for operators
B) To prevent equipment from overheating
C) To maintain the correct pressure inside the room
D) To filter dust and other particles

10. Which of the following best defines "live footage" in the context of CCTV?

A) Footage that is being recorded for later review
B) Footage that is being viewed in real-time as events occur
C) Footage that has been exported to an external drive
D) Footage that has been backed up to a cloud server

11. Which of the following types of CCTV monitor is a type of flat panel display that produces its own light? Select one

A) LCD Monitor
B) LED Monitor
C) OLED Monitor
D) Plasma Monitor

B. Answer the following statements by TRUE if the statement is correct otherwise FALSE.

1. Live footage in a CCTV system refers to video that is recorded and stored for later review.
2. LED monitors are commonly used for viewing live footage in a CCTV system due to their clarity and brightness.
3. The term "Export" in CCTV operations refers to the process of reviewing recorded footage on a monitor.
4. Air conditioning is an important environmental condition in a CCTV control room to prevent equipment from overheating.
5. CCTV system logs are used primarily for recording user actions and system events.
6. A control room operator is responsible for installing new CCTV cameras and configuring their settings.
7. "Playback" refers to the process of viewing live footage as it is captured by the cameras.
8. When generating an incident report, it is critical to ensure that the footage used is from the correct time frame and location.
9. The primary purpose of a CCTV control room is to store recorded footage for archival purposes.
10. Environmental conditions like ambient lighting and background music are crucial for maintaining a CCTV control room.

11. Incident report promoted hazards and threats Communication and Awareness….
12. Incident is defined as any event, condition, or situation which attracts positive media attention or a positive profile for the worksite……
13. Sentinel events are unexpected occurrences that resulted in serious physical or psychological injury or death…….
14. All data to be included in incident report must be clear and specific

Practical assessment

MAKORO Company Ltd located in Western Province, NYABIHU District, recruit a CCTV system installer team to facilitate its works. You are recruited in this team and you are assigned you to install CCTV cameras in the building located in MUKAMIRA Sector. You will need to perform the following tasks:

1. Export and playback required information
2. Export recorded video surveillance footage.
3. Play the exported videos.



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Indicative contents
5.1. Check deployed CCTV system
5.2. Selection of testing tools
5.3. Performing connectivity testing
5.4. Performing Functional testing
5.5. Testing performance
5.6. Elaboration of testing report

Key Competencies for Learning Outcome 5: Test CCTV system.

Knowledge	Skills	Attitudes
• Identification of installed CCTV materials and Equipment, • Differentiation of hardware and software testing tools, • Identification of the Items to be tested for physical connectivity of CCTV System, • Description of the functionality of CCTV hardware components, • Description of the functionality of CCTV software, • Identification of the items to check when testing the performance of	• Checking CCTV deployed materials and equipment, • Selecting of hardware and software testing tools, • Selecting of site based on site survey, • Performing physical and logical connectivity testing, • Performing hardware and software functionality testing, • Testing the CCTV System performance,	• Be a Self-Motivator, • Be a Teamwork player, • Have a Curiosity spirit, • Have a Self-confidence, • Be a Problem solver, • Be a Critical thinker.

CCTV Camera System, ● Description of the steps for generating CCTV Camera testing report, ● Identification of the items to be included in CCTV Camera testing report.	● Elaborating the testing report.	



Duration: 10 hrs

Learning outcome 5 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Check correctly deployed CCTV system based on CCTV system.
2. Select testing tools correctly based on type of testing.
3. Perform correctly connectivity testing according to the CCTV system installed.
4. Perform correctly functional testing according to the CCTV system installed.
5. Perform correctly performance testing according to the CCTV system installed.
6. Elaborate properly report based on test results.



Resources

Equipment	Tools	Materials
● Computer ● Monitor ● Power Supply ● Switch ● Storage devices ● Cameras	● Crimping tool, ● Wire stripper, ● Drilling machine, ● Hammer, ● Ladder, ● Pliers,	● Internet bundles ● Connectors ● Cable ● Power extension

• Recording Devices (NVR/DVR)	• Screw drivers • Cable Tester •	

Indicative content 5.1: Check deployed CCTV system

Duration: 1 hrs

Theoretical Activity 5.1.1: Identification the CCTV materials and equipment deployed.

Tasks:

1: In small groups, you are requested to answer the following questions related to the identification the CCTV materials and equipment deployed.

i. What are the CCTV camera system materials?
ii. What are the equipments used for deploying CCTV camera system?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class.

4: For more clarification, read the key readings 5.1.1

5: In addition, ask questions where necessary.

Key readings 5.1.1.:

The increased use of sophisticated security system has led the need of simple tests that such systems are operational. The followings are the materials and equipments to be checked.

- **Equipments**

A CCTV (Closed-Circuit Television) camera system comprises various equipments designed to capture, transmit, record, and display video footage for surveillance and security purposes.

Equipments typically found in a CCTV camera system:

- Cameras: CCTV cameras are the primary components of the system responsible for capturing video footage. They come in various types, including dome cameras, bullet cameras, PTZ (Pan-Tilt-Zoom) cameras, and specialty cameras for specific applications like license plate recognition or thermal imaging.
- Mounting Hardware: Mounting hardware includes brackets, housings, and mounts used to install and position CCTV cameras in desired locations. This hardware ensures secure and stable placement of cameras for optimal surveillance coverage.
- Cabling: Cabling is used to connect CCTV cameras to power sources, video recorders, and monitoring devices. Common types of cables include coaxial cables for analog cameras and Ethernet cables (e.g., Cat5e, Cat6) for IP cameras. Power-over-Ethernet (PoE) cables can provide both power and data connectivity for IP cameras.

- Power Supplies: Power supplies provide electrical power to CCTV cameras, typically using AC (alternating current) or DC (direct current) power sources. Power supplies may include individual adapters for each camera or centralized power distribution units for multiple cameras.
- **Video Recorders:** Video recorders are devices used to store and manage recorded video footage from CCTV cameras. They include DVRs (Digital Video Recorders) for analog cameras and NVRs (Network Video Recorders) for IP cameras. These recorders often feature built-in storage drives (HDDs or SSDs) for storing recorded video data.
- **Storage Media:** Storage media, such as hard disk drives (HDDs) or solid-state drives (SSDs), are used to store recorded video footage. The capacity of the storage media determines the amount of video data that can be stored and retained for playback and archival purposes.
- **Network Equipment**: Network equipment includes routers, switches, and network cables used to establish connections between IP cameras, video recorders, and monitoring devices. These components facilitate the transmission of video data over local area networks (LANs) or the internet.
- **Monitoring Devices:** Monitoring devices, such as monitors, displays, and video management software (VMS), are used to view live video feeds and recorded footage from CCTV cameras. Monitors and displays provide real-time monitoring capabilities, while VMS software offers advanced features for managing multiple cameras and accessing recorded video data.
- **Accessories and Consumables:** Various accessories and consumables may be required for installation, maintenance, and operation of CCTV camera systems. This includes cable connectors, power adapters, surge protectors, weatherproof enclosures, and cleaning supplies for camera lenses.

- **Materials and tools**

The list of Materials and tools that shall be used for CCTV testing:

- Electrician tool kit
- Drill machine tools and spanner sets
- Cramping tool with crimping bits for all required sizes
- Insulation Resistance Tester
- Power Tools 220V
- Mobile Scaffold Tower
- Aluminum Step ladder
- Line Tester
- Multi-meter
- Internet Bundles
- Internet bundles
- Connectors and cables

Practical Activity 5.1.2: Checking the CCTV materials and equipment deployed.

Task:

1: Referring to previous activity (5.1.1) you are requested to perform the given task. The task should be done **individually**.

As a CCTV system installer, you are asked to check the deployed materials and equipment necessarily to see if the installed CCTV camera system is well performing its tasks.

2: List out procedures to select the CCTV materials and equipment deployed to perform the given tasks

3: Referring to the procedures provided from task 2, Perform the given task.

4: Present your work to the trainer and whole class.

5: Read key reading 5.1.2 and ask clarification where necessary.

Key readings 5.1.2

- **Checking the CCTV materials and equipment deployed.**

To check the materials and equipment deployed in a CCTV (Closed-Circuit Television) system, you'll need to perform a thorough inspection of all components to ensure they are installed correctly and functioning properly. Step-by-step guide:

1. **Review Equipment List**:

✓ Start by reviewing the inventory or equipment list provided for the CCTV system. This should include details of all cameras, recording devices, storage devices, cabling, power supplies, and accessories deployed in the system.

2. **Inspect Cameras**:

✓ Check each CCTV camera installed in the system to ensure they are positioned correctly and securely mounted. Verify that the cameras have clear lines of sight to the areas they are monitoring and are angled appropriately for optimal coverage.

✓ Inspect the physical condition of the cameras, looking for any signs of damage, tampering, or weather-related wear and tear. Ensure that any protective enclosures or housings are intact and functioning properly.

3. **Check DVR/NVR and Storage Devices**:

✓ Examine the DVR or NVR unit and any associated storage devices (e.g., hard drives, NAS devices) to ensure they are properly connected and powered on.

- ✓ Check the status indicators on the recording devices for any error messages or warnings. Verify that the recording devices have sufficient storage capacity and are recording video footage as expected.

4. **Verify Power Supply and Cabling**:

- ✓ Inspect the power supply units and electrical wiring used to power the CCTV cameras and recording devices. Ensure that power connections are secure and that cables are properly routed to prevent tripping hazards or damage.
- ✓ Check for any signs of cable damage, such as fraying or exposed wires, and replace or repair as necessary. Use cable testers or multimeters to verify continuity and proper voltage levels.

5. **Test Network Connections**:

- ✓ If the CCTV system includes IP cameras or network-connected recording devices, test the network connections to ensure they are functioning correctly. Check network cables, connectors, and switches for proper connectivity.
- ✓ Use network diagnostic tools to ping cameras and recording devices, check IP addresses, and verify network settings. Ensure that remote access to the CCTV system is configured securely and is accessible as required.

6. **Inspect Mounting Hardware and Accessories**:

- ✓ Check the mounting hardware and accessories used to install CCTV cameras, including brackets, mounts, and weatherproof housings. Ensure that all components are securely fastened and properly aligned.
- ✓ Inspect any additional accessories such as infrared illuminators, microphones, and PTZ controllers to ensure they are installed correctly and functioning as intended.

7. **Test System Functionality**:

- ✓ Conduct functional tests of the CCTV system to verify that all components are working together effectively. Test live video feeds from each camera, playback recorded footage, and check for motion detection and alarm triggers.
- ✓ Use monitoring software or remote viewing applications to access the CCTV system from different devices and locations to ensure remote access capabilities are functioning correctly.

8. **Document Findings**:

- ✓ Document any observations, issues, or corrective actions taken during the equipment check. Note any components that require maintenance, repair, or replacement, and schedule follow-up actions accordingly.

Points to Remember

- **An Analog CCTV System equipments deployed are** Camera, DVR, Hard Disk, Wiring for video and power, Power Supply and Connector - BNC or RJ45.
- **An IP CCTV Camera System Equipments deployed are** IP Camera, NVR, Hard disk, Wiring for Video (CAT5 / CAT5E / CAT6), Wiring for power. Power Supply or PoE, Switch / PoE switch, Connector – RJ45.
- **Materials and tools deployed are** Electrician tool kit, Drill machine tools and spanner sets, Cramping tool with crimping bits for all required sizes, Insulation Resistance Tester, Power Tools 110V, Mobile Scaffold Tower, Aluminium Step, Line Tester, Multi-meter**,** Internet bundles, Connectors and cables.
- Steps to check the materials and equipment deployed in a CCTV system are Review Equipment List, Inspect Cameras, Check DVR/NVR and Storage Devices, Verify Power Supply and Cabling, Test Network Connections, Inspect Mounting Hardware and Accessories, Test System Functionality and Document Findings

Application of learning 5.1.

MARANA Company Ltd located in Northern Province, Burera District, recruit a CCTV system installer team to facilitate its works. You are recruited in this team and you are assigned you to install CCTV cameras in the building located in KINYABABA Centre.

Task:

Before making installation, you are requested to elaborate and check materials and equipments required that will facilitate the installation process and after installation you are requested to check if every equipments and materials are working properly.

Indicative content 5.2: Selection of testing tools

Duration: 2 hrs

Theoretical Activity 5.2.1: Selection of hardware and software testing .

Tasks:

1: In small groups, you are requested to answer the following questions related to the Selection of hardware and software testing tools

i. What are the hardware testing tools?
ii. What are the software testing tools?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key **readings 5.2.1**

5: In addition, ask questions where necessary.

Key readings 5.2.1.:

✓ **Hardware testing tools**

In CCTV (Closed-Circuit Television) camera systems, various hardware testing tools are used to diagnose, troubleshoot, and maintain the system's components. These tools help technicians ensure that cameras, cables, connectors, and other hardware are installed correctly and functioning properly.

Common hardware testing tools used in CCTV camera systems:

1. **Multimeter**: A multimeter is a versatile tool used to measure voltage, current, and resistance in electrical circuits. Technicians use multimeters to test power supplies, check continuity in cables, and verify proper electrical connections in CCTV systems.
2. **Cable Tester**: Cable testers are devices used to check the integrity and functionality of network cables, coaxial cables, and other types of wiring used in CCTV installations. Cable testers can identify wiring faults such as open circuits, short circuits, and miswiring, helping technicians troubleshoot connectivity issues quickly.
3. **Video Signal Tester**: Video signal testers, also known as CCTV testers or camera testers, are handheld devices designed specifically for testing CCTV cameras and video signals. These testers can display live video feeds from cameras, measure video signal quality, adjust camera settings, and troubleshoot image quality issues.
4. **Power Supply Tester**: Power supply testers are tools used to diagnose issues with power supplies used in CCTV systems. These testers can verify voltage levels, check

for voltage fluctuations, and detect power supply failures or inconsistencies that may affect the operation of cameras and recording devices.

5. **Tone Generator and Probe Kit**: A tone generator and probe kit is a tool used to trace and identify cables in CCTV installations. The tone generator injects a signal onto a cable, while the probe detects the signal along the length of the cable, allowing technicians to locate and identify specific cables within a wiring network.
6. **Network Analyzer**: Network analyzers are advanced diagnostic tools used to analyze and troubleshoot network communications in IP-based CCTV systems. These tools can monitor network traffic, detect network errors, identify bandwidth issues, and diagnose connectivity problems affecting IP cameras and recording devices.
7. **Power Over Ethernet (PoE) Tester**: PoE testers are specialized tools designed to test Power over Ethernet connections used to power and transmit data to IP cameras and other network devices. These testers can verify PoE voltage levels, detect PoE compatibility issues, and troubleshoot problems related to power delivery over Ethernet cables.
8. **Crimping Tool**: A crimping tool is used to terminate and crimp connectors onto the ends of coaxial cables, Ethernet cables, and other types of wiring used in CCTV installations. Properly terminated connectors ensure secure connections and reliable transmission of signals between devices.
9. **Digital Inspection Camera**: Digital inspection cameras are small, portable cameras equipped with LED lights and a flexible probe, allowing technicians to visually inspect hard-to-reach areas within CCTV installations. These cameras can help identify cable routing issues, inspect connections, and locate hidden cameras or components.
10. **Infrared Thermometer**: An infrared thermometer is a non-contact temperature measurement device used to detect hotspots, overheating components, and thermal anomalies in CCTV equipment. Technicians can use infrared thermometers to identify potential issues with cameras, power supplies, and other hardware that may be affected by heat-related problems.

✓ **Software testing tools**

Software testing tools play a crucial role in ensuring the reliability, functionality, and performance of CCTV (Closed-Circuit Television) camera systems. These tools help identify defects, verify system behavior, and validate software functionality to ensure that the CCTV system meets its requirements.

✓ **Common software testing tools categories used in the CCTV camera industry:**

1. **CCTV Management Software**: Many CCTV camera manufacturers provide proprietary management software for configuring, monitoring, and managing CCTV systems. This software typically includes features for live viewing, video

playback, camera configuration, motion detection settings, alarm management, and user management.

2. **Video Management Software (VMS)**: Video Management Software (VMS) solutions are comprehensive platforms designed to manage video surveillance systems of all sizes. VMS software offers advanced features such as video analytics, event management, remote access, multi-camera viewing, and integration with third-party devices and systems.
3. **Network Protocol Analyzers**: Network protocol analyzers such as Wireshark or tcpdump are used to analyze network traffic between CCTV cameras, recording devices, and other networked components. These tools help identify communication issues, network errors, and potential security vulnerabilities by capturing and analyzing network packets.
4. **Camera Emulators**: Camera emulators simulate the behavior of CCTV cameras to test software applications, recording devices, and management systems without the need for physical cameras. Emulators can simulate different camera models, resolutions, video streams, and camera settings to validate software functionality.
5. **Video Playback and Analysis Tools**: Video playback and analysis tools allow users to review recorded footage, analyze video quality, and extract metadata from video streams. These tools provide features for frame-by-frame analysis, motion detection verification, object tracking, and forensic investigation of video recordings.
6. **Load Testing Tools**: Load testing tools such as Apache JMeter or LoadRunner are used to simulate high volumes of concurrent users or traffic on CCTV system components, including recording servers, network infrastructure, and management software. Load testing helps assess system performance, scalability, and resilience under heavy loads.
7. **Event Management Systems**: Event management systems are software platforms designed to handle and process events generated by CCTV cameras, sensors, and other devices. These systems provide features for event correlation, alarm notification, event escalation, and automated response actions based on predefined rules and policies.
8. **Video Analytics Software**: Video analytics software utilizes advanced algorithms to analyze video streams and extract actionable insights from CCTV footage. These tools offer features such as object detection, facial recognition, license plate recognition, crowd monitoring, and behavior analysis to enhance situational awareness and security intelligence.
9. **Mobile Viewing Apps**: Mobile viewing apps provided by CCTV camera manufacturers or third-party developers allow users to access live video feeds, view recorded footage, and manage CCTV systems from smartphones and tablets.

These apps typically support remote monitoring, push notifications, and control of camera settings on mobile devices.

- ✓ **Some software testing tools examples are described below:**

1. NetCam Studio.

There is a free version of NetCam Studio, but if you want something more powerful, then you can buy one of their licenses. These licenses allow you to use the software for as long as you like, and they will update the software for up to two years after purchase.

After the two-year period, you can keep the software running as normal, but if you want upgrades, then you will have to buy them. Upgrades shouldn't be needed unless you buy a new piece of camera hardware and it requires a new type of camera software.

The NetCam Studio Features

If you go for the free version of NetCam Studio, then you get to use a camera feed from two sources. There is a watermark added to the camera images, and you get five-day notifications. There is also community support.

The paid features include up to 64 sources, no watermarks, 250-day notifications, and you get priority support. After the 250 days is up, you may buy upgrades in order to keep the notification system going.

Pros

- The free version does not expire in any way besides the notifications
- You are able to use this software with a wide variety of camera brands
- Updates are frequent enough to tackle most system-breaking bugs

Cons

- The interface needs updating to make it more user friendly
- They need to better explain the “Add Source” procedure
- Troubleshooting audio problems is very difficult

2. Security Eye.

The reason Security Eye ranks so highly is that it is a professional piece of software that somebody decided should be free. It is not open source security camera software, but the developers have decided that it should be free. There used to be a free version where you would get 2 hours of free surveillance per day for free, but in 2019 the company removed its payment system. Will they keep it up or go back to paid subscriptions? Who knows, but you should get it while it is still free (if you can).

Security Eye Features

You can use a timer to determine when the video feed should start. You can capture images as JPEGs and then compress them. The system also allows you to receive alerts via a loudspeaker. It works with motion-detection devices too.

Pros

- It comes with the Xvid MPEG-4 Codec
- The full version has no nag screen or recording limit
- The Security Eye interface is pretty modern for what is essentially a free piece of software

Cons

- Demands 1GB of RAM and 2GB of HHD space
- Requires at least an Intel Pentium 1.8 processor
- The icons should be more intuitive

3. Perfect Surveillance.

There are three types of free security camera software for you to choose from. There is one for IP cameras, one for web cameras, and there is a piece of viewing software that people may use if they do not have something like VLC video viewer installed.

Perfect Surveillance Features

Simple icons and a clear interface is a great benefits. Determine when the cameras start monitoring and set up your own layouts to suit your viewing style.

Pros

- A simple program that works in a very uncomplicated manner
- Most of its icons are intuitive to where you can guess what they do
- The Perfect Surveillance system has motion detection and allows you to monitor up to devices.

Cons

- Installing seems overly long and bloated
- There are not enough instructions on how to get your cameras connected
- They do not explain how to capture and export screenshots

4. iVideon.

There are times when a piece of free software simply does its job good enough to warrant no complaints, and that is exactly what iVideon offers. There are plenty of ways it can be improved, but there is nothing about it that demands harsh criticism.

iVideon Features

One of its most advertised features is facial recognition, but it is not very accurate. It does allow for very good mobile usage. You can view live and recorded footage on your desktop or mobile device. It is also available for Mac OS X and Linux.

Pros

- Online home use and access is completely full-featured and freeware
- It has smart notifications and has its own desktop and mobile apps
- it doesn't take long to learn how to use the iVideon interface

Cons

- Seems rather light on features
- Installing and connecting cameras requires a better tutorial

- Embedding live video on your website is trickier than it first seems

5. Xeoma.

This program has plenty of features, but most of them are locked off to free users. It takes over too long to learn how to use the system, but it may be worth it if you have many cameras. The paid version has all sorts of perks from SMS texts to system health monitoring.

Xeoma Features

The lite version is the free version. It allows you to add four cameras with up to six modules in each chain. You can test out and use the lite version as long as you like, or you can get a free trial of the larger fully-featured program.

Pros

- Record and monitor from four cameras
- Updates are very frequent
- This software works with mostly every type of camera

Cons

- Around 90% of features are locked off to free users.
- Xeoma is difficult to learn and set up
- Connecting many cameras is difficult because the system is difficult to learn as a whole

6. Genius Vision.

The Genius Vision system is a fairly advanced piece of software that is probably best used for businesses. If used at home, then there are many features that may never be used. It also works on mobile devices too.

Genius Vision Features

There are plenty of features to play with, including a massive array of settings to fit the many types of camera that this program can be used with. You can even support the company through Patreon and get more extensions on the program.

Pros

- Works in-depth with many types of camera
- The tools are very sophisticated and accurate
- The drop-down menus are complicated while also being thoughtfully laid out.

Cons

- Getting paid upgrades is trickier than just buying on their website
- Setting up the remote features is difficult
- You have to figure out most of how to use the program based on which camera you have connected

7. Deskshare IP Camera Viewer.

The Deskshare IP Camera Viewer company has a great piece of security camera software that they repackage into many different names and then give away on their website. They do this because the software comes with lots of unwanted

bloatware that the company is paid to promote. The sad part is that many other websites will replace this bloatware with Trojan viruses, which means it is difficult to get a clean version of this program.

Deskshare IP Camera Viewer Features

Live recording, playback, and split-screen is all available. You can use up to 4 cameras at a time, and it works with 2200 different camera models. If your hardware allows it, then you can pan, tilt and zoom your camera.

Pros

- It is compatible with many types of camera brand
- You can set up floating surveillance windows if you want
- Adjust the properties of each camera individually

Cons

- Installing is very difficult to do correctly
- Comes with bloatware and/or unwanted content
- The IP software has the least features of all the Deskshare camera software

8. Anycam.

The Anycam system is a very average piece of software, but it works with most any type of camera. The interface is basic but easy to use and easy to set up. You will discover that the free version is only limited in terms of record time, you can still use all its features.

Anycam Features

Anycam is mostly featureless. It works with motion-detection cameras. It allows for video recording and playback. Plus, you can watch a live feed. Other than that, there is very little you can do with this basic-but-effective piece of software.

Pros

- The program seems to work with any type of camera
- Anycam has a paid version where you only make a one-time payment and get free lifetime updates
- Installing and setup is very easy

Cons

- Trying to find specific events on the timeline is very fiddly
- The free version only allows two cameras to be linked
- The recording is very limited with the free version

9. ContaCam

In the strictest terms, this piece of software is not free. The developers ask you for a donation, but there is nothing forcing you to give them one. Plus, you can try the software, and then if it turns out to be brilliant, then you can always go back and donate to them.

ContaCam Features

Despite seeming very complicated, the interface is actually very self-explanatory. At first glance, it looks overly condensed and full of features, but each explains itself very well. You can configure the cameras very precisely to the point where you can even specify within which seconds it starts and stops.

Pros

- A rather advanced piece of software
- The interface layout is very condensed without being confusing
- You can use web servers, the Internet, or intranet to receive live feeds

Cons

- The interface has no aesthetic decoration at all
- ContaCam takes a long time to learn through trial and error testing
- The developers say this will never work with mobile devices

10. Agent DVR

The great thing about Agent DVR is that it is completely free because it is freeware. The sad part is that it was created by a developer who went on to finer things and left the program lodged on a few high-use servers. It is now being passed around freeware websites, which means it is sometimes difficult to find a clean version. Nevertheless, it is a fine piece of camera software that still works for many types of cameras.

Agent DVR Features

The Agent DVR software allows for video surveillance with a live feed, and for recording. You can use it with motion-detection software, and it still works for many different types of cameras. It uses a reliable SSL algorithm for its signal encryption, and it is very lightweight, which means it is not a big drain on your device's CPU.

Pros

- A very simple and lightweight piece of software
- Works all the way back to Windows 7
- It still works with motion detection systems

Cons

- Agent DVR will become more obsolete as newer cameras appear over the coming years
- Updates will never occur beyond Windows 10
- The interface looks very dated compared to the modern camera software.



Points to Remember

- **Hardware testing tools are** IP camera cable Tester, Camera Tester, UTP Cable tester , Cable fault finder, Micro-scanner cable verifier, Multi-functional CCTV Tester, Multi-meter, Security Camera Tester and Network low voltage tester
- **Software testing tools categories that used in the CCTV camera are** CCTV Management Software, Video Management Software (VMS), Network Protocol Analyzers, Camera Emulators, Video Playback and Analysis Tools, Load Testing Tools, Event Management Systems, Video Analytics Software and Mobile Viewing Apps.
- **Software testing tools examples are** Net Cam Studio, Security Eye, Perfect Surveillance**,** iVideon, Xeoma, Genius Vision, Deskshare IP Camera Viewer, Anycam, ContaCam, Agent DVR.



Application of learning 5.2

MAKS Company Ltd located in Northern Province, Gakenke District, recruit a CCTV system installer team to facilitate its works. You are recruited in this team and you are assigned you to install CCTV cameras in the building located in MUGUNGA Centre.

Task:

Before making installation, you are requested to select hardware and software testing tools that will facilitate the installation process.

Indicative content 5.3: Performing connectivity testing.

Duration: 2 hrs

Theoretical Activity 5.3.1: Identify Physical and Logical connectivity testing to be done.

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of different types of switches.

i. What is physical connectivity?
ii. What is logical connectivity?

2: Provide the answers for the asked questions and write them on papers.
3: Present the findings/answers to the whole class
4: For more clarification, read the key readings 5.3.1
5: In addition, ask questions where necessary.

Key readings 5.3.1.:

- ✓ **Identification of Physical and Logical connectivity testing to be done.**
  Testing CCTV camera is to verify proper functioning of the equipment and system after installation.
- ✓ **Physical connectivity.**
  Physical connectivity in a CCTV (Closed-Circuit Television) camera system refers to the physical interconnection of various components within the surveillance infrastructure. It involves the deployment of cables, connectors, and other hardware to establish reliable communication pathways between CCTV cameras, recording devices, power sources, and other system components. Physical connectivity is essential for transmitting video signals, power, and data between different elements of the CCTV system.
- ✓ **Key aspects of physical connectivity in a CCTV camera system:**

1. **Camera Installation:**

- CCTV cameras are installed in strategic locations to monitor specific areas or premises. During installation, cameras are physically mounted onto walls, ceilings, poles, or other structures using mounting brackets or fixtures. Proper positioning and alignment of cameras are critical to ensuring optimal surveillance coverage.

2. **Cabling Infrastructure:**

- Cables are used to establish connections between CCTV cameras, recording devices (e.g., DVR, NVR), power supplies, and other system components. Common types of cables used in CCTV installations include coaxial cables (for analog cameras), Ethernet cables (for IP cameras), and power cables.
- The cabling infrastructure may include cables for video transmission, power supply (e.g., DC power cables), and network connectivity (e.g., Ethernet cables for IP-based systems). Cables are routed and terminated according to the system design and installation requirements.

3. **Connectors and Terminations:**

- Connectors are used to terminate cables and establish connections between different components of the CCTV system. Common types of connectors used in CCTV installations include BNC connectors (for analog cameras), RJ45 connectors (for network connections), and power connectors (e.g., DC power plugs).
- Proper termination of cables and connectors is crucial to ensure reliable signal transmission and minimize signal loss or interference. Connectors should be securely attached and properly insulated to prevent damage and maintain signal integrity.

4. **Power Distribution:**

- Power distribution involves supplying electrical power to CCTV cameras, recording devices, and other system components. Power sources may include AC (mains power) or DC (low-voltage power) supplies, depending on the voltage requirements of the equipment.
- Power distribution units (PDUs), power supplies, and power over Ethernet (PoE) switches may be used to distribute power to multiple cameras and devices. Power cables are connected to cameras and recording devices to provide electrical power for operation.

5. **Grounding and Surge Protection:**

- Grounding and surge protection measures are implemented to protect CCTV equipment from electrical surges, voltage fluctuations, and electromagnetic interference. Grounding rods, surge protectors, and shielded cables may be used to minimize the risk of equipment damage and ensure system reliability.
- Proper grounding ensures that electrical currents are safely discharged to the ground, reducing the risk of electrical hazards and equipment malfunctions.

6. **Physical Security:**

- Physical security measures are implemented to protect CCTV equipment from tampering, vandalism, and unauthorized access. This may include installing protective enclosures, housings, and tamper-resistant mounting hardware to secure cameras and recording devices.

- Access control measures such as locks, security cages, and restricted access to equipment rooms help prevent unauthorized tampering or theft of CCTV equipment.

✓ **General physical connectivity check to be done:**

- Check all the cables are identified.
- Check the continuity of the cables.
- CCTV location should be properly shown in Drawings.
- The work area shall be inspected for completion and safe to start the testing and commissioning work.
- Check the field devices cable identification and termination as per the manufacturer's instructions and also check the right power supply provided.
- Ensure LAN cable (CAT6A) is connected between the CCTV to the Network Switch.
- Now connect the workstation PC to the LAN
- Ensure permanent power supply must be provided to CCTV switches with back up by UPS as indicated in the specifications.

- **Logical connectivity**

The system is called "closed-circuit" because the cameras, monitors and/or video recorders communicate across a proprietary coaxial cable run or wireless.

CCTV works by the camera or cameras taking a constant sequence of images that are then transmitted by cable or wirelessly (depending on the chosen system type) to the recording device and then on to the display monitor, which enables an individual to see the sequence of images as video footage.

Logical connectivity in a CCTV (Closed-Circuit Television) camera system refers to the establishment of logical or virtual connections between different components of the surveillance infrastructure. Unlike physical connectivity, which involves the physical interconnection of devices using cables and connectors, logical connectivity focuses on the configuration of network settings, protocols, and software interfaces to facilitate data transmission, communication, and control within the CCTV system.

✓ **Key aspects of logical connectivity in a CCTV camera system:**

1. **Network Configuration:**

- CCTV cameras, recording devices (e.g., DVRs, NVRs), and other components may be connected to a local area network (LAN) or a wide area network (WAN) to facilitate data transmission and remote access. Network configuration involves assigning IP addresses, subnet masks, gateway addresses, and DNS settings to each device to enable communication over the network.
- Network configuration settings are typically configured using management software, web interfaces, or command-line interfaces (CLI) provided by the

manufacturer. Proper network configuration ensures that devices can communicate with each other and with remote monitoring stations effectively.

2. **Protocol Configuration:**

- Network protocols such as TCP/IP, UDP, HTTP, RTSP, and ONVIF are used to facilitate data transmission, video streaming, and device control within the CCTV system. Protocol configuration involves configuring protocol settings, ports, and communication parameters to ensure compatibility and interoperability between different devices and software components.
- Configuration of network protocols may include enabling protocols for video streaming (e.g., RTSP for live video feeds), remote access (e.g., HTTP or HTTPS for web-based access), and device discovery (e.g., ONVIF for interoperability with third-party devices).

3. **Remote Access and Monitoring:**

- Remote access functionality allows authorized users to access live video feeds, view recorded footage, and manage CCTV cameras and recording devices from remote locations using desktop computers, laptops, smartphones, or tablets. Logical connectivity enables remote access by configuring network settings, security protocols, and authentication mechanisms.
- Virtual private network (VPN) connections, secure sockets layer (SSL) encryption, and two-factor authentication (2FA) may be used to secure remote access and protect against unauthorized access to the CCTV system.

4. **Integration with Third-Party Systems:**

- Logical connectivity enables integration with third-party systems and software applications such as access control systems, alarm systems, and video management software (VMS). Integration may involve configuring APIs (application programming interfaces), SDKs (software development kits), or middleware to exchange data and events between different systems.
- Integration allows for centralized management, event correlation, and automated response actions based on predefined rules and policies across multiple systems.

5. **Data Transmission and Video Streaming:**

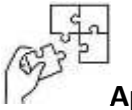
- Logical connectivity facilitates data transmission and video streaming between CCTV cameras, recording devices, and monitoring stations. Video streams are encoded, transmitted, and decoded using network protocols such as RTSP, H.264, H.265, or MJPEG. Bandwidth management, quality of service (QoS), and network optimization techniques may be employed to ensure smooth and reliable video transmission over the network.

6. **Event Notification and Alerting:**

- Logical connectivity enables the configuration of event notification and alerting mechanisms to notify users of critical events or alarms detected by CCTV cameras or sensors. Alerts may be sent via email, SMS, or push notifications to notify users

of security breaches, motion detection events, camera tampering, or system errors.

- Event notification settings are configured using management software or monitoring platforms, allowing users to define alert triggers, recipient lists, and escalation procedures for handling security incidents.



Application of learning 5.3

A new CCTV system has been installed in a large shopping mall, and you, as the technician, are tasked with performing connectivity testing to ensure the system operates correctly. You are asked to:

- Explain the physical connectivity testing to be performed.
- Explain the logical connectivity testing to be performed.



Practical Activity 5.3.2: Perform physical and logical connectivity testing.



Task:

1: Referring to previous activities **(5.3.1**) you are requested to perform the given task. The task should be done **individually**.

As a CCTV Camera installer, you are requested to test the physical connectivity of the installed CCTV camera system.

2: Draw the connectivity diagram to be used to perform the given tasks,

3: Referring to the diagram in task2, perform the given tasks

4: Present your work to the trainer and whole class.

5: Read key reading 5.3.2 and ask clarification where necessary

6: Perform the task provided in application of learning 5.3



Key readings 5.3.2

- **CCTV System connectivity testing**

  Testing CCTV camera connectivity is to verify proper functioning of the equipment and system after installation.

- **Physical connectivity**

  Testing the physical connectivity of CCTV (Closed-Circuit Television) cameras

ensures that all components are properly installed, connected, and functioning as expected. The procedures for conducting physical connectivity testing in a CCTV camera system:

1. **Visual Inspection**:

- Start by visually inspecting each CCTV camera, recording device (e.g., DVR, NVR), power supply, and cabling to ensure they are installed correctly and securely mounted. Check for any signs of damage, loose connections, or improper positioning.

2. **Cable Continuity Testing**:

- Use a cable tester or multimeter to check the continuity of all cables used in the CCTV system, including coaxial cables (for analog cameras) and Ethernet cables (for IP cameras). Verify that there are no open circuits, short circuits, or cable faults that could affect signal transmission.
- Test each cable individually by connecting one end of the cable to the tester and the other end to the corresponding camera or recording device. Ensure that the tester indicates a successful continuity test for each cable.

3. **Power Supply Testing**:

- Test the power supply units (PSUs) used to power CCTV cameras, recording devices, and other components to ensure they are providing the correct voltage and current. Use a multimeter to measure the output voltage of the power supplies and verify that it matches the rated voltage specified by the manufacturer.
- Check for any fluctuations or irregularities in the power supply output that could indicate a faulty power supply or inadequate power delivery to the devices.

4. **Connector Inspection**:

- Inspect all connectors used in the CCTV system, including BNC connectors (for analog cameras), RJ45 connectors (for network connections), and power connectors. Ensure that connectors are securely attached, properly terminated, and free from damage or corrosion.
- Check for any bent pins, loose connections, or signs of wear that could affect signal transmission or power delivery. Replace or repair connectors as necessary to maintain reliable connectivity.

5. **Camera Functionality Testing**:

- Test the functionality of each CCTV camera by connecting it to a monitor or recording device and verifying that the camera is capturing video footage correctly. Check for any image distortion, color issues, or focusing problems that could indicate camera malfunctions.
- Activate any additional features or capabilities of the cameras, such as infrared (IR) night vision, motion detection, or PTZ (Pan-Tilt-Zoom) functionality, and ensure that they are working as expected.

6. **Recording Device Testing**:

- Test the recording devices (e.g., DVR, NVR) by connecting them to the CCTV cameras and verifying that they are recording video footage properly. Check the recording settings, storage capacity, and playback functionality of the devices to ensure they are functioning correctly.
- Playback recorded footage from each camera to verify that the recording device is capturing and storing video footage accurately without any gaps or errors.

7. **Signal Quality Testing**:

- Use a CCTV signal tester or monitor to check the quality of video signals transmitted by the cameras. Verify that the video signals are clear, stable, and free from interference or noise.
- Adjust camera settings, cable routing, or positioning as needed to optimize signal quality and minimize signal degradation over long cable runs.

8. **Weatherproofing Inspection**:

- If CCTV cameras are installed outdoors or in harsh environments, inspect weatherproofing seals, gaskets, and enclosures to ensure they are intact and providing adequate protection against moisture, dust, and debris.
- Check for any signs of water ingress, condensation buildup, or physical damage to the camera housings that could compromise their weatherproofing capabilities.

- **Logical connectivity testing**

  Logical connectivity in a CCTV camera system testing refers to establishing logical or virtual connections between different components of the surveillance infrastructure. Unlike physical connectivity, which involves the physical interconnection of devices using cables and connectors, logical connectivity focuses on the configuration of network settings, protocols, and software interfaces to facilitate data transmission, communication, and control within the CCTV system.

  Structured approach to perform this testing:

  1. Preparation:

✓ **Documentation:** Gather network diagrams, IP addresses, device configurations, and user manuals.

✓ **Tools:** Ensure access to network testing tools such as a laptop with network diagnostic software and ping tools.

2. Network Device Verification:

✓ **Switches and Routers:** Verify configurations, ensure VLANs (if used) are properly set up, and routing rules are correctly implemented.

✓ **Network Cabling:** Check that all network cables are properly connected and undamaged.

3. IP Address Configuration:

✓ **Camera Verification:** Ensure each IP camera has a unique static IP address within the correct subnet.

- ✓ **Camera Setup:** Ensure analog cameras are properly connected to DVRs using coaxial cables and that power supplies are correctly connected.
- ✓ **DVR/NVR Configuration:** Verify DVRs/NVRs have static IP addresses and are configured to communicate with the cameras.
- ✓ **DVR Configuration:** Ensure DVRs are configured to receive and record footage from the connected analog cameras.

4. Connectivity Testing:

- ✓ **Ping Test:** Ping each camera from a computer on the same network to ensure they respond. Ping the DVR/NVR to verify its connectivity.
- ✓ **Traceroute Test:** Run a traceroute to each device to check for unexpected hops or delays in the network path.

5. Software Integration:

- ✓ **Monitoring Software Configuration:** Confirm all cameras and DVR/NVRs are correctly added to the monitoring software.
- ✓ **Live View Test:** Check the live feed from each camera in the monitoring software to ensure it is visible and there is no significant lag.
- ✓ **Recording Verification:** Ensure the DVR/NVR is receiving and recording footage from each camera.

6. Data Transmission Check:

- ✓ **Bandwidth Monitoring:** Use network monitoring tools to check the bandwidth usage of the CCTV system, ensuring the network can handle the data load without significant packet loss or delay.
- ✓ **Quality of Service (QoS) Settings:** Verify QoS settings are properly configured to prioritize CCTV traffic if needed.

7. Security Verification:

- ✓ **Firewall Settings:** Check firewall rules to ensure they allow necessary traffic for CCTV operation while blocking unauthorized access.
- ✓ **Access Control:** Verify only authorized devices and users have access to the CCTV system.

8. Error and Alert Testing:

- ✓ **Simulate Disconnection:** Temporarily disconnect a camera and check if the monitoring software detects and reports the issue.
- ✓ **Alarm Verification:** Test alarm or alert features to ensure they are triggered appropriately and notifications are sent as expected.

9. Redundancy and Backup:

- ✓ **Failover Testing:** If the system has redundant paths or failover mechanisms, test them to ensure they activate correctly in case of a primary path failure.
- ✓ **Backup Verification:** Ensure backup procedures for configurations and footage are functioning correctly.

10. Documentation and Reporting:

- ✓ **Record Findings:** Document the results of all tests, noting any issues and steps taken to resolve them.
- ✓ **Report:** Prepare a comprehensive report summarizing the connectivity status and any recommendations for improvement.

  By following these steps, you can ensure that the logical connectivity of your CCTV system is robust, reliable, and ready for effective surveillance operations.



Points to Remember

- **The materials and components to be tested for physical connectivity of CCTV Camera System include** Check all the cables, Check the continuity of the cables, Check the field devices cable termination, check power supply, Ensure LAN cable (CAT6A) is connected between the CCTV to the Network Switch, etc…
- **Things to be tested for logical connectivity of CCTV Camera System are** Network Configuration, Protocol Configuration, Remote Access and Monitoring, Integration with Third-Party Systems, Data Transmission and Video Streaming, Event Notification and Alerting.
- Steps Perform physical connectivity testing.
- Steps to Perform Logical Connectivity Testing.



Application of learning 5.3

XYZ School system want to install the CCTV camera system in their workshops for better management of equipment and consumables. As one of the system installer recruited by this school, you are asked to install CCTV cameras in the workshops of the school and you will also need**:**

- Test CCTV system installation logical connectivity and provide the results.
- Test CCTV system installation logical connectivity and provide the results.

Indicative content 5.4: Performing Functional testing

Duration: 2 hrs

Theoretical Activity 5.4.1: Provide the hardware and software functional testing to be done.

Tasks:

1: In small groups, you are requested to answer the following questions related to the description of Provide the hardware and software functional testing to be done.

i. What are the hardware functional testing processes to be done?
ii. What are the software functional testing processes to be done

2: Provide the answers for the asked questions and write them on papers.
3: Present the findings/answers to the whole class
4: For more clarification, read the key readings 5.4.1
5: In addition, ask questions where necessary.

Key readings 5.4.1.:

- **Hardware functional testing to be done.**

  Hardware functional testing in CCTV (Closed-Circuit Television) camera systems involves verifying that all hardware components, including cameras, recording devices, power supplies, and accessories, are functioning correctly and meeting performance requirements.

  The key processes involved in hardware functional testing for CCTV cameras:

1. **Power On/Off Testing:** Verify that all CCTV cameras, recording devices (e.g., DVRs, NVRs), and power supplies are capable of powering on/off properly. Test the power switches or buttons on each device to ensure they function as expected and can reliably control the power state of the equipment.
2. **Camera Initialization:** Power on each CCTV camera and observe its startup sequence. Verify that the camera initializes correctly without any errors or abnormal behavior. Check for any indicator lights or LEDs on the camera that indicate power status, network connectivity, and camera activity.
3. **Image Capture and Quality Testing:** Test the image capture capabilities of each CCTV camera by observing the quality of the video footage captured by the camera. Ensure that the camera produces clear, sharp images with accurate colors and minimal distortion. Verify that the camera's field of view covers the intended monitoring area effectively.
4. **Day/Night Mode Switching:** If CCTV cameras feature day/night mode functionality with infrared (IR) cut filters, test the automatic switching between day and night

modes. Verify that the camera switches to night mode and activates IR illumination when ambient light levels decrease below a certain threshold, and vice versa.

5. **Pan-Tilt-Zoom (PTZ) Functionality Testing:** If CCTV cameras are equipped with PTZ functionality, test the pan, tilt, and zoom capabilities of the cameras. Use the camera's PTZ controls or management software to adjust the camera's position and zoom level and verify that the camera responds accurately to commands.
6. **Motion Detection Testing:** If CCTV cameras support motion detection, configure the motion detection settings and test the functionality by moving objects within the camera's field of view. Verify that the camera detects motion events accurately and triggers recording or alerts as configured.
7. **Recording and Playback Testing:** Test the recording and playback capabilities of the recording devices (e.g., DVRs, NVRs) by configuring recording schedules, motion-triggered recording, or continuous recording modes. Verify that recorded footage is stored correctly and can be played back smoothly without any gaps or errors.
8. **Remote Access and Control Testing:** Test remote access and control functionality by accessing the CCTV system from remote locations using desktop computers, laptops, smartphones, or tablets. Verify that users can view live video feeds, access recorded footage, and control camera settings remotely via web interfaces or mobile apps.
9. **Network Connectivity Testing:** Test the network connectivity of CCTV cameras and recording devices by verifying that they can communicate with each other and with remote monitoring stations over the network. Use network diagnostic tools to check IP addresses, ping cameras, and monitor network traffic for any issues.
10. **Alarm and Event Trigger Testing:** Test alarm and event trigger functionality by simulating various events such as motion detection, camera tampering, or input/output (I/O) triggers. Verify that alarms are triggered correctly, and event notifications are sent as expected to alert users of security incidents or system events.
11. **Environmental Testing:** Conduct environmental testing to ensure that CCTV cameras and recording devices can withstand operating conditions such as temperature extremes, humidity, dust, and vibration. Test cameras and recording devices in different environmental conditions to verify their durability and reliability.
12. **Documentation and Reporting:** Document the results of hardware functional testing, including any observations, issues, or corrective actions taken during the testing process. Prepare a detailed report summarizing the testing procedures, findings, and recommendations for follow-up actions.

    Provide the testing report to stakeholders, system administrators, or clients for review and approval before finalizing the CCTV installation.

- **Software functional testing to be done.**

  Software functional testing in CCTV (Closed-Circuit Television) camera systems involves verifying that all software components, including management software, firmware, and applications, are functioning correctly and meeting the specified requirements.

  The key processes involved in software functional testing for CCTV cameras:

1. **User Interface Testing**: Test the user interface (UI) of CCTV management software, web interfaces, and mobile apps to ensure they are intuitive, user-friendly, and easy to navigate. Verify that users can access all features and functionalities efficiently without encountering usability issues or navigation problems.
2. **Configuration Testing**: Test the configuration settings of CCTV cameras, recording devices (e.g., DVRs, NVRs), and management software to ensure they can be configured correctly according to user requirements. Verify that users can customize camera settings, recording schedules, motion detection parameters, and network settings as needed.
3. **Live View Testing**: Test the live view functionality of CCTV management software and web interfaces to ensure users can view live video feeds from CCTV cameras in real-time. Verify that live video streams are displayed correctly, with minimal latency, and that users can switch between multiple camera views efficiently.
4. **Recording and Playback Testing**: Test the recording and playback functionalities of CCTV recording devices (e.g., DVRs, NVRs) and management software to ensure they can record video footage accurately and play back recorded footage smoothly. Verify that recorded footage is stored securely and can be searched, sorted, and exported as needed.
5. **Motion Detection and Alarm Testing**: Test the motion detection and alarm functionalities of CCTV cameras and recording devices by simulating motion events or triggering alarms manually. Verify that motion detection events are detected accurately, and alarms are triggered promptly to alert users of security incidents.
6. **Remote Access Testing**: Test remote access functionality by accessing the CCTV system from remote locations using desktop computers, laptops, smartphones, or tablets. Verify that users can log in securely, view live video feeds, access recorded footage, and control camera settings remotely via web interfaces or mobile apps.
7. **Event Notification and Alert Testing**: Test event notification and alerting mechanisms by simulating various events such as motion detection, camera tampering, or system errors. Verify that users receive timely notifications and alerts via email, SMS, or push notifications to alert them of security incidents or system events.
8. **Integration Testing**: Test integration with third-party systems and software applications such as access control systems, alarm systems, and video

management software (VMS). Verify that data exchange, event correlation, and automated response actions work seamlessly between different systems.

9. **Network Connectivity Testing**: Test network connectivity by verifying that CCTV cameras, recording devices, and management software can communicate with each other and with remote monitoring stations over the network. Use network diagnostic tools to check IP addresses, ping cameras, and monitor network traffic for any issues.
10. **Security Testing**: Test the security features and protocols implemented in CCTV management software and devices to ensure they protect against unauthorized access, data breaches, and cyber threats. Verify that encryption, authentication, access controls, and audit trails are configured correctly and comply with security standards.
11. **Performance Testing**: Test the performance of CCTV management software and recording devices under normal and peak load conditions to ensure they can handle the expected number of cameras, users, and video streams without degradation in performance. Verify that response times, video playback speeds, and system responsiveness meet performance requirements.
12. **Compatibility Testing**: Test compatibility with different operating systems, web browsers, mobile devices, and hardware configurations to ensure that CCTV management software and applications work correctly across various platforms and environments.
13. **Documentation and Reporting**: Document the results of software functional testing, including any observations, issues, or corrective actions taken during the testing process. Prepare a detailed report summarizing the testing procedures, findings, and recommendations for follow-up actions.



Practical Activity 5.4.2: Perform hardware and software functional testing.



Task:

1: Referring to previous activities (5.4.2) you are requested to perform the given task. The task should be done **individually**.

As a CCTV system installer who perform logical and physical connectivity testing, you are asked to perform all hardware and software functional testing of the given CCTV installation and provide the report of the testing.

2: List out procedures of functional testing to be used to perform the given tasks

3: Referring to procedures provided on task 2, Perform the given tasks

4: Present your work to the trainer and whole class.

5: Read key reading of practical activity **5.4** for more clarifications.



Key readings 5.4.1

- **Process of hardware functional testing:**
  - ✓ Ensure all installation works are complete as per approved method statement for installation of CCTV system. Also ensure that IT data network is ready and commissioned.
  - ✓ Make sure all cables/wires for CCTV system has been tested for continuity.
  - ✓ First Step involved in pre-commissioning is to establish LAN network between camera and commissioning laptop.
  - ✓ Connect IP camera network port to the portable POE network switch from the patch panel in respective communication room. Plug the commissioning laptop to the other port of the portable network switch.
  - ✓ Set the IP address of commissioning laptop to the required range.
  - ✓ From the commissioning laptop run "Configuration Manager" application. Then press search.
  - ✓ Fixed dome IP camera with default IP address will be discovered in IP Installer windows.
  - ✓ Click "Manual Set" button on the IP Installer application and set the IP address of the camera as per your own IT network plan.
  - ✓ Now change the IP address of commissioning laptop same as of IT network plan and again discover the camera through IP Installer application.
  - ✓ Once the camera is discovered double click on it, web browser for the camera will open "Live Monitoring View" screen will open. Go to setup then camera setup then focus setup.
  - ✓ Adjust zoom and focus for the camera lens to the desired field of view
  - ✓ Go to system setup then time and date setup.
  - ✓ Check Synchronize with the time of PC define IP address of the related communication server.
  - ✓ Pre-commissioning of camera is complete.
- **Process of software functional testing:**

  Dome IP Camera
  - Open the CCTV application on the desktop client.
  - Login - using the user name and password.
  - Create the site tree e.g. levels 02, level 3 or floor 1 etc…
  - Go to "System Tool Box" then "All devices" then "CCTV System".
  - From CCTV System go to IP camera.
  - Define Camera name, CG, Site name, Device Type, Time zone, IP address.
  - Now click on the NVR configuration and define IP address of NVR, Video

data backup path, Retention period. Click ok.

- IP camera is created with its NVR association.
- Go to the device browser and right click on the IP camera then press discover. H264 and MPEG profile of the camera will be discovered.
- Now right click on IP camera and press edit. Select live video profile as H264 and NVR profile as MPEG.
- Open the Video Viewer from software i.e. Fusion toolbar. Drag the IP camera from device
- browser and drop it into any pane. Live video of camera will be displayed.

Analogue with encoder

• Open the CCTV application on the desktop client.
• Login using the user name and password
• Create the site tree i.e. floor 01,02, or level 02, level 03
• Go to "System Tool Box" then "All devices" then "CCTV System".
• From CCTV System, go to IP encoder.
• Define Encoder name, CG, Site name, Device Type Time zone, IP address. Now click on the NVR configuration and define IP address of NVR. Video data backup path, Retention period. Click ok
• Video encoder is created with its NVR association.
• Go to the device browser and right click on the encoder then press add analogue camera. Analogue camera will be attached to this encoder.
• Now right click on IP encoder and press edit. Select live video profile as H264 and NVR profile as MPEG.

Open the Video Viewer from Fusion toolbar. Drag the analogue camera from device browser and drop it into any pane. Live video of camera will be displayed



Points to Remember

- Hardware Functional Testing Processes
  1. Power-on Test
  2. Camera Functionality
  3. Connectivity Test
  4. Recording Device Check
  5. Video Feed Quality
  6. Peripheral Devices Test
- Software Functional Testing Processes
  1. Login and Authentication
  2. Camera Configuration
  3. Recording and Playback
  4. Alerts and Notifications

5. System Logs
6. Backup and Export

- Steps to Perform Hardware functional testing
  1. Power On Devices
  2. Test Camera Functionality
  3. Verify Physical Connections
  4. Check Recording Devices
  5. Assess Video Quality
  6. Test Peripheral Devices
  7. Simulate Power Failure
- Steps to Perform Software Functional Testing
  1. Login and User Access
  2. Configure Camera Settings
  3. Test Recording and Playback
  4. Test Alerts and Notifications
  5. Review System Logs
  6. Backup and Export Data



Application of learning 5.4

MARXT Company Ltd located in Northern Province, Burera District, recruit a CCTV system installer team to facilitate its works. You are recruited in this team and you are assigned you to install CCTV cameras in the building located in KINYABABA Centre.

Task:

- ✔ Test CCTV Camera hardware functionality and generate testing report.
- ✔ Test CCTV Camera software functionality and generate testing report.

Indicative content 5.5: Testing performance

Duration: 2 hrs

Theoretical Activity 5.5.1: Test the CCTV Camera System Performance

Tasks:

1: In small groups, you are requested to answer the following questions related to the testing of CCTV System system performance

i. What is CCTV Visual quality?
ii. What is CCTV Latency?
iii. What is CCTV Audio quality?
iv. What is CCTV NVR/DVR throughput?

2: Provide the answers for the asked questions and write them on papers.
3: Present the findings/answers to the whole class
4: For more clarification, read the key readings 5.5.1
5: In addition, ask questions where necessary.

Key readings 5.5.1.:

- **CCTV Visual quality.**

  CCTV visual quality refers to the clarity, detail, color accuracy, and overall fidelity of the video footage captured by Closed-Circuit Television (CCTV) cameras. It encompasses various aspects of image quality that determine how effectively the camera can capture, record, and reproduce visual information for surveillance and security monitoring purposes. Visual quality is a crucial factor in ensuring that CCTV systems can accurately capture and analyze events, objects, or individuals within the monitored area.

  Key elements of CCTV visual quality include:

1. **Resolution:** The resolution of CCTV cameras determines the level of detail captured in the video footage. Higher resolutions result in sharper, more detailed images, making it easier to identify objects, individuals, or events within the scene.
2. **Color Accuracy:** Accurate color reproduction is essential for distinguishing between different objects, identifying individuals, and assessing the environment accurately. Cameras with good color accuracy can reproduce colors faithfully, helping users make informed decisions based on the visual information provided.
3. **Low-Light Performance:** The ability of CCTV cameras to capture clear and detailed images in low-light conditions is critical for surveillance applications, especially

during nighttime or in poorly lit environments. Cameras with advanced low-light performance features can produce better-quality images in challenging lighting conditions.

4. **Dynamic Range:** Dynamic range refers to the camera's ability to capture details in both bright and dark areas of the scene simultaneously. Cameras with a wide dynamic range (WDR) or high dynamic range (HDR) capabilities can produce well-balanced images with accurate color reproduction and reduced overexposure or underexposure in high-contrast scenes.
5. **Frame Rate**: The frame rate of CCTV cameras determines how many frames per second (fps) of video footage are captured and displayed. Higher frame rates result in smoother motion and more fluid video playback, which is important for capturing fast-moving objects or events accurately.
6. **Compression Quality:** Video compression algorithms are used to reduce the size of video files for storage and transmission. The choice of compression algorithm can impact the visual quality of the footage, with more efficient algorithms preserving more detail and image clarity.
7. **Image Enhancements:** Some CCTV cameras offer built-in image enhancement features such as digital noise reduction (DNR), image stabilization, and image sharpening to improve overall visual quality and reduce artifacts or distortions.
8. **Environmental Adaptability:** CCTV cameras should be able to adapt to varying environmental conditions such as lighting changes, weather conditions, and scene complexity. Properly configuring camera settings and adjusting parameters such as exposure, white balance, and sensitivity can help optimize image quality under different conditions.

- **CCTV Latency**

  CCTV latency refers to the delay or lag between the time an event occurs and when it is displayed on the monitor or recorded by the CCTV camera system.

  It is the time it takes for the video signal to be captured by the camera, processed, transmitted, and displayed on the monitor or recording device. Latency can occur at various stages of the CCTV system, including the camera, transmission network, and display device, and it can impact the effectiveness of surveillance and security monitoring.

  CCTV latency refers to the delay or lag in the transmission of video signals from CCTV cameras to monitoring or recording devices.

  Excessive latency can affect the real-time nature of surveillance and monitoring applications and may impact the effectiveness of security measures. Reducing latency is crucial for ensuring prompt response to events and maintaining the integrity of CCTV systems.

  Key points to understand about CCTV latency:

1. **Causes of Latency:** Latency in CCTV systems can be caused by several factors, including the processing time of the camera sensor, encoding and compression of the video signal, transmission delays over the network, decoding and playback processing, and display latency on the monitor or screen.
2. **Camera Latency:** The camera itself can introduce latency due to the time it takes for the image sensor to capture the video frame and process it before transmitting it to the recording device or monitor. Higher-resolution cameras or cameras with advanced features may have longer processing times, leading to increased latency.
3. **Transmission Latency:** Latency can occur during the transmission of video signals over the network or cables connecting the cameras to the recording device or monitor. Network congestion, bandwidth limitations, and packet delays can all contribute to transmission latency, especially in large-scale CCTV systems or systems with multiple cameras.
4. **Encoding and Decoding Latency:** Video encoding and compression algorithms used to reduce the size of video files for storage and transmission can introduce latency as the video signal is processed and encoded by the camera and decoded by the recording device or monitor. The complexity of the encoding algorithm and the processing power of the devices can affect encoding and decoding latency.
5. **Display Latency:** The monitor or display device used to view the CCTV footage can also introduce latency due to processing delays, input lag, or refresh rates. High-resolution displays or displays with advanced image processing features may have longer response times, leading to increased display latency.
6. **Impact on Surveillance:** Latency in CCTV systems can impact the effectiveness of surveillance and security monitoring by delaying the display of critical events or activities. High latency can result in delayed response times, making it difficult for operators to react quickly to security incidents or emergencies.
7. **Reducing Latency:** To minimize latency in CCTV systems, it is essential to use cameras, recording devices, and network infrastructure with low latency specifications. Optimizing camera settings, using high-speed network connections, and selecting display devices with low response times can help reduce overall system latency.

- **CCTV Audio quality**

CCTV audio quality refers to the clarity, fidelity, and overall sound reproduction capabilities of the audio captured by CCTV (Closed-Circuit Television) camera systems. While the primary focus of CCTV systems is typically on video surveillance, the inclusion of audio can provide valuable additional information for security monitoring purposes.

Important key aspects of CCTV audio quality:

1. **Microphone Quality:** The quality of the microphone used in the CCTV camera significantly impacts audio quality. High-quality microphones with sensitive and

accurate audio capture capabilities can produce clear and crisp sound recordings, making it easier to understand conversations or detect sounds within the monitored area.

2. **Frequency Response:** The frequency response of the microphone determines the range of audio frequencies it can capture. A wider frequency response range allows the microphone to capture a broader spectrum of sounds, including low-frequency bass tones and high-frequency treble tones, resulting in more natural and realistic audio reproduction.
3. **Noise Reduction:** Effective noise reduction techniques can help minimize background noise and unwanted interference, improving the clarity and intelligibility of recorded audio. Noise reduction algorithms or hardware filters can help suppress background noise such as wind, traffic, or machinery, allowing for clearer audio recordings.
4. **Echo Cancellation:** In environments with reflective surfaces or reverberant acoustics, echo cancellation techniques can help reduce echo and reverberation in audio recordings. Echo cancellation algorithms analyze the audio signal and remove unwanted echo artifacts, resulting in cleaner and more intelligible audio playback.
5. **Bitrate and Compression:** The bitrate and compression settings used for audio recording can affect audio quality. Higher bitrate audio recordings preserve more detail and fidelity, while lower bitrate recordings may result in reduced audio quality and potential loss of detail. However, higher bitrates also require more storage space.
6. **Synchronization with Video:** In CCTV systems where audio is recorded alongside video, synchronization between the audio and video streams is essential to maintain coherence between the two. Proper synchronization ensures that audio events align accurately with corresponding video footage, allowing for more effective analysis and incident review.
7. **Legal and Privacy Considerations:** When implementing audio recording in CCTV systems, it is essential to comply with legal regulations and privacy standards governing the use of audio surveillance. In some jurisdictions, audio recording may be subject to stricter regulations than video recording, requiring explicit consent or notification from individuals being recorded.
8. **Maintenance and Calibration:** Regular maintenance and calibration of CCTV audio equipment are crucial for ensuring consistent audio quality over time. Periodic checks of microphone sensitivity, audio levels, and signal-to-noise ratios can help identify and address any issues that may affect audio quality.

- **CCTV NVR/DVR throughput.**

  CCTV NVR/DVR throughput refers to the maximum amount of data that the network video recorder (NVR) or digital video recorder (DVR) can process, store,

or transmit within a given period, typically measured in bits per second (bps), megabits per second (Mbps), or gigabits per second (Gbps). Throughput is a critical performance metric for NVRs and DVRs as it determines their ability to handle the video streams from connected cameras, record them to storage, and transmit them for viewing or remote access.

Important key aspects of CCTV NVR/DVR throughput:

1. **Recording Throughput:** Recording throughput refers to the maximum data rate at which the NVR or DVR can record video streams from connected cameras to its internal storage or external storage devices such as hard disk drives (HDDs) or solid-state drives (SSDs). Higher recording throughput allows the device to capture and store more video data simultaneously without dropping frames or losing video quality.
2. **Playback Throughput:** Playback throughput refers to the maximum data rate at which the NVR or DVR can retrieve and play back recorded video footage from its storage devices. Higher playback throughput enables smooth and uninterrupted playback of video recordings, even when accessing multiple video streams concurrently or fast-forwarding/rewinding through footage.
3. **Live Viewing Throughput:** Live viewing throughput refers to the maximum data rate at which the NVR or DVR can transmit live video streams from connected cameras to client viewing devices such as monitors, computers, or mobile devices over a network connection. Higher live viewing throughput ensures low latency and high-quality live video streaming, allowing users to monitor surveillance feeds in real-time.
4. **Remote Access Throughput:** Remote access throughput refers to the maximum data rate at which the NVR or DVR can transmit video streams to remote client devices over the internet or a network connection. Higher remote access throughput enables smooth and responsive remote viewing of live or recorded video footage, even over bandwidth-constrained networks or mobile connections.
5. **Network Bandwidth Considerations:** The throughput capabilities of the NVR or DVR must align with the available network bandwidth to ensure optimal performance. Insufficient throughput relative to network bandwidth may result in video buffering, stuttering, or degraded image quality during live viewing or remote access.
6. **Compression and Encoding:** Video compression and encoding technologies used by the NVR or DVR can impact throughput requirements. More efficient compression algorithms (e.g., H.264, H.265) reduce the amount of data transmitted and stored, allowing for higher throughput while maintaining video quality.
7. **Scalability and Capacity:** The NVR or DVR's throughput should be scalable to accommodate future expansion and increased camera counts. Higher-throughput

devices with ample storage capacity and processing power can support larger CCTV systems with a greater number of cameras and higher-resolution video streams.



Practical Activity 5.5.2: Test the CCTV Camera System Performance



Task:

1: Referring to previous activities **(5.5.1)** you are requested to perform the given task. The task should be done **individually**.

As a CCTV camera system installer, you are asked to test the CCTV NVR/DVR throughput, latency, audio/video quality and provide testing reports of the given CCTV installation.

2: List out the procedures to perform the given tasks,

3: Referring to the procedures provided in task2, perform the given tasks

4: Present your work to the trainer and whole class.

5: Read key reading 5.5.2 and ask clarification where necessary

6: Perform the task provided in application of learning 5.5



Key readings 5.5.2

- **Steps of testing NVR/DVR throughput**

  Testing the throughput of a Network Video Recorder (NVR) or Digital Video Recorder (DVR) involves assessing its ability to handle and process video data streams, both during recording and playback, as well as during live viewing and remote access.

  The steps to test the throughput of an NVR/DVR:

1. **Determine Testing Methodology:** Decide on the specific aspects of throughput to test, such as recording, playback, live viewing, and remote access. Determine the metrics to measure, including data rate, latency, and video quality.
2. **Select Test Equipment:** Choose appropriate test equipment, including network traffic generators, video playback software, network analyzers, and monitoring

tools. Ensure that the test equipment can accurately measure data throughput and latency.

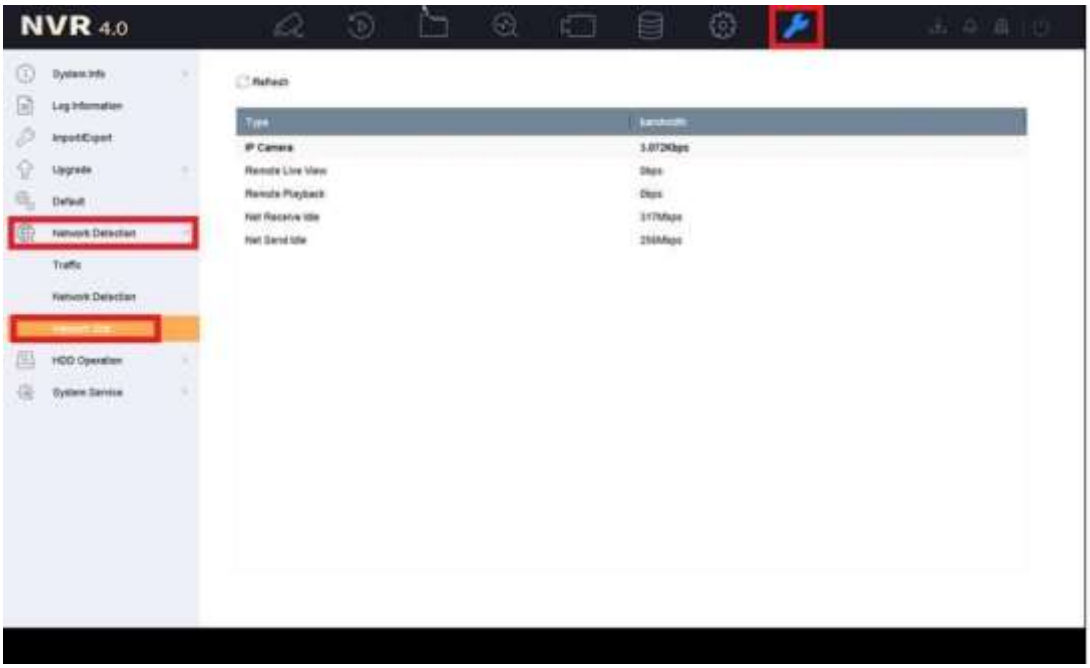
3. **Prepare Test Environment:** Set up a test environment that closely resembles the production environment of the NVR/DVR system. Connect the cameras to the NVR/DVR and configure the network infrastructure, including switches, routers, and cables.
4. **Configure NVR/DVR Settings:** Configure the NVR/DVR settings to match the desired test parameters, including video resolution, frame rate, compression settings, and recording modes. Ensure that the NVR/DVR is set up to record and transmit video streams according to the test scenario.
5. **Perform Recording Throughput Test:** Initiate a recording throughput test by simulating multiple camera streams at different resolutions and frame rates. Monitor the NVR/DVR's ability to record and store video data in real-time without dropping frames or experiencing buffer overflows.
6. **Conduct Playback Throughput Test:** Playback recorded video footage from the NVR/DVR while simultaneously monitoring data throughput, latency, and video quality. Test various playback scenarios, including single-camera playback, multi-camera playback, and fast-forward/rewind operations.
7. **Evaluate Live Viewing Throughput:** Test the NVR/DVR's live viewing throughput by accessing live video streams from connected cameras through client viewing devices. Monitor the latency and smoothness of live video playback, especially during periods of high network traffic or multiple concurrent connections.
8. **Assess Remote Access Throughput:** Test remote access throughput by accessing the NVR/DVR's video streams remotely from client devices over the internet or a network connection. Evaluate the performance of remote viewing, including latency, buffering, and video quality, under different network conditions.
9. **Measure Network Bandwidth Utilization:** Use network monitoring tools to measure the network bandwidth utilization during the throughput tests. Monitor network traffic patterns, data rates, and packet loss to identify potential bottlenecks or performance issues.
10. **Analyze Test Results:** Analyze the test results to determine the NVR/DVR's throughput capabilities and identify any performance limitations or areas for improvement. Compare the measured throughput against the device's specifications and recommended best practices.
11. **Document Findings:** Document the test procedures, results, and observations in a comprehensive test report. Include details such as test configurations, equipment used, test duration, and any issues encountered during testing.
12. **Optimize NVR/DVR Configuration:** Based on the test results and findings, optimize the configuration settings of the NVR/DVR to improve throughput performance.

Adjust recording settings, network parameters, and video compression settings as needed to achieve optimal performance.

- **Steps to watch the bandwidth usage on NVR**

  Step 1. Login into your NVR by entering your username and password.

  Step 2. Enter the Maintenance interface and select **Network Detection** in the drop-down list on the top-left side.



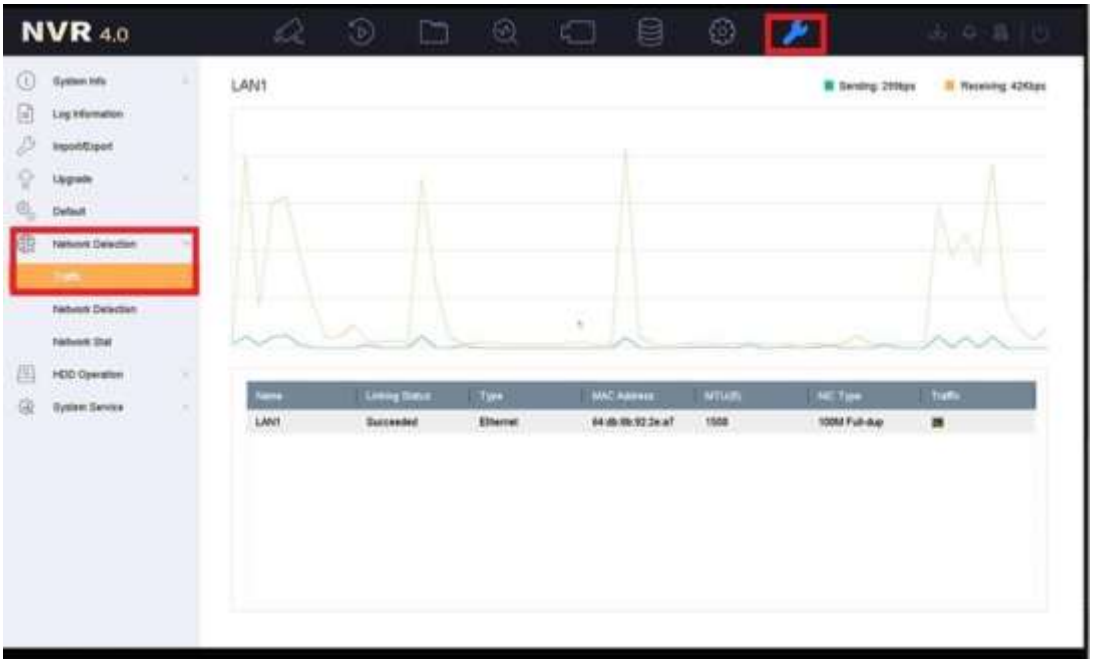
- ✓ The **Total Receiving Bandwidth** = IP Camera + Net Receive Idle.
- ✓ The **Total Sending Bandwidth** = Remote Live View + Remote Playback + Net Send Idle.
- ✓ The Net HDD such as IP SAN or NAS will use bandwidth but it's not counted in the total bandwidth.
- ✓ The maximum IP cameras' number depends on Receiving Bandwidth and the maxsupporting-channel of NVR.
- ✓ The remote live view depends on Sending Bandwidth and the number of remote connections.

- **Steps to watch Real-time Bandwidth usage in NVR**

  Step 1. Login directly into your NVR.

  Step 2. Enter the Maintenance Interface and select **Network Detection** in the drop-down list on the top-left side.

  Step 3. Select **Traffic** in the following menu.



- ✓ The Total Receiving Bandwidth = IP Camera + Net Receive Idle.
- ✓ The Total Sending Bandwidth = Remote Live View + Remote Playback + Net Send Idle.

- **CCTV Visual Quality Testing:**
- **Check Resolution:** Verify that cameras are set to the appropriate resolution and compare live footage against the camera's specifications.
- **Assess Clarity and Detail:** Observe live and recorded footage for image clarity, sharpness, and detail in various lighting conditions.
- **Test Color Accuracy:** Ensure that color reproduction is accurate and consistent.
- **Review for Artifacts:** Look for any visual artifacts like blurriness, distortion, or pixelation in both live and recorded video.
- **CCTV Latency Testing:**
- **Measure Delay:** Use network tools to measure the time delay between the live camera feed and the displayed video on the monitoring system.
- **Perform Real-Time Actions:** Test the response time of live video updates when changes or movements occur in the camera's field of view.
- **Check Recording Delay:** Verify that there is minimal delay between live events and their recording on NVR/DVR.
- **CCTV Audio Quality Testing:**
- **Check Audio Input:** Verify that microphones (if used) are properly capturing audio and transmitting it to the CCTV system.
- **Assess Clarity and Volume:** Evaluate the clarity and volume of recorded and live audio to ensure it is clear and not distorted.
- **Test Synchronization:** Ensure audio is synchronized with the video feed for accurate recording and playback.



Points to Remember

- **Visual quality is** the clarity and resolution of the image captured by a security camera. The higher the video quality, the clearer and more detailed the image will be. Improving video quality involves increasing both clarity and resolution.
- **Latency** is the time between the instant a frame is captured and the instant that frame is displayed. Latency is usually expressed in time units, e.g., seconds or milliseconds (ms).
- **Audio quality:** CCTV system is fully capable of recording sound. Most cameras are either equipped with microphones or have an input for an external mic. CCTV microphones can record quality audio up to six meters away.
- **NVR/DVR throughput:** This is the bandwidth usage on NVR/DVR. The more bandwidth of the NVR or DVR, the more the CCTV system perform fast.
- Steps to Perform CCTV Performance Testing of CCTV Visual Quality, CCTV Latency, CCTV Audio Quality and CCTV NVR/DVR Throughput.



Application of learning 5.5

XYZ Complex market want to install CCTV Camera System market buildings for better management of the security. As one of the team of System installer recruited by the company, you are asked to install CCTV cameras in of those buildings and you will also need to test performance of devices.

Task:

- ✓ Test CCTV Camera visual/Audio quality and provide the result.
- ✓ Test CCTV Camera latency and provide the result.
- ✓ Test CCTV Camera NVR/DVR throughput and provide the result.

Indicative content 5.6: Elaboration of testing report

Duration: 1 hr

Theoretical Activity 5.6.1: Elaboration of CCTV Testing report

Tasks:

1: In small groups, you are requested to answer the following questions related to the Elaboration of CCTV Testing report

i. What is testing report?
ii. What are the items to be included in CCTV Camera testing report?
iii. What are the steps for generating CCTV Camera testing report?
iv. How can you interpret CCTV Camera testing report?

2: Provide the answers for the asked questions and write them on papers.
3: Present the findings/answers to the whole class
4: For more clarifications, read the key readings activity 5.6.1
5: In addition, ask questions where necessary.

Key readings 5.6.1.:

- **Description of Testing report**

  Testing report provides the relevant information with a detailed account of the overall test results and the defects found. It aims to formally summarize the results of the entire testing process.

  Testing Report is an important document that is prepared at the end of a Testing, or rather after the Testing cycle has been completed. The prime objective of this document is to explain the details of the Testing activities performed for testing done.

  While the Daily Status Reports only deliver the daily test status and results on the overall testing activities performed in the cycle.

- **What to include in testing report?**

✓ Test objectives
✓ Areas covered
✓ Areas not covered
✓ Testing approach
✓ Defect report
✓ Platform details
✓ Overall summary

- **Interpret testing report**

A CCTV test report can capture a wide range of information, such as the camera IP address which is programmed during installation and shows the network address of the camera and whether this is fixed (static) or assigned by the network (dynamic).

The report also captures network details, which encompasses the complete network configuration of the camera including gateway, DNS, subnet information, and more to ensure the camera is connected to the intended network. The camera MAC address, which is the unique, unalterable hardware ID of the camera, can also be captured.

The IP CCTV testing process will also show where the camera is installed and what it is aimed at, along with a screen capture to show what the camera is monitoring. This is key because it shows the camera's field of view and that it is properly focused. Reports should allow multiple images to demonstrate both day and night image quality.

To prove the camera is recording at the resolution required by the client, the resolution of the video stream can be tested. The report can also show the frame rate, which is the number of images captured per send (fps). Most cameras will allow up to 25-30 fps for smooth video, but the frame rate can be reduced to lessen the impact on network bandwidth.

Bandwidth should also be tested in the report, to show the average amount of data streaming from the camera in the current video configuration settings, measured in megabits per second (Mb/s).



Practical Activity 5.6.2: Elaborate the CCTV Camera System Performance report

Task:

1: Referring to previous activities **(5.6.1.**) you are requested to perform the given task. The task should be done **individually**.

As a trainee who learned the CCTV System installation, you are asked to elaborate a CCTV system testing reports of the given CCTV installation.

2: List out the procedures to perform the given tasks.

3: Referring to the procedures to elaborate a CCTV system testing reports in task2, then perform the given tasks.

4: Present your work to the trainer and whole class.

5: Read key reading 5.6.2. and ask clarification where necessary

6: Perform the task provided in application of learning 5.6



Key readings 5.6.2

For generating testing report, use the following steps.

Step 1: Capture the purpose of the document: short description of the purpose of the document.

Step 2: **Capture an overview of the product in test**: Overview of the application tested in brief.

Step 3: Capture the Testing Scope: Areas/modules that are in Scope, Out of Scope and also the areas that were not tested due to any constraints or dependencies.

Step 4: Capture the Metrics: the test execution results, status of the cases as well as defects

Step 5: Capture the types of testing performed: various types of Testing that were performed.

Step 6: Capture the Test Environment and the Tools used: details about the Test Environment in which the testing activities were mostly carried out and the details of any tools that were used.

Step 7: Learnings during the test activity: Describe the critical issues that were faced and the solutions that were implemented

Step 8: Recommendations or Suggestions, if any: recommendations or suggestions for the relevant the planning for the next cycle

Step 9: Exit Criteria: Test Completion when certain conditions are fulfilled

Step 10: Sign Off



Points to Remember

- **Testing report:** The Interpretation (or Discussion) section of the systematic review helps readers interpret the main findings of the review, brought together in the synthesis step.
- **The items to be included in CCTV Camera testing report include** Test objectives, Areas covered, Areas not covered, Testing approach, Defect report, Platform details and Overall summary
- Steps of generating testing report

Application of learning 5.6

XYZ Complex market want to install CCTV Camera System market buildings for better management of the security. As one of the team of System installer recruited by the company, you are asked to install CCTV cameras in of those buildings and you will also need to elaborate the performance report of the installation.

Learning outcome 5 end assessment

Theoretical assessment

A. Select the letter corresponding to the correct answers.

1. Which material is typically checked when deploying a CCTV system?

A) Software updates
B) Installation manuals
C) Cable ties
D) User access credentials

2. What is a key component of hardware testing tools for CCTV systems?

A) Patch panels
B) Video analysis software
C) Network cable testers
D) System logs

3. When performing connectivity testing, which of the following is considered a form of physical connectivity?

A) Checking network IP configuration
B) Verifying cable connections between cameras and NVR/DVR
C) Analyzing system performance metrics
D) Reviewing recorded footage

4. Which testing method assesses the functionality of CCTV camera hardware?

A) Software testing
B) Performance testing
C) Functional testing
D) Connectivity testing

5. What aspect of performance testing evaluates the clarity of the video feed?

A) Latency
B) Visual quality
C) Audio quality
D) NVR/DVR throughput

6. Which tool would be used for software testing in a CCTV system?

A) Cable crimper
B) Voltage meter
C) Network simulation software
D) Signal generator

7. In functional testing, what would be tested under software?

A) Camera resolution
B) Recording schedules and settings
C) Cable integrity
D) Power supply

8. What is a critical aspect of testing the latency in a CCTV system?

A) Measuring the delay between camera feed and display
B) Assessing video storage capacity
C) Checking the quality of recorded audio
D) Evaluating the physical condition of cables

9. What should be included in a testing report for a CCTV system?

A) System installation procedures
B) Results from testing performance and connectivity
C) User manual
D) Manufacturer's warranty information

10. What is the purpose of generating a testing report for a CCTV system?

A) To document installation steps
B) To outline the costs associated with the system
C) To provide a summary of the testing results and any issues found
D) To review user training materials

B. Answer True or False

1. Material such as cable ties is typically checked when deploying a CCTV system.
2. A video analysis software is a key component of hardware testing tools for CCTV systems.
3. Verifying cable connections between cameras and NVR/DVR is considered a form of physical connectivity.

4. Functional testing assesses the functionality of CCTV camera hardware.
5. Visual quality is an aspect of performance testing that evaluates the clarity of the video feed.
6. Network simulation software would be used for hardware testing in a CCTV system.
7. Recording schedules and settings would be tested under software in functional testing.
8. Measuring the delay between camera feed and display is a critical aspect of testing latency in a CCTV system.
9. Results from testing performance and connectivity should be included in a testing report for a CCTV system.
10. The purpose of generating a testing report for a CCTV system is to provide a summary of the testing results and any issues found.

Practical assessment

MONDAY Complex market want to install CCTV Camera System market buildings for better management of the security. As one of the team of System installer recruited by the company, you are asked to Test the CCTV camera system in of those buildings and you will also need to elaborate the performance report of the installation.



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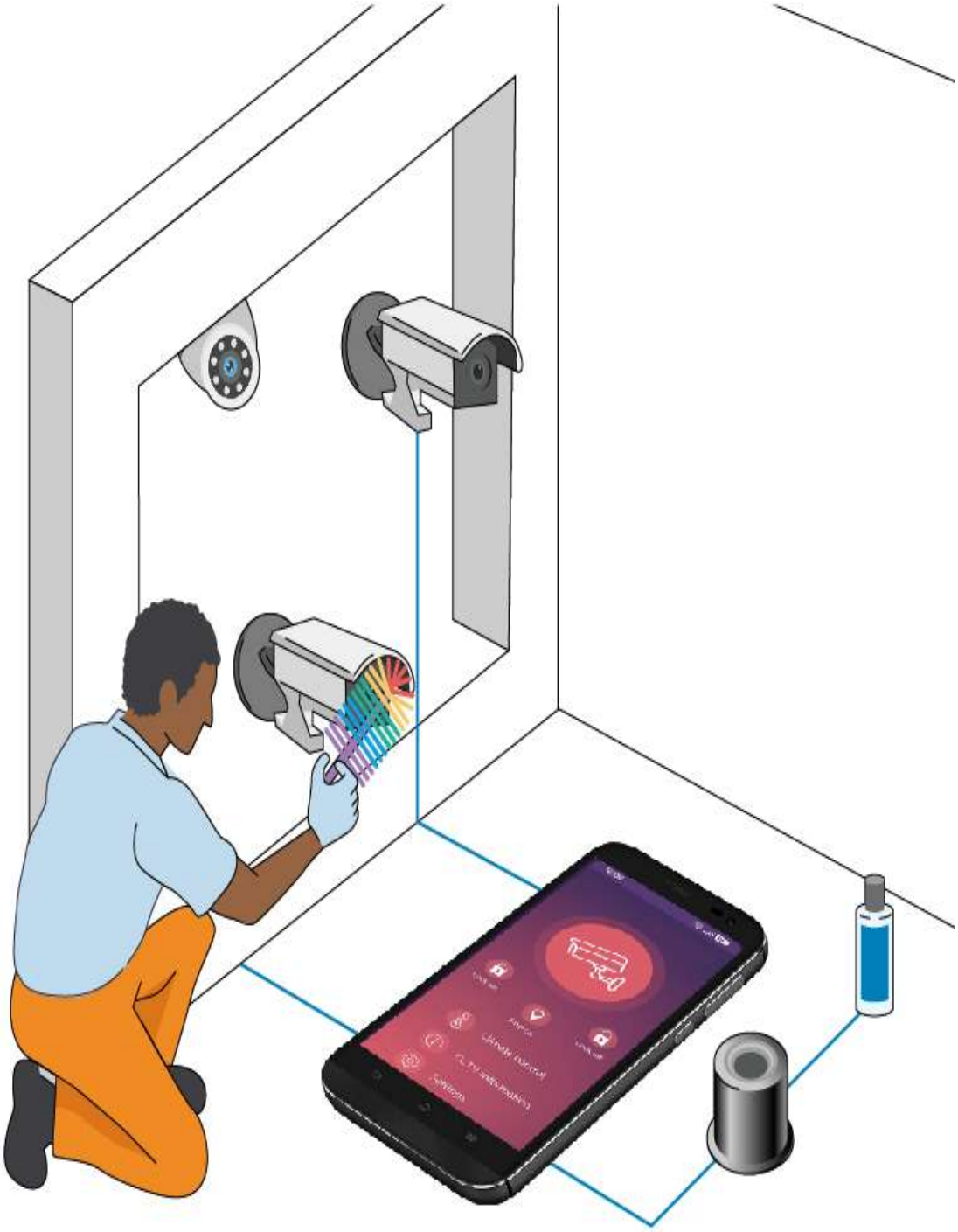
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Indicative contents
6.1. Performing preventive maintenance
6.2. Performing corrective maintenance
6.3. Elaboration of maintenance report

Key Competencies for Learning Outcome 6: Maintain CCTV system.

Knowledge	Skills	Attitudes
• Description of the types of periodic backup, • Identification of the best CCTV camera position, • Identification of the measures for Security camera (logical & physical), • Identification of common hardware and software problems of CCTV Camera Systems, • Identification of the tips for maintaining the CCTV Security system, • Description of the elements of maintenance report	• Performing corrective maintenance, • Performing periodic Backup, • Positioning CCTV components and materials, • Deciding the measures of CCTV system security, • Performing corrective maintenance, • Elaborate maintenance report.	• Be a Self-Motivator, • Be a Teamwork player, • Have a Curiosity spirit, • Have a Self-confidence spirit, • Be an innovator, • Be a problem solver • Have a Creativity spirit, • Be a Critical thinker



Duration: 10 hrs

Learning outcome 6 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Set correctly periodic backup based on the client requirements.
2. Check correctly CCTV camera positioning based on coverage area.
3. Check properly CCTV security measures of CCTV system on system diagnosis
4. Handle properly CCTV system issues-based system diagnosis.
5. Elaborate efficiently maintenance report base on CCTV system status.



Resources

Equipment	Tools	Materials
● Computer ● Monitor ● UPS ● Power Supply ● Switch ● Storage devices ● Cameras ● Recording Devices (NVR/DVR) ● Air Conditioner	● Crimping tool, ● Wire stripper, ● Drilling machine, ● Hammer, ● Ladder, ● Pliers, ● Screw drivers, ● Cable Tester.	● Internet bundles ● Connectors ● Cables ● Power extension

Indicative content 6.1: Performing preventive maintenance

Duration: 4 hrs

Theoretical Activity 6.1.1: Perform CCTV system preventive maintenance.

Tasks:

1: In small groups, you are requested to answer the following questions related to Perform CCTV system preventive maintenance

i. What is periodic backup?
ii. What are the types of periodic backup?
iii. List the advantages of periodic backup.
iv. What is scheduled backup?
v. What is camera positioning?
vi. What are the logical and physical measures that must be taken for camera security?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key readings 6.1.1

5: In addition, ask questions where necessary.



Key readings 6.1.1.

- **Set periodical backup**

  Periodic backup in a CCTV (Closed-Circuit Television) camera system refers to the process of automatically creating backups of recorded video footage at regular intervals. This backup strategy helps ensure that important surveillance footage is consistently and reliably preserved, reducing the risk of data loss due to hardware failures, system errors, or security incidents.

  Important keys aspects of periodic backup in CCTV camera systems:

1. **Automatic Backup Schedule:** Periodic backups are scheduled to occur automatically at predefined intervals, such as daily, weekly, or monthly. The backup schedule is configured within the CCTV system's settings or management software, allowing administrators to define the frequency and timing of backup operations.
2. **Backup Retention Policy:** A backup retention policy determines how long backup copies of video footage are retained before being overwritten or deleted. Administrators can specify the retention period based on organizational requirements, compliance regulations, or storage capacity constraints.
3. **Incremental or Full Backups:** Periodic backups may be performed as incremental backups, which only capture changes made since the last backup, or as full backups, which capture all data regardless of changes. Incremental backups are typically faster and require less storage space, while full backups provide complete data redundancy but consume more resources.
4. **Offsite Storage:** To enhance data resilience and disaster recovery preparedness, periodic backups are often stored offsite in secure locations separate from the primary CCTV system. Offsite storage protects backup copies of video footage from localized events such as theft, vandalism, or natural disasters affecting the primary surveillance site.
5. **Backup Verification:** After each backup operation, administrators may perform verification checks to ensure the integrity and completeness of the backup data. Verification processes may include comparing checksums, conducting test restores, or using backup validation tools to confirm the backup's accuracy.
6. **Notification and Monitoring:** The CCTV system may generate notifications or alerts to inform administrators of backup status, completion, or any errors encountered during the backup process. Monitoring tools may also be used to track backup performance metrics and identify potential issues that require attention.
7. **Compliance and Audit Requirements:** Periodic backups play a crucial role in meeting regulatory compliance requirements and audit standards for surveillance data retention and protection. Organizations operating in sectors such as law

enforcement, finance, or healthcare may be subject to specific regulations mandating the implementation of backup and archival practices.

8. **Integration with Backup Solutions:** CCTV camera systems may integrate with dedicated backup solutions or network-attached storage (NAS) devices to facilitate automated backup processes. Integration allows for seamless data transfer, centralized management, and scalability to accommodate growing storage needs.

- **Types of periodic backup**

1. Full backups

A full backup is exactly what the name implies: It is a full copy of your entire data set.

Although full backups arguably provide the best protection, most organizations only use them on a periodic basis because they are time-consuming and often require a lot of disk or tape capacity.

Because not every organization has the time or storage space for frequent full backups, other types are often necessary.

2. Incremental backups

Incremental backups only back up the data that has changed since the previous backup.

For example, suppose that you created a full back up on Monday and used incremental backups for the rest of the week. Tuesday's scheduled backup would only contain the data that has changed since Monday. Wednesday's backup would only contain the data that has changed since Tuesday, and so on.

Incremental backups were introduced as a way to decrease the amount of time and storage space that it takes to do a full backup.

3. Differential backups

A differential backup is similar to an incremental backup in that it starts with a full backup and subsequent backups only contain data that has changed. The difference is that while an incremental backup only includes the data that has changed since the previous backup, a differential backup contains all of the data that has changed since the last full backup.

- **Benefits of periodic backup**

- Data Security
- Data Recovery and Replication.
- Retaining multiple copies of data provides the insurance and flexibility to restore to a point in time not affected by data corruption or malicious attacks.
- A lot fewer files are backed up daily between full backups allowing for shorter backup windows.
- It provides a solution for long term storage or tape retention of your data.
- Quick access to the files.
- Maintaining Standards of Compliance

- **Schedule backups**

  The file that is exported by the backup process can be replayed using the video player application that is built into the integrated remote station software that the DVR comes with. The backup system is designed to export minutes' worth of video. The system is NOT meant to be an archiving service to export hours and days of video to an external drive.

  To do this, follow the steps below:

- From the left hand side select device -> camera or cameras.
- Select Record type.
- Select Stream Type (Main Stream or Sub Stream).
- Select Date and time -> Select OK -> Select Search.
- Choose a time frame and click on time clip.
- Choose the beginning and end of time clip you want to backup.
- Click on time clip again and select the folder to back up to and select the exported file type.
- Click Ok
- It should start downloading the back.
- You should now have the file backed up at the specified folder.

- **Checking of camera positioning**

  When utilizing CCTV cameras to protect your home, ensuring you know where to position the CCTV cameras is essential. What does that mean? Simply put, placing security cameras in the right place means that every part, corner and angle of your home is in the field of view of a security camera, ensuring there are no blind spots and areas that cannot be seen and observed by the security system. Whether it's the front driveway or the back door, it's important that it's covered and protected.

  Where to place your security cameras at home?

  1. Start with the Front Door

  Sounds straightforward, but the most important places you could place security cameras starts with the front door. Most break-ins and burglaries occur simply via the front door, and if you're protecting your home, ensuring the front door is under around-the-clock surveillance is the first place to go. In the case of a business or other facility, unauthorized persons are most likely to enter just by waltzing in the front door.

  With remote monitoring and viewing, a security camera positioned at the front door will also allow you to see who's knocking or ringing the doorbell when you're not home.

  2. Point Security Cameras at The Back Door of Your Home

  Next, you can move on to positioning a CCTV camera to point at the back door of your home. This is one of the most likely locations for break-ins (as well as less

violent security breaches) to occur, after the front door and it's just as vital to cover as the front door.

To make the most of the placement of your security cameras, and to prevent them from being damaged or vandalized in case of a burglary, be sure to place them out of reach. A second floor height is often ideal, provided the camera lens and picture are capable of seeing up close from such a height.



Camera Position.

3. Secure First Floor Windows

After the front and back doors, windows on the first floor of your home are some of the most likely spots for break-ins and intrusions to occur and should be high on your priorities for CCTV camera placement. Smaller cameras could be placed above the window frame, but larger, professional-grade security cameras will likely need to be placed on the side or corner of the building and pointed towards the window at an angle. In some situations, they could also be placed inside and pointed out, towards the window.

4. Place CCTV Cameras on the Corners of your Home

Positioning CCTV cameras in the right places requires complete perimeter security. To achieve this, surveillance cameras should be placed on all corners of the property, as well as corners of your home, to achieve 360-degree coverage while eliminating blind spots. This requires either an extensive array of security cameras facing in all directions and at all angles, place strategically to overlap and eliminate blind spots, or makes use of security cameras with wide, expansive field-of-view – sometimes up to 180 degrees.

In some cases, it may be more feasible to opt for PTZ cameras, which can simply be controlled and moved to point in different directions and see exactly what you want it to see. If you have a fence or barrier protecting the property, you will also want to place CCTV cameras covering the front gate and the perimeter of the fence. These cameras can be placed on the fence itself or can be placed inside the perimeter (on trees or posts, for example) pointed outwards, at the proper angles.

5. Don't Forget Interior Security Cameras

Finally, don't forget the interior of your home; it's just as important as the perimeter when it comes to CCTV camera position, in case of a break-in or security breach. Locations inside your home most vital for surveillance include stairwells, the front entrance and foyer, and of course the back door. You will also most likely want to monitor garages and loading docks. You may also want to position CCTV cameras in the basement, especially if the basement has an entrance from the outside; a night vision or IR camera will likely be necessary in such locations.



Interior Security camera

Further tips for Positioning CCTV Cameras in your Home

It can be more difficult to hide CCTV cameras placed outdoors, as there is often less to cover them up, especially when placed on the side of a building or home. Be sure to place them high enough to avoid being tampered with, as well as at the right angles to avoid blind spots. It's also a good idea to make sure your outdoor cameras are waterproof and weatherproof and avoid placing them where direct sunlight can blind them and wash out or disrupt the video. This is not only bad for surveillance, but bad for the camera over time, too.

If you're unsure of how to properly place security cameras to ensure the correct position, or need help scoping out your property and finding the best locations and angles to properly cover every angle, you may want to consult with a security systems integrator. They'll be able to help you identify exactly where to position CCTV cameras around your home.

Recommended placement of CCTV Camera is above the door and pointing away from light sources or windows, to ensure the best quality images**.**

- **Check security measures of Physical security**

Physical security describes security measures that are designed to deny unauthorized access to facilities, equipment and resources and to protect personnel and property from damage or harm.

Physical security systems for protected facilities are generally intended to:

- ✓ Deter potential intruders (e.g. Warning signs, security lighting and perimeter markings);
- ✓ Detect intrusions and monitor/record intruders (e.g. Intruder alarms and CCTV systems); and
- ✓ Trigger appropriate incident responses (e.g. By security guards and police).

Logical Security is the use of software and systems to control and limit access to information. Anti-virus software, passwords, and encryption are all examples of logical security mechanisms.

Logical Security consists of software safeguards for an organization's systems, including user identification and password access, authenticating, access rights and authority levels. These measures are to ensure that only authorized users are able to perform actions or access information in a network or a workstation

Logical Security can include items such as passwords, encryption, logical access controls, firewalls, and intrusion detection systems.



Points to Remember

- **Periodic Backup:** Regularly scheduled process of copying and storing data to secondary storage media, Ensures data redundancy and minimizes the risk of data loss in case of system failures or disasters.
- **Types of Periodic Backup** are Full Backup, Incremental Backup, Differential Backup.
- **Advantages of Periodic Backup are** Safeguards data against loss, Provides a recovery point.
- **Scheduled Backup:** Backup process initiated automatically at predefined intervals.
- **Camera Positioning:** Placement and orientation of CCTV cameras to achieve optimal coverage of the monitored area, Consider factors such as field of view, angle of coverage, and potential obstructions.
- **Logical Measures:** Implementing access controls, encryption, and authentication mechanisms to protect camera feeds and recordings from unauthorized access.
- **Physical Measures:** Installing cameras in secure locations, using tamper-resistant enclosures, and implementing physical barriers to prevent unauthorized tampering or removal of cameras.



Application of learning 6.1

MARANA Company Ltd located in Northern Province, Burera District, recruit a CCTV system installer team to facilitate its works. You are recruited in this team and you are assigned you to install CCTV cameras in the building located in KINYABABA Centre.

Task:

After installation, you are asked to perform the backup of the system, inspect camera placements to ensure complete coverage and perform physical security measures.

Indicative content 6.2: Performing corrective maintenance

Duration: 4 hrs

Theoretical Activity 6.2.1: Perform corrective maintenance

Tasks:

1: In small groups, you are requested to answer the following questions related to Perform corrective maintenance

i. What are the common hardware problems of CCTV Camera System?
ii. What are the common software problems of CCTV Camera System?
iii. What are the tips for maintaining your CCTV Security system?

2: Provide the answers for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key readings 6.2.1

5: In addition, ask questions where necessary.

Key readings 6.2.1.:

- **Hardware maintenance**

  Prevention is better than cure, and frequently monitoring your surveillance system allows you to maintain the functionality of the cameras. Maintaining your CCTV system is essential to avoid premature breakdown, which could ultimately leave your construction site vulnerable and leave you with substantial replacement bills.

- **Identifications of common problems**

  Once everything is connected and powered on, you should see the camera images on the monitor. It will be a split screen, so if you have four cameras you will see four images, one for each camera.

  If you don't see any camera images, then the first thing to check is that the box power supply is switched on and connected properly.

  If you see images for some of the cameras but not for others then check the connections and network clips, RJ45 connectors for IP cameras and BNC connectors for analogue HD cameras. If there is still no image for a camera then check the power cables are connected correctly, are positive and negative connected the right way around?

  If it's still not working, then check if the cable is damaged, test with a video test monitor with the continuity function. The camera can be powered up with the video test monitor itself and then the video cable of the camera can be connected directly to the video test monitor, so if a camera is powered up and you can see image on the video test monitor that means that the camera is working fine. If the

camera doesn't power up and there is no image, then the camera is faulty, so you need to replace it.

If you do see an image on the video test monitor for the camera in question where you cannot see the image on the DVR monitor, that means the fault is in the video or power cable to the DVR. If it's an IP camera you can use a network test tool or sometimes a network test tool is built into the video test monitor itself. If the cable doesn't work, then re-crimp both ends. If it still doesn't work, then replace the cable. If using an analogue HD camera shotgun cable, then you can test with a coax cable tester and if faulty then re-crimp both ends and if still not working then replace the cable. Then all CCTV camera images should be visible.

- **Repair/replace damaged devices and cables**

Follow through the steps on the checklist to safeguard and maintain your surveillance system appropriately and to ensure that your CCTV system is running on an operational status:

✓ **Camera Lens**

The main thing to check for and avoid with the camera lens is causes and potential causes of obstruction, which will limit the view of the camera.

- Check that nothing is obscuring the field of view of the lens and adjust if necessary
- Keep the camera clean to prevent any build-up of dust that could prohibit the device performance
- Check the focus and ensure that the camera's functions are working (i.e. pan our and zoom)
- Clean any marks off, as build-up will lead to eventual view obstruction
- Ensure that any foliage is trimmed back
- Check that the cameras are firmly attached to the post or wall
- If your CCTV has motion detection, check that they are working

✓ **Wiring**

Conduct a visual inspection of all major components on your CCTV, including connections & cabling for signs of deterioration or damage.

- Check for any loose wires, exposed or fraying wires and cables, and general wear and tear
- Check the image quality and ensure that you are getting a clear transmission of the pictures and not hearing any distortion with the sound
- Check that the power supply to the device is working

✓ **Control Equipment**

Check that all the elements of the CCTV control equipment, such as monitors, switches, VCR, DVR, and Multiplexer, are functioning sufficiently.

- Check that the quality of the image on the monitor is clear and the brightness and contrast settings are adjusted correctly
- Remove any grime and dust build-up on the monitors and control panels

- Check that cables on the control equipment are not damaged, deteriorating or have any weak connection points
- Check that the monitor is displaying the correct date and time stamp

✓ **Check the power source**

Checking the power source may seem like a simple step in troubleshooting a malfunctioning security camera, but it's a necessary one!

Most current surveillance cameras are powered through Power Over Ethernet (PoE), which sends power down the same cable as the network connection. If this is the case, verify that your PoE source has power and that it is configured to send power to the cameras.

If the surveillance camera is powered electrically through a plug, first check to make sure it is plugged in. If it's plugged in, check to make sure that the camera is receiving the appropriate wattage that is necessary for that particular device. Some advanced surveillance cameras require a higher wattage than what is traditionally pulled out of the outlet.

If the surveillance camera is battery operated, check to make sure the battery is charged and offering enough power to the camera. It is important to note that all batteries have a life span and may need to be replaced in older devices.

How to tell if your surveillance camera is on: Most security cameras have LED lights that indicate their power status. These lights may also be an indication of whether or not the camera is connected to the security network and actually transmitting data. However, each security camera is different so we recommend referring to the manual for your camera's specific power indicators.

✓ **Check to see if cameras work from inside or outside the network.**

Many problems are related to your internet connection or internet router changing. Subscribing to a new internet service, changing a router due to a failure, or software updates requiring additional configuration are often the cause of an external connection problem.

If you can connect to the cameras and system from within your network (while at your office/work location) the issue is usually related to the external connectivity. Some things to check:

Have you changed your internet provider? If so, your outside IP address may have changed. This can also be the case if you don't have a 'static IP' internet service.

Has your router been replaced? If so, you will need to configure firewall or port configuration settings on the new router.

Does your recorder/camera system have access to the internet still? This could be caused by configuration changes on the camera system itself such as a new IP address, incorrect IP gateway or other configuration changes.

✓ **Check the cabling**

Another simple yet effective step you can do while troubleshooting your business's security camera is to check the cabling.

You'll want to look for things like:

- Tangled or kinked cables
- Frayed wires
- Loose connections.
- It is also valuable to double-check that your wires are properly and securely connected to the proper ports.

✓ **TRADITIONAL CAMERA: CHECK THE FOLLOWING ITEMS**

1. Clean lens and enclosure glass plate.
2. Remove dirt, moisture or other foreign substances.
3. Clean camera enclosure filter.
4. Clean enclosure fan assembly.
5. Confirm operation of enclosure fan.
6. Confirm operation of heater strips.
7. Check for damage inside the enclosure, including insect damage, lightning damage, or other mechanical failure.
8. Inspect cables and connectors and cable protection between pole and pan/tilt unit or camera enclosure for abrasions, cracks, or deterioration.
9. Inspect concrete poles for hairline cracks wider than 1mm, and metal poles for any cracks, mechanical damage or corrosion.
10. Verify operation of pan/tilt, zoom and focus using appropriate tool or via RESCU Control Room Operators.
11. Check enclosure grounds are securely connected.
12. Report any visible damage.

✓ **CCTV CAMERA: DOME CAMERA -> CHECK THE FOLLOWING ITEMS**

1. Clean camera eyeball.
2. Clean inside and outside of dome cover.
3. Remove dirt, moisture or other foreign substances.
4. Clean enclosure fan assembly.
5. Confirm operation of fan.
6. Confirm operation of heater.
7. Check for damage inside the enclosure, including insect damage, lightning damage or other mechanical failure.
8. Inspect cables and connectors and cable protection between pole and mounting base for abrasions, cracks, or deterioration.
9. Inspect concrete poles for hairline cracks wider than 1mm, and metal poles for any cracks, mechanical damage or corrosion.
10. Check housing grounds are securely connected.
11. Inspect camera lowering device for damaged and functionality. (if applicable)

12. Report any visible damages

- **Tips for maintaining your CCTV Security system.**

✓ **Regularly Clean Camera Lenses**

Check your indoor and outdoor cameras regularly to ensure that the lenses are clean. Dust, dirt, and fingerprints can all interfere with the quality of your video footage.

Use a soft, dry cloth to wipe them down to clean your camera lenses. If there are any stubborn stains, you can use a mild cleaning solution (like soap and water) to remove them. Just be sure to avoid using any harsh chemicals, as these can damage your camera lens.

✓ **Inspect Cables and Connections**

Next, take a look at all of the cables and connections to make sure that they are secure. Loose cables can cause video quality issues, so it's important to check them on a regular basis and tighten any loose connections.

✓ **Test Your System Regularly**

It's important to test your CCTV system regularly to ensure that it works properly. This includes testing the cameras, DVR, and monitor to make sure that they are all functioning correctly. You should also test your system's recording capabilities to ensure that it is recording high-quality video footage.

✓ **Back-Up Your Video Footage**

Make sure to regularly back up your video footage to avoid losing important data. This is especially important if you are using a cloud-based storage system. In addition, backing up your footage will help you avoid any potential issues if your system fails or if your footage is accidentally deleted.

✓ **Keep Your Software Up-To-Date**

One of the most important CCTV maintenance tips is to keep your security camera software up-to-date. Security camera manufacturers regularly release updates for their products, and these updates often include new features and bug fixes. By keeping your software up-to-date, you can ensure that your system is always running at its best.

✓ **Monitor Your System Remotely**

If you have a remote monitoring system in place, make sure to regularly check in on it to ensure that everything is working properly. This includes checking the DVR, cameras, and monitor to ensure that they are all online and functioning correctly.

✓ **Check the Power Supplies**

Another important CCTV maintenance task is to check the power supplies for your security cameras regularly. This includes checking the batteries (if your system uses them) and making sure that the cameras are connected to a reliable power source.

If you notice any issues with the power supply, be sure to contact a professional for help.

✓ **Check the Lighting Conditions**

When checking your security cameras, pay attention to the lighting conditions around them. So f you notice that the camera is not getting enough light, you may need to adjust the camera's position or add additional lighting. Poor lighting can cause video quality issues, so it's important to make sure that your cameras are properly lit.

✓ **Check the Hard Drives Periodically**

Don't forget to check the hard drives for your security cameras regularly. This includes checking for any errors or issues and making sure that they have enough free space. If you notice any problems with the hard drive, be sure to contact a professional for help.

✓ **Hire a Professional**

Speaking of which, if you don't feel comfortable maintaining your CCTV system yourself, you can always hire a professional to do it for you. Security camera maintenance is a critical part of owning a security camera system, and it's important to make sure that it is done properly. Hiring a professional to do it for you can help ensure that your system is always running smoothly.

- **Software maintenance**

Software maintenance is the process of **changing**, modifying, and **updating software** to keep update with the needs.

✓ **Identification of common problems**

These are just the most common problems that might happen to security cameras.

- No video even when the LED lights are on.
- It's not recording any video.
- The security camera can't connect to the Wi-Fi.
- It works in the daytime but not at night.
- The security camera playback is not working.
- Can't remotely view the security cameras.
- The night vision does not work.
- The infrared is not working
- There are green and red lights that are continuously flashing on the DVR or NVR
- The video has no color, blurred, or not responding

✓ **Regular Change of controller device credentials**

Change Passwords Regularly. Regularly change the credentials to your devices to help ensure that only authorized users are able to access the system.

Guidelines to ensure the security of passwords on Hikvision DVR, NVR and IP Cameras:

- Complex passwords; Upper Case Letters + Lower Case Letters + Numbers

- Do not use the same password on other websites, systems and devices
- Log into the camera through the web browser
- Follow pop-ups to install the plug-in or allow it to run on your browser
- Configuration >> Basic Configuration >> Security
- Highlight the desired user and click the 'Modify' button
- Enter the new password (strong password is recommended)

Steps to change default password on Hikvision DVR/NVR

- Go to Main Menu >> Configuration >> User.
- Select the desired user and click 'Edit' icon.
- Input the old password.
- Check change password box.
- Input the new password (complex password is highly recommended). Confirm the new password.

✓ **Installing updates and features of firmware and software for devices**

➢ **Reboot the system**

If you've ever watched *The I.T. Crowd*, you'll be familiar with the question: "Did you try turning it off and on again?" This classic line is truly a staple with any tech troubleshooting. If your security camera is on the fritz, first try giving it a reboot. **Be sure to wait about 15 seconds before turning it back on.**

You might be asking: "*How do I reboot my CCTV camera?*" The simple answer is that every CCTV camera is different and will have a different method for being rebooted. We recommend referring to your system's manual for specific instructions.

➢ **Know your login details**

Before you do any troubleshooting with your surveillance system, we strongly recommend you know the login details. This means all usernames, email addresses, passwords, codes, answers to security questions, etc.

If you jump straight to rebooting or unplugging the device and you don't have your login details handy, you may be adding unnecessary stress to your day.

➢ **Check for an IP conflict**

Sometimes the reason for a surveillance camera is acting up can be because of an IP address conflict. During install, each security camera is assigned a unique IP address. If there is another device using the same IP address, this will cause an IP conflict and a malfunctioning security camera.

First, you'll need to locate your camera's IP address. Then you will need to ping the camera. **How to locate your security camera's IP:** Locate the camera's IP address by checking the network page on the security camera software. This page will display all of your camera IP address information.

How to ping your security camera: Type “cmd” in the control screen’s search function. This will open a DOS command prompt where you can “ping” command to try and connect to the camera.

How to check for IP conflicts: Type “arp-a” into the control screen. If there is a conflict, change the camera's IP address or assign it a new one.

How to see if the camera is connected to the network: Type” ping cameraIPaddress-t”. If you receive an “Unchangeable” or “Times Out” notification, it means the camera is not connected.

Pro tip: Enable the DHCP function so you can find the camera’s IP address on the network. Some brands will require this to be done manually on the camera settings page.

- **Check ARP tables**

  Your security camera’s ARP (Address Resolution Protocol) tables will list both the IP address and the MAC address of the camera. Use the IP address (from step 5) and the MAC address to cross-reference the ARP table to ensure all addresses match up.

  How to find a camera’s MAC address: Most security camera MAC addresses can be found on the product label directly on the camera itself. It will likely be located near a barcode. If there is no product label on the camera, login to the network system and search for the camera’s information page.

  If the IP and MAC addresses do not match, update them in the control settings.

  Pro tip: Keep an organized record of all security camera IP and MAC addresses.

- **Update software & firmware**

  Most surveillance camera manufacturers will release software and firmware updates to help with common issues, glitches, or to add helpful new features. We recommend checking the manufacturer’s website for software and firmware releases and updating your system to the latest version.

  Pro tip: Make sure you’re downloading the right firmware and software for your specific system. If you download the wrong versions, this may cause more problems.

- **Reset to factory settings**

  We know this is no one’s favorite option. Starting over can be time-consuming as it often means the loss of your settings and preferences. But, it can also be incredibly effective in getting your company’s security cameras back up and running!

  For cabling cameras: The factory reset is often a button along one of the cables.

  For wireless cameras: The factory reset often occurs if you press and hold the power button for longer than it takes to simply reboot it.

  It is important to note that every camera manufacturer will have a different factory reset process so it is best to confirm with the user manual.

- **Call your technology provider (Like Paladin Technologies)**
  If you have a service provider for your technology needs, they may have information to help your situation.
- Most service providers, such as Paladin Technologies, will be able to provide you with assistance and service support including hardware failures, configuration changes and complete system recovery after a failure.



Practical Activity 6.2.2: Perform CCTV system corrective maintenance.



Task:

1: Referring to previous activities **(6.2.1**) you are requested to perform the given task. The task should be done **individually**.

According to the CCTV system that you have, Perform the following tasks: As a trainee who taught how to Repair/replace damaged devices and cables, Regular Change of controller device credentials and Installing updates and features of firmware and software for devices, you are asked to Repair/replace damaged devices and cables, Regular Change of controller device credentials and Installing updates and features of firmware and software for devices of the given CCTV installation.

2: Identify the steps required to perform the given tasks.
3: Referring to the previous task given, perform the given tasks
4: Present your work to the trainer and whole class.
5: Read key reading **6.2.2** and ask clarification where necessary
6: Perform the task provided in application of learning 6.2



Key readings 6.2.2

- **General process and steps for Performing Corrective Maintenance:**
- ✓ **Hardware Maintenance:**
- **Identification of Common Problems:**
  - Check if the CCTV system displays images correctly.
  - Ensure the power supply is connected and functioning.
  - Verify connections and cables (e.g., RJ45 for IP cameras, BNC for analog cameras).
  - Test with a video test monitor to determine if the camera is faulty.
  - Inspect cables for damage and re-crimp or replace as necessary.
- **Repair/Replace Damaged Devices and Cables:**
  - Replace or repair faulty cameras, cables, or connectors.

- Clean camera lenses and enclosures, and check focus and functionality.
- Inspect and repair wiring, power supplies, and control equipment.
- Confirm the operation of hardware components, including camera mounts and power supplies.

✓ **Software Maintenance:**

Identification of Common Problems:

- Troubleshoot issues like no video, recording failures, connectivity problems, or playback issues.
- Verify if the camera is recording and connecting to the network properly.
- Check for software errors or performance issues.

Regular Change of Controller Device Credentials:

- Update passwords regularly to enhance security.
- Use complex passwords and ensure they are not reused across different systems.
- Access device settings and change credentials through the device's web interface or control panel.

Installing Updates and Features of Firmware and Software:

- Check for firmware and software updates from the manufacturer.
- Download and install updates to address bugs, enhance features, and improve security.
- Reboot the system after updates to ensure changes take effect.
- Test the system to confirm that updates have been applied correctly and that the system functions as expected.

➢ **Regular Change of controller device credentials**

✓ **Steps to change the controller devices credentials are:**

1) Login to your CCTV system via computer browser (Chrome, Firefox, or IE), or the CMS software you have installed.

2) If you log in using browser, go to "configuration" -> "System" -> "User Management", you will see user details over there.

3) Select the user you want to modify, and change the password by clicking "Modify". Remember to click "Ok" after you have changed the password.

4) If you log in using CMS software, go to "Device Management". Select the device in there, and click "Remote Configuration".

5) Click "User" on the left side. Then select the user you want to modify, and change the password by clicking "Modify". Remember to click "Save" after you have changed the password.

6) You are done! Please remember to update the password in the CMS software, and in the Safe Trolley CCTV App of your mobile devices. Here is the instruction on how to configure your app on mobile devices.

7) Last by not least. **Please remember your new password**. It is a bit troublesome to reset this admin password, if you have forgotten it.

- **The process of upgrading CCTV camera firmware**

  The process of upgrading CCTV camera firmware can vary depending on the manufacturer and model of the camera.

  General steps that may help guide you through the firmware upgrade process:

  1. Identify the Manufacturer and Model: Determine the manufacturer and model of your CCTV camera. This information is usually printed on the camera itself or can be found in the camera's documentation or settings menu.

  2. Check for Firmware Updates: Visit the manufacturer's official website or support portal to check if there are any available firmware updates for your camera model. Look for a "Downloads" or "Support" section on their website.

  3. Download the Firmware: Locate the firmware update specifically intended for your camera model. Ensure that you select the correct firmware version to avoid compatibility issues. Follow the instructions provided by the manufacturer to download the firmware file to your computer.

  4. Prepare for the Upgrade: Before proceeding with the firmware upgrade, it is crucial to ensure that your camera is connected to a stable power source to prevent any interruptions during the process. Review the firmware update instructions provided by the manufacturer to understand any specific requirements or precautions.

  5. Access the Camera's Web Interface: Open a web browser on your computer and enter the IP address of the CCTV camera into the address bar. This should bring up the camera's web interface.

  6. Log In to the Camera: Enter the appropriate login credentials to access the camera's settings. This typically involves entering the camera's IP address, username, and password.

  7. Locate the Firmware Upgrade Section: Navigate to the settings menu within the camera's web interface and look for the firmware upgrade section. The exact location and terminology may vary depending on the camera's manufacturer and model. Refer to the camera's user manual or manufacturer's documentation for specific instructions on finding the firmware upgrade section.

  8. Perform the Firmware Upgrade: Follow the instructions provided by the manufacturer to select the downloaded firmware file from your computer and initiate the upgrade process. The firmware upgrade process may take several minutes, during which the camera may restart and temporarily become inaccessible.

  9. Verify the Firmware Upgrade: Once the upgrade process is complete, log back into the camera's web interface and check the firmware version to confirm that the upgrade was successful. It's also a good idea to test the camera's functionality to ensure everything is working as expected.

  Important Note: It is essential to carefully follow the manufacturer's instructions

when performing a firmware upgrade, as incorrect procedures or interruptions during the process could potentially damage the camera or render it inoperable. If you are unsure or uncomfortable performing the upgrade yourself, consider contacting the manufacturer's support team or consulting a professional technician for assistance.



Points to Remember

- Common Hardware Problems of CCTV Camera System.
    1. Power Supply Issues
    2. Loose or Damaged Cables
    3. Faulty Camera Lenses
    4. Weather Damage
    5. Recording Device (NVR/DVR) Malfunction
    6. Infrared (IR) Lighting Failure
- Common Software Problems of CCTV Camera System.
    1. Login Issues
    2. Recording and Playback Failures
    3. Motion Detection Failures
    4. Firmware Issues
    5. Network Connectivity Issues
    6. Incorrect Time and Date Settings
- Tips for Maintaining Your CCTV Security System.
    1. Regularly Clean Camera Lenses
    2. Inspect and Test Hardware
    3. Update Software and Firmware
    4. Ensure Proper Power Supply
    5. Check Recording Devices
    6. Monitor Network Connectivity
    7. Backup Footage
    8. Perform Routine System Audits.
- Steps to perform CCTV System Corrective Maintenance.
    1. Identify the Issue
    2. Power Cycle
    3. Check Connections
    4. Inspect Cameras
    5. Replace Faulty Components
    6. Test the System



Application of learning 6.2

XYZ Hotel has installed the CCTV camera systems in their buildings for better management of security but the system indicates the system software out of date and the system is indicating other hardware faults. As CCTV system installer, you are hired by the Hotel handle this problem and also you are asked to:

- Perform Hardware Maintenance by Conduct a physical inspection of cables, connectors, cameras, and power supplies and Replace/repair damaged cables and connectors, and install new devices as needed.
- Perform Software Maintenance like Firmware Updates, Software Updates and Changing Device Credentials.

Indicative content 6.3: Elaboration of maintenance report

Duration: 2 hrs

Theoretical Activity 6.3.1: Elaboration of CCTV maintenance report

Tasks:

1: In small groups, you are requested to answer the following questions related to Elaboration of CCTV maintenance report

i. What is maintenance report?
ii. What are the elements of maintenance report?
iii. What can be suggested to be included in maintenance report?
iv. What are the recommendations that must be given to the CCTV Camera.

2: Provide the answer for the asked questions and write them on papers.
3: Present the findings/answers to the whole class
4: For more clarification, read the key readings 6.3.1
5: In addition, ask questions where necessary.

Key readings 6.3.1.:

- **Introduction to Maintenance report**

  A maintenance report is a detailed document that covers any routine or emergency maintenance services performed on a specific CCTV Camera System. It serves as a guide outlining the proper care of the equipment in your workplace and can help your company avoid duplicate service requests.

  Maintenance reports play a crucial role in several ways such as maintenance planning, forecasting, and maintenance budgeting. Now if reports are not accurate or not complete then it will impact all these ways as well. That is why accuracy and complete data are so important.

  For any report to be useful, it has to be:

✓ **Concise**: the report should be brief but comprehensive; presented in a way that makes complex data easy to digest.
✓ **Fact-based**: accurate; up-to-date; objective, based on hard data rather than opinions and speculations.
✓ **Reviewed in the proper context**: was downtime prolonged because a supplier delayed the shipment of an important spare part or because we didn't order it in time – context is crucial for decision making.

- **Elements of Maintenance report**

At the conclusion of testing, produce a test report, that addresses the following elements:

- ✓ **Background**
  Explain what is being tested and why the testing is being done
- ✓ **Test team**
  Identify all people who were involved with the testing and their roles
- ✓ **Requirements summary**
  Briefly describe the CCTV Camera requirements and reference the requirements document for more details.
- ✓ **Test methods and tools**

Describe how the testing was accomplished and the tools that were used to collect the data.

- ✓ **Test results and analysis**
  Include all applicable test data. Many test tools put data in a format that you can include in your test report. If this is too cumbersome for inclusion directly within the test report, reference applicable test files. Also, explain the results, including any underlying issues that might be causing problems.
- ✓ **Recommendations**
  Explain what changes should be made to the network to counteract issues found during testing.


Thus, test documentation becomes a vital part of CCTV Camera. Managers and support staff can refer back to test reports in the future to better understand why changes were made to the network and what might be useful to fix future problems. As a result, be certain to fully document any testing that you do.

This depends entirely on what you are making the report for. It is impossible to give a unified template. In general, an official maintenance report might include:

- information about the organization (name, logo, address, contact information…)
- name/type of the report
- timeframe for which the report was generated
- report details (the main part of the report that outlines all of the data)
- any additional information that might be useful (considered on a case-to-case basis)

- **Tips for maintenance report**

When making maintenance report, includes the followings:

- ✓ Status before maintenance
- ✓ Activity Procedures
- ✓ Tools, materials and equipments used
- ✓ Status after maintenance

- **Recommendations to the CCTV System**

- It is advisable to monitor the entire site of the high-security truck park using a CCTV system which provides full coverage of all areas.

- The entirety of the fencing including all the pedestrian entrances and exits should be monitored using a CCTV system which makes it possible to view all parts of the fencing via monitors. The cameras must be installed on their masts in such a way that their areas of coverage overlap.
- It is preferable to install panning and tilting CCTV cameras which permit the periodic surveillance of the spaces between the parking bays during both daytime and night time.
- The check-in/check-out area for arriving or departing trucks should be subject to permanent CCTV surveillance offering suitable high-resolution images in order to ensure the unambiguous subsequent identification of individuals and vehicles. The CCTV pictures must be recorded.
- If permitted, CCTV surveillance should also be used outside of the dual access gate system for truck drivers and crew waiting to be checked in
- Baffle gate systems for pedestrians should be subject to permanent CCTV surveillance offering suitable high-resolution images in order to ensure the unambiguous subsequent identification of individuals. The CCTV pictures must be recorded.
- The digital video recorders should be positioned in such a way that it is not possible for the control personnel to interfere with the CCTV recordings.
- It is recommended that the digital video recorders be installed in an intrusion-resistant, alarm protected room in accordance with DIN ENV 1627 (Requirements for and classification of intrusion-resistant doors and windows), class WK5
- The surveillance monitors should be installed in the gate area. The monitors are only activated if an alarm is issued or as necessary.
- The illumination systems must respond to the differing camera requirements for both daytime and nighttime operation.
- The CCTV system should be planned and installed in such a way that the intended surveillance area can be observed at the monitors both day and night and independently of weather conditions.

Points to Remember

- **Maintenance report:** A document detailing the status of CCTV systems, including issues, repairs, and maintenance tasks, serving as a record for tracking performance.
- **Elements of maintenance report:** Identification of cameras, date/time of maintenance, issue descriptions, repair details, and recommendations for improvements.
- **Suggestions for report inclusion:** Summary of system performance, analysis of issues, spare parts inventory, and recommendations for optimizing performance.
- **Recommendations for maintenance:** Regular inspection, lens cleaning, testing functions, firmware updates, staff training, and implementing a maintenance schedule.



Application of learning 6.3

XYZ Hotel has installed the CCTV camera systems in their buildings for better management of security. As CCTV system installer, you are hired by the Hotel to elaborate maintenance report showing all details of maintenance performed.



Learning outcome 6 end assessment

Theoretical assessment

Part I. Read carefully the following statements, answer by TRUE if the statement is correct otherwise FALSE.

1. Periodic backups are performed on a set schedule, rather than event-driven, such as whenever new data is saved to a hard drive or existing data files are modified……
2. While an incremental backup only includes the data that has changed since the previous backup, a differential backup contains all of the data that has changed since the last full backup….

Part II. Select the letter corresponding to the correct answer

3. Which of the followings is not a benefit of periodic backup
   A. Data Recovery and Replication.
   B. Data Security
   C. Maintaining Standards of Compliance
   D. Quick access to the files.
   E. Virus elimination.
4. When choosing the best camera position, remember to place CCTV Cameras on the Corners of your Home.
   A. False
   B. True
5. It is recommended to place CCTV Camera above the door and pointing away from light sources or windows, to ensure the best quality images.
   A. False
   B. True
6. If you do see an image on the video test monitor for the camera in question where you cannot see the image on the DVR monitor, that means the fault is in the video or power cable to the DVR.
   A. False
   B. True
7. These are just the most common problems that might happen to security cameras, EXCEPT:
   A. Can't remotely view the security cameras.
   B. It's not recording any video.
   C. The night vision does not work.
   D. The security camera playback

8. At the conclusion of testing, produce a test report, that addresses the following elements, EXCEPT:
   A. Conclusions
   B. Requirements summary
   C. Test methods and tools
   D. Test results and analysis

Practical assessment

PORODO Investment Group wants to maintain the CCTV camera systems installed in their buildings for better management of security. As CCTV system installer, you are hired by this Investment Group to maintain this CCTV system by maintaining Hardware and software faults, by also doing data backup and perform security measures. After work, you are also requested to report the work done to your Supervisor.



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